



US012414962B2

(12) **United States Patent**
Havranek

(10) **Patent No.: US 12,414,962 B2**

(45) **Date of Patent: Sep. 16, 2025**

(54) **METHOD FOR TREATING *C. acnes* BACTERIA-ASSOCIATED DISEASES**

(71) Applicant: **ELIGO BIOSCIENCE**, Paris (FR)

(72) Inventor: **Jan Havranek**, Paris (FR)

(73) Assignee: **Eligo Bioscience**, Paris (FR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/782,965**

(22) Filed: **Jul. 24, 2024**

(65) **Prior Publication Data**

US 2025/0032528 A1 Jan. 30, 2025

Related U.S. Application Data

(60) Provisional application No. 63/515,223, filed on Jul. 24, 2023.

(51) **Int. Cl.**

A61K 31/7088 (2006.01)

A61P 17/10 (2006.01)

A61P 31/04 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 31/7088** (2013.01); **A61P 17/10** (2018.01); **A61P 31/04** (2018.01)

(58) **Field of Classification Search**

CPC **A61K 31/7088**; **A61P 17/10**; **A61P 31/04**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,113,163 B2 10/2018 Liu et al.
2015/0166980 A1 6/2015 Liu et al.

FOREIGN PATENT DOCUMENTS

WO 2017/136738 A2 8/2017
WO 2017/141173 A2 8/2017
WO 2018/236548 A1 12/2018
WO 2020/181178 A1 9/2020
WO 2020/181180 A1 9/2020
WO 2020/181193 A1 9/2020
WO 2020/181195 A1 9/2020
WO 2020/181202 A1 9/2020

OTHER PUBLICATIONS

Abudayyeh et al. RNA targeting with CRISPR-Cas13a, *Nature*, 2017, 550(7675), 280-284.

Anzalone et al. Search-and-replace genome editing without double-strand breaks or donor DNA, *Nature*, 2019, 576 (7785), 149-157.

Barnard et al. The balance of metagenomic elements shapes the skin microbiome in acne and health. *Scientific Reports*, 2016, 6(39491), 1-12.

Barnard et al. Porphyrin Production and Regulation in Cutaneous Propionibacteria, *mSphere*, 2020, 5(1), e00793-19, 1-10.

Bhate and Williams et al. Epidemiology of acne vulgaris, *British Journal of Dermatology*, 2013, 168, 474-485.

Briers et al. A standardized approach for accurate quantification of murein hydrolase activity in high-throughput assays, *J. Biochem. Biophys. Methods*, 2007, 70, 531-533.

Bruggemann et al. A Janus-Faced Bacterium: Host-Beneficial and Detrimental Roles of *Cutibacterium acnes*, *Frontiers in Microbiology*, 2021, 12(673845), 1-22.

Chen et al. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34, 2018, i884-i890.

Chen et al. Precise and programmable C:G to G:C base editing in genomic DNA. *bioRxiv*, 2020, 213827, 1-19.

Chen et al. Programmable C:G to G:C genome editing with CRISPR-Cas9-directed base excision repair proteins. *Nature Communications*, 2021, 12:1384, 1-7.

Chernomor et al., Terrace Aware Data Structure for Phylogenomic Inference from Supermatrices, *Syst. Biol.* 2016, 65(6), 997-1008.

Choby and Skaar. Heme Synthesis and Acquisition in Bacterial Pathogens, *J Mol Biol.* 2016, 428(17), 3408-3428.

Ding et al. panX: pan-genome analysis and exploration. *Nucleic Acids Research*, 2018, 46(1), 1-12.

Donovan et al. Lysis of staphylococcal mastitis pathogens by bacteriophage phi11 endolysin, *FEMS Microbiol Lett*, 2006, 265, 133-139.

Edgar. Muscle: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Research*, 2004, 32(5), 1792-1797.

Farzadfard et Lu. Genomically Encoded Analog Memory with Precise In vivo DNA Writing in Living Cell Populations. *Science*. 2014, 346(6211), 1256272, 1-18.

Fitz-Gibbon et al. Propionibacterium acnes Strain Populations in the Human Skin Microbiome Associated with Acne. *Journal of Investigative Dermatology*, 2013, 133, 2152-2160.

Fonfara et al. Phylogeny of Cas9 determines functional exchangeability of dual-RNA and Cas9 among orthologous type II CRISPR-Cas systems. *Nucleic Acids Research*, 2014, 42(4), 2577-2590.

Gaudelli et al. Programmable base editing of A•T to G•C in genomic DNA without DNA cleavage, *Nature*, 2017, 551 (7681), 464-471.

Grunewald et al. A dual-deaminase CRISPR base editor enables concurrent adenine and cytosine editing, *Nat Biotechnol*, 2020, 38(7), 861-864.

Huang et al. Circulating Antibodies to Skin Bacteria Detected by Serological Lateral Flow Immunoassays Differentially Correlated With Bacterial Abundance, *Frontiers in Microbiology*, 2021, 12, 709562, 1-10.

Jinek et al. A programmable dual RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*, 2012, 337 (6096), 816-821.

Johnson and Cummins. Cell Wall Composition and Deoxyribonucleic Acid Similarities Among the Anaerobic Coryneforms, Classical Propionibacteria, and Strains of *Arachnia propionica*, *Journal of Bacteriology*, 1972, 109(3), 1047-1066.

(Continued)

Primary Examiner — J. E. Angell

(74) *Attorney, Agent, or Firm* — Arrigo, Lee, Guttman & Mouta-Bellum LLP

(57) **ABSTRACT**

The present invention concerns a method for treating or preventing a *Cutibacterium acnes* bacteria-associated disease in a subject, said method comprising modulating, in particular reducing, in said subject, the ratio of the amount of *C. acnes* bacteria of a newly identified category called α -type to the amount of *C. acnes* bacteria of newly identified category called β -type, the α -type category efficiently defining acne-associated strains and the β -type defining non acne-associated strains.

19 Claims, 8 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

OTHER PUBLICATIONS

- Johnson et al. Strain-Level Differences in Porphyrin Production and Regulation in *Propionibacterium acnes* Elucidate Disease Associations, *mSphere*, 2016, 1(1), e00023-15, 1-12.
- Karberg et al. Group II introns as controllable gene targeting vectors for genetic manipulation of bacteria, *Nature Biotechnology*, 2001, 19, 1162-1167.
- Kelhala et al. Isotretinoin and lymecycline treatments modify the skin microbiota in acne, *Exp Dermatol.*, 2018, 27 (1), 30-36.
- Kolar et al. *Propionibacterium acnes*-induced immunopathology correlates with health and disease association, *JCI Insight*. 2019;4(5):e124687, 1-12.
- Komor et al. Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. *Nature*. 533(7603), 420-424.
- Koonin et al. Diversity, classification and evolution of CRISPR-Cas systems. *Current Opinion in Microbiology*, 2017, 37, 67-78.
- Kosiol and Goldman. Different Versions of the Dayhoff Rate Matrix, *Molecular Biology and Evolution*, 2005, 22(2), 193-199.
- Kurt et al. CRISPR C-to-G base editors for inducing targeted DNA transversions in human cells, *Nat Biotechnol.*, 39 (1), 41-46.
- Laclaverie et al. Development and characterization of a 3D in vitro model mimicking acneic skin. *Exp Dermatol*, 2020, 30, 347-357.
- Langmead and Salzberg. Fast gapped-read alignment with Bowtie 2, *Nature methods* 2012, 9(4), 357-360.
- Lessin and Leyden, *Cutibacterium acnes* (formerly *Propionibacterium acnes*) In-Vivo Reduction Assay: A Pre-Clinical Pharmacodynamic Assay for Evaluating Antimicrobial/ Antibiotic Agents in Development for Acne Treatment Skin Microbiome Handbook: From Basic Research to Product Development, 2020, 289-302.
- Li. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. *arXiv, Quantitative Biology—Genomics*, 2013, 1-3.
- Li et al. Targeted, random mutagenesis of plant genes with dual cytosine and adenine base editors, *Nature Biotechnology*, 2020, 38, 875-882.
- Lomholt and Kilian. Population Genetic Analysis of *Propionibacterium acnes* Identifies a Subpopulation and Epidemic Clones Associated with Acne. *PLOS One*, 2010, 5(8), e12277, 1-10.
- Mayslich et al. *Cutibacterium acnes* as an Opportunistic Pathogen: An Update of Its Virulence-Associated Factors, *Microorganisms* 2021, 9, 303, 1-21.
- McCoy et al. Skin Ecology during Sebaceous Drought—How Skin Microbes Respond to Isotretinoin. *Journal of Investigative Dermatology*, 2019, 139, 732-735.
- McDowell et al. *Propionibacterium acnes* Types I and II Represent Phylogenetically Distinct Groups. *Journal of Clinical Microbiology*, 2005, 43(1), 326-334.
- McDowell et al. A new phylogenetic group of *Propionibacterium acnes*. *Journal of Medical Microbiology*, 2008, 57, 218-224.
- McDowell et al. A novel multilocus sequence typing scheme for the opportunistic pathogen *Propionibacterium acnes* and characterization of type I cell surface associated antigens. *Microbiology*, 2011, 157, 1990-2003.
- McDowell et al. Is *Cutibacterium* (previously *Propionibacterium*) *acnes* a potential pathogenic factor in the aetiology of the skin disease progressive macular hypomelanosis? *J EADV*, 2021, 35, 338-344.
- Minh et al. IQ-TREE 2: New Models and Efficient Methods for Phylogenetic Inference in the Genomic Era. *Mol. Biol. Evol.* 2020, 37(5), 1530-1534.
- Needleman and Wunsch. A General Method Applicable to the Search for Similarities in the Amino Acid Sequence of Two Proteins, *J. Mol. Biol.*, 1970, 48, 443-453.
- Nolan et al. Treatment response to isotretinoin correlates with specific shifts in *Cutibacterium acnes* strain composition within the follicular microbiome. *Experimental Dermatology*. 2023, 32, 955-964.
- Oh et al. Biogeography and individuality shape function in the human skin metagenome. *Nature*, 2014, 514, 59-64.
- Oprica et al. Clinical and Microbiological Comparisons of Isotretinoin vs. Tetracycline in *Acne Vulgaris*. *Acta Derm Venereol*. 2007, 87, 246-254.
- Page et al. Roary: rapid large-scale prokaryote pan genome analysis. *Bioinformatics*, 2015, 31(22), 3691-3693.
- Pearson and Lipman. Improved tools for biological sequence comparison. *PNAS*, 1988, 85, 2444-2448.
- Ravin et al. The anti-immunity system of phage-plasmid N15: identification of the antirepressor gene and its control by a small processed RNA. *Molecular Microbiology*, 1999, 34(5), 980-994.
- Rees and Liu. Base editing: precision chemistry on the genome and transcriptome of living cells. *Nat Rev Genet*. 2018, 19(12), 770-788.
- Schaller et al. Induction of a chemoattractive proinflammatory cytokine response after stimulation of keratinocytes with *Propionibacterium acnes* and coproporphyrin III. *British Journal of Dermatology*, 2005, 153, 66-71.
- Scholz et al. A Novel High-Resolution Single Locus Sequence Typing Scheme for Mixed Populations of *Propionibacterium acnes* In Vivo. *PLOS One*, 2014, 9(8), e104199, 1-8.
- Sharon et al. Functional genetic variants revealed by massively parallel precise genome editing. *Cell*. Oct. 4, 2018; 175(2): 544-557.e16.
- Simon et al. Retrons and their applications in genome engineering, *Nucleic Acids Research*, 2019, 47(21), 11007-11019.
- Simonart and Dramaix. Treatment of acne with topical antibiotics: lessons from clinical studies, *British Journal of Dermatology*, 2005, 153, 395-403.
- Smith and Waterman. Comparison of Biosequences. *Advances in Applied Mathematics*, 1981, 2, 482-489.
- Spittaels et al. Porphyrins produced by acneic *Cutibacterium acnes* strains activate the inflammasome by inducing K⁺ leakage, *iScience*, 2021, 24, 102575, 1-16.
- Stamatakis. RAxML version 8: a tool for phylogenetic analysis and post-analysis of large phylogenies. *Bioinformatics*, 2014, 30(9), 1312-1313.
- Teramoto et al. Classification of *Cutibacterium acnes* at phylotype level by MALDI-MS proteotyping. *Proc. Jpn. Acad., Ser.*, 2019., 95, 612-623.
- Wang et al. The Anti-Inflammatory Activities of *Propionibacterium acnes* CAMP Factor-Targeted Acne Vaccines, *Journal of Investigative Dermatology*, 2018, 138, 2355-2364.
- Wannier et al. Improved bacterial recombineering by parallelized protein discovery, *PNAS*, 2020, 117(24), 13689-13698.
- Webster and Cummins. Use of Bacteriophage Typing to Distinguish *Propionibacterium acnes* Types I and II, *Journal of Clinical Microbiology*, 1978, 7(1), 84-90.
- Wood, et al. Improved metagenomic analysis with Kraken 2, *Genome Biology*, 2019, 20(257), 1-13.
- Yan et al. Cas13d Is a Compact RNA-Targeting Type VI CRISPR Effector Positively Modulated by a WYL-Domain-Containing Accessory Protein, *Molecular Cell*, 2018, 70, 327-339.
- Zhao et al. New base editors change C to A in bacteria and C to G in mammalian cells. *Nature Biotechnology*, 2021. 39, 35-40.

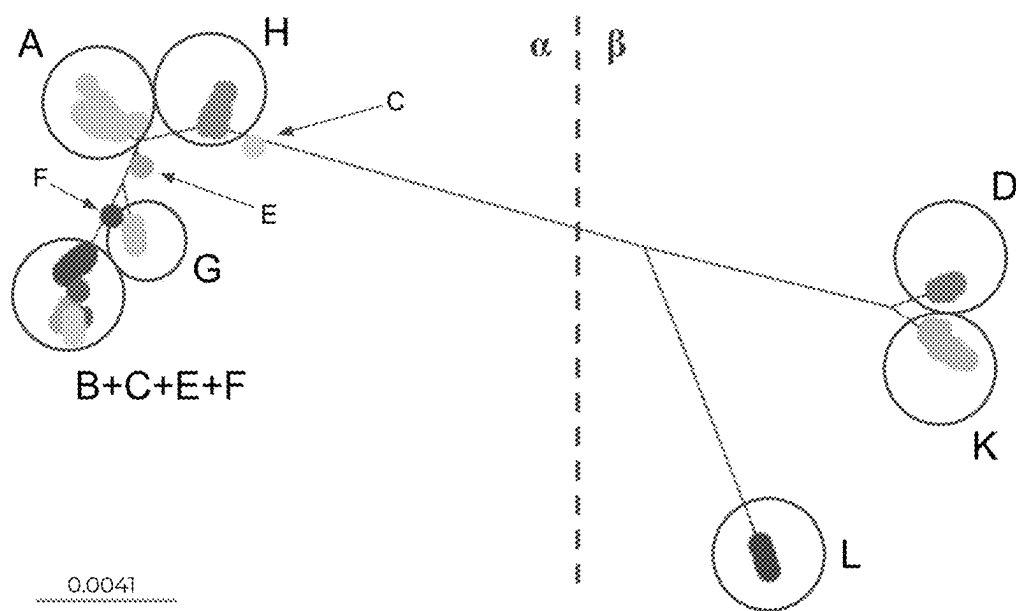


FIGURE 1

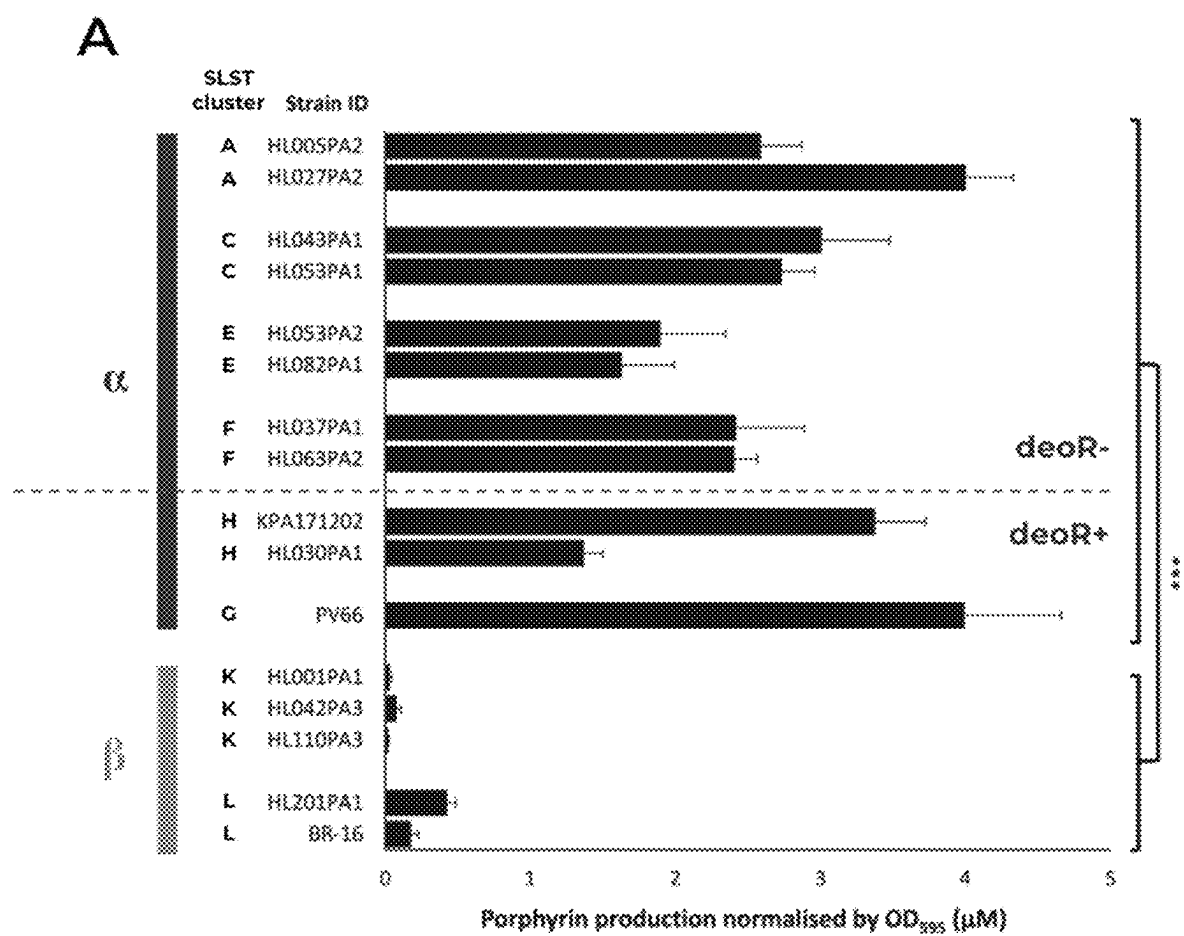


FIGURE 2A

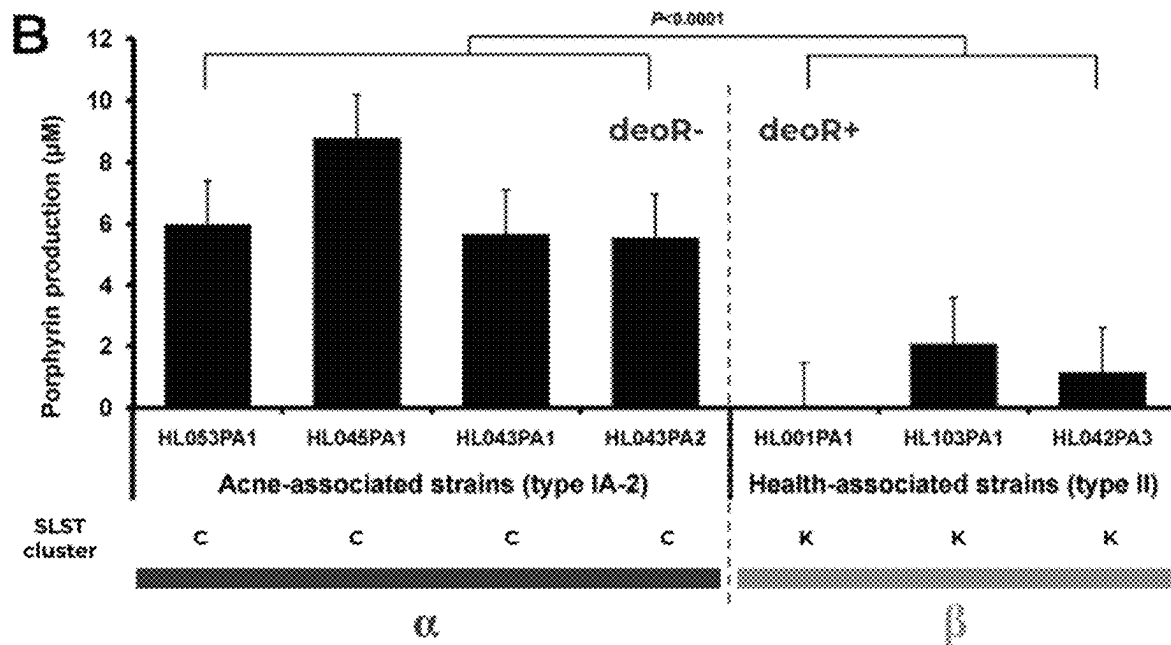


FIGURE 2B

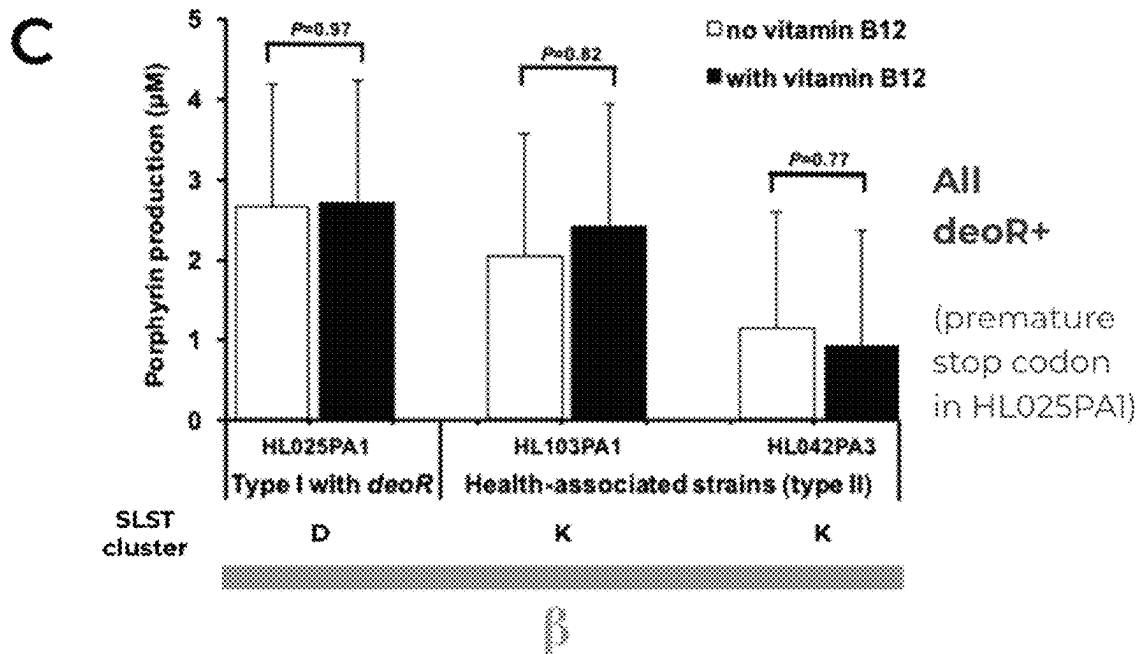


FIGURE 2C

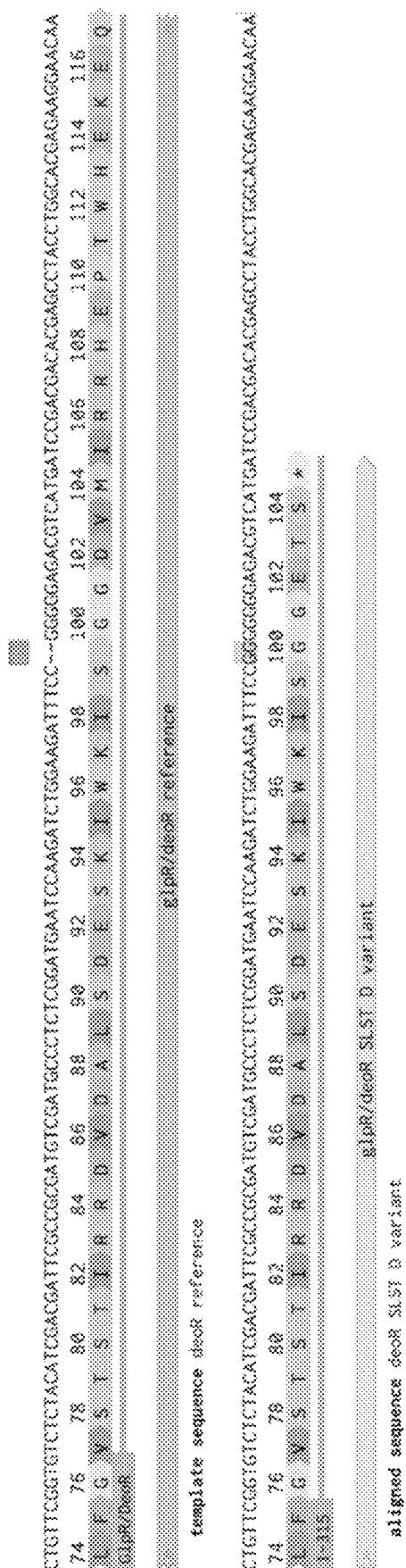
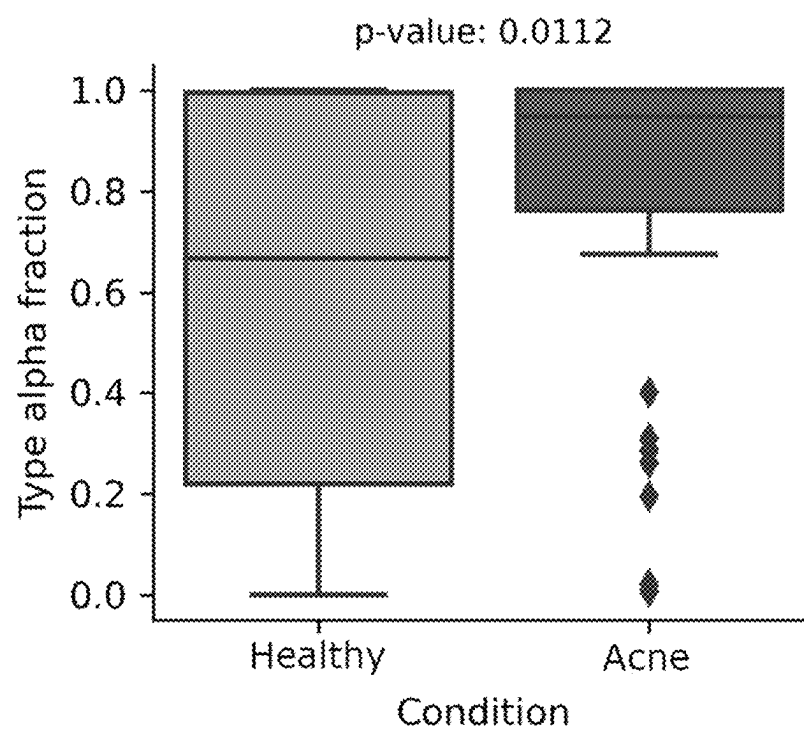


Figure 3

**FIGURE 4**

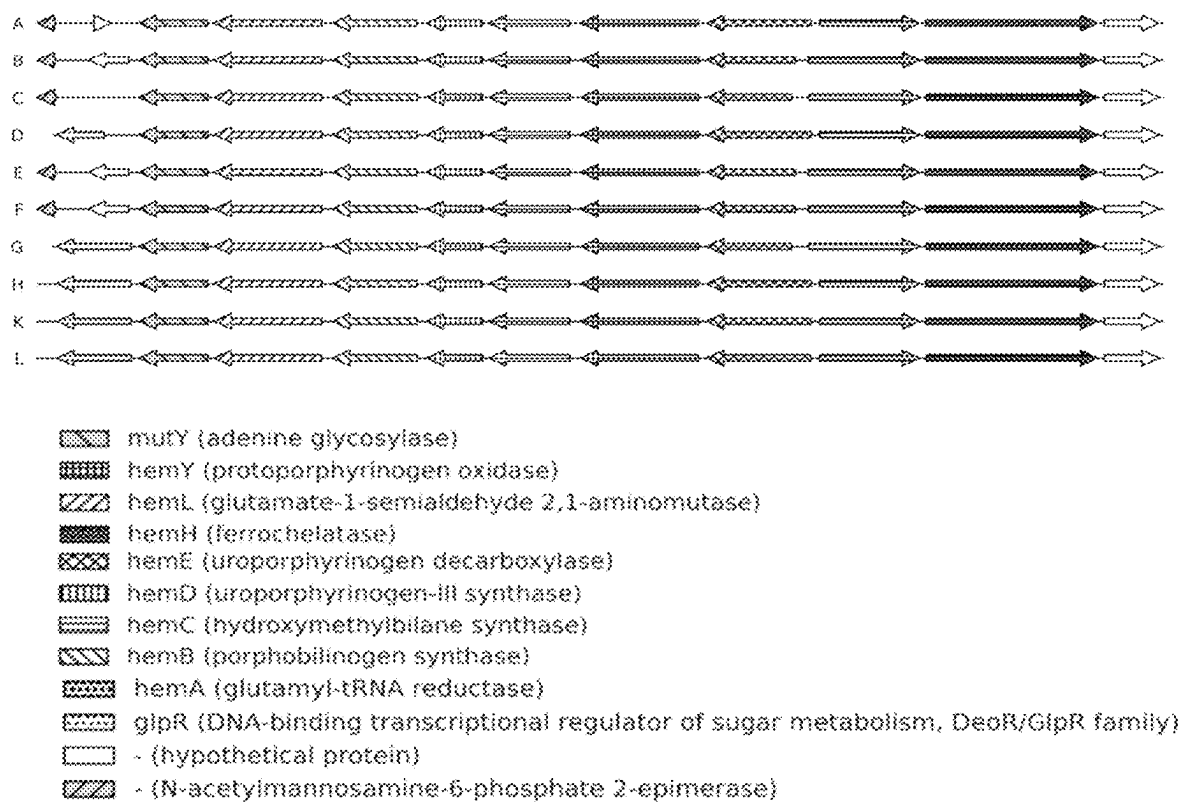


Figure 5

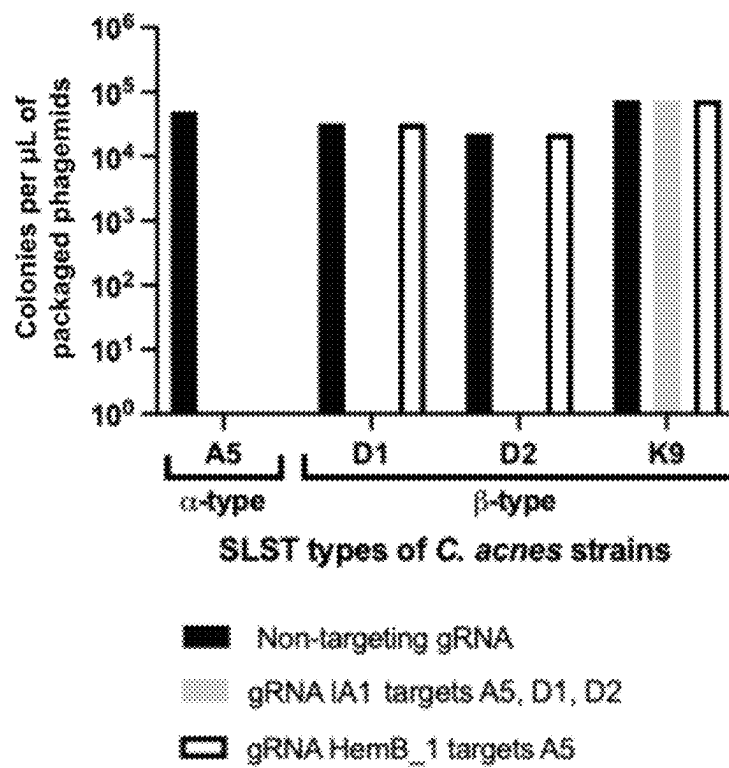


Figure 6A

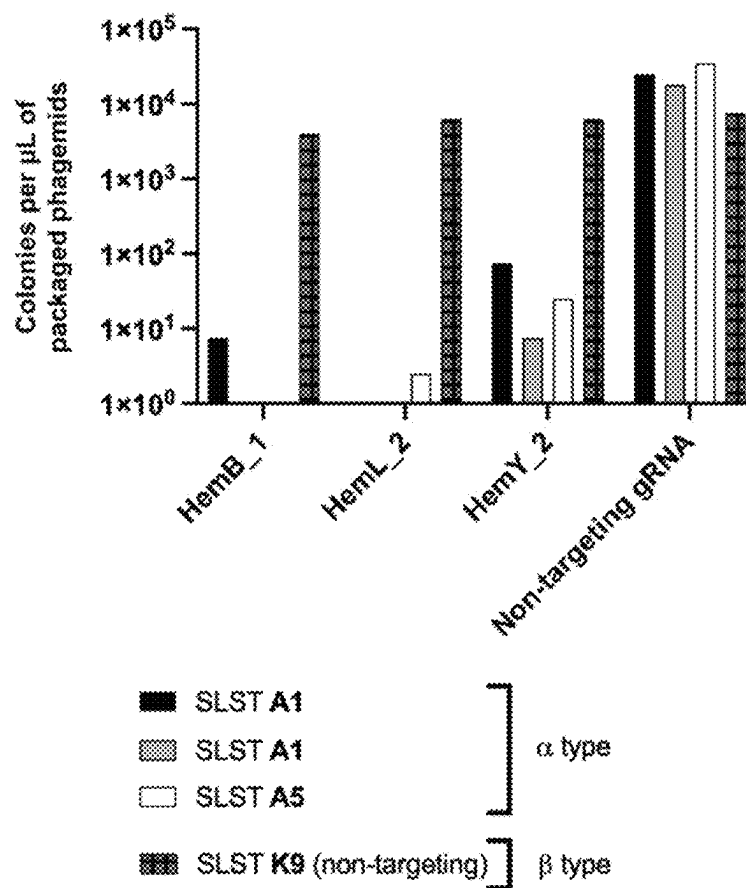


Figure 6B

1

METHOD FOR TREATING *C. acnes* BACTERIA-ASSOCIATED DISEASES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application 63/515,223, filed on Jul. 24, 2023, the entire disclosure of which is incorporated by reference herein.

REFERENCE TO SEQUENCE LISTING SUBMITTED

This application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on May 6, 2025, is named SequenceListing-1726241190261 and is 52,429 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present invention relates to methods for treating *C. acnes* bacteria-associated diseases, in particular for treating acne, more particularly acne vulgaris, in a subject.

BACKGROUND

Cutibacterium acnes is a major commensal bacterium of the skin microbiome, representing the most abundant bacteria in the hair follicle. While having a strong role in maintaining skin health, it has also been implicated in several diseases and infections, including acne vulgaris (Bruggemann et al. (2021) Front Microbiol. 12:673845) or progressive macular hypomelanosis (McDowell et al. (2021) J. Eur. Acad. Dermatol. Venereol. 35:338-344).

Acne vulgaris is a chronic inflammatory disorder of the hair follicle, for which *C. acnes* has been described as one of the etiological factors. Indeed, previous studies have shown (i) that antimicrobial treatment efficacy correlated with *C. acnes* load reduction (Lessin et al. (2020) Skin Microbiome Handbook: From Basic Research to Product Development, Ed. Nava Dayan, p289-302), (ii) that isotretinoin, the most efficient drug available so far to treat acne, led to a reduction in *C. acnes* load (Oprica et al. (2007) Acta Derm Venerol 87:246-254; Nolan et al. (2023) Exp. Dermatol. 32:955-964; McCoy et al. (2019) J Invest. Dermatol. 139:732-735; Kelh  la et al. (2017) Exp. Dermatol. 27:30-36), (iii) that antibiotic inefficacy correlated with the presence of *C. acnes* antibiotic resistant strains (Simonart et al. (2005) British J. Dermatol. 153:395-403); (iv) that *C. acnes* triggered acne-like inflammation in different in vitro (Lacloserie et al. (2021) Exp. Dermatol. 30:347-357) and in vivo models (Kolar et al. (2019) J CI Insight 4(5):e124687), and (v) that patients suffering from acne vulgaris had anti-*C. acnes* antibodies (Wang et al. (2018) J. Invest. Dermatol. 138:2355-2364; Huang et al. (2021) Front. Microbiol. 12:709562).

Since *C. acnes* also plays an important role in maintaining skin health, it is of major importance to identify those *C. acnes* types that are involved in inflammatory disorders (e.g. acne vulgaris) and distinguish them from non-inflammatory disorders-associated *C. acnes* types.

Different studies tried to achieve such identification but with a poor outcome. In 2013, Fitz-Gibbon et al. compared the skin microbiome at the strain level and genome level of

2

C. acnes between acne patients and healthy individuals. They showed a significant enrichment of specific *C. acnes* ribotypes in acne patients (namely RT4, RT5, RT8 and RT10 ribotypes) and a significant enrichment of another specific *C. acnes* ribotype in healthy subjects (namely RT6 ribotype) (Fitz-Gibbon et al. (2013) J. Invest. Dermatol. 133:2152-2160). However, not all acne patients showed a clear presence of *C. acnes* bacteria belonging to an acne-associated ribotype. Therefore, the distinction of *C. acnes* bacteria between different ribotypes did not enable the effective identification of *C. acnes* bacteria involved in acne.

Based on the whole genome sequencing of *C. acnes* clones belonging to different ribotypes, Fitz-Gibbon et al. also showed that 3 loci (referred to by the authors as loci 1, 2 and 3) were specific to ribotypes RT4 and RT5, which are enriched in acne patients (Fitz-Gibbon et al. (2013) J. Invest. Dermatol. 133:2152-2160). However, once again the presence of these loci does not enable the specific identification of acne-associated strains, since *C. acnes* strains belonging to the RT8 ribotype (which is enriched in acne patients) are not enriched in these 3 loci, and *C. acnes* strains belonging to the RT1 ribotype (abundant in both acne patients and healthy subjects) can carry loci 1, 2 and 3.

Later, the same lab performed a metagenomic profiling of acne and healthy follicular plugs and showed that the RT4, RT5 and RT8 ribotypes were more abundant and prevalent in acne patients, the RT2 and RT6 ribotypes were more abundant and prevalent in healthy subjects, and that there was an enrichment of specific genes contained in the previously identified loci 1, 2 and 3 (Barnard et al. (2016) Scientific Reports 6:39491). However, a large fraction of acne metagenomic samples did not carry loci 1, 2 and 3, and, inversely, some healthy metagenomic samples carried locus 1 or 3. This confirms that ribotypes and loci 1, 2 and 3 are not suitable to efficiently identify acne-associated strains.

Other different typing methods (including multi-locus sequence typing (MLST), single-locus sequence typing (SLST), Belfast scheme using 7 target genes, multiplex touchdown PCR, Aarhus scheme using 9 target genes, MALDI-TOF MS-based typing, and MLVA typing using the polymorphism of 13 VNTRs) have been used to try to discriminate acne-associated strains and non acne-associated strains, but the types defined using those methods are inconsistent when compared with each other, and none of them enables a clear and efficient identification of acne-associated strains (Mayslich et al. (2021) Microorganisms 9:203).

There is thus an important need for tools enabling a clear and efficient identification of acne-associated strains, so that these strains can be specifically targeted while maintaining non acne-associated strains alive and in good shape.

Some studies reported a higher production of certain porphyrins (mainly coproporphyrin III) by acne-associated strains (Johnson et al. (2016) mSphere 1(1):e00023-15; Barnard et al. (2020) mSphere 5:e00793-19). Porphyrins were considered to be of interest because they were shown, when extracted from acne-associated strains, to activate the inflammasome, offering an explanation for *C. acnes* role in inflammation (Spittaels et al. (2021) iScience 24:102575). Therefore, using the difference in porphyrin production could be a way of discriminating acne-associated strains from non acne-associated strains.

Enzymes involved in the porphyrin pathway are encoded by 8 hem genes in a locus that is conserved across *C. acnes* strains and represented on FIG. 5. Additionally, the Huiying Li team reported the presence of a putative repressor from the deoR family (also known as glpR) in the proximity of

this locus in non acne-associated strains (Johnson et al. (2016) mSphere 1(1):e00023-15; Barnard et al. (2020) mSphere 5:e00793-19), and therefore speculated that this deoR gene could be of interest in the diagnosis and/or treatment of acne vulgaris (WO2017/136738). However, as highlighted by the authors, “[s]ince deoR was found to be expressed in both high- and low-level-porphyrin-producing strains, [they] sought to determine if there were differences in the deoR gene sequence that could explain variations in regulation and porphyrin production among the strains. (. . .) [However, they] did not identify any single nucleotide polymorphisms (SNPs) that were uniquely shared by strains from clades IB-3 and IC and that could potentially explain the increased porphyrin production in these lineages” (see Barnard et al. (2020) mS phere 5:e00793-19).

Therefore, there is still an important need for tools enabling a robust and efficient identification of acne-associated strains, so that these strains can be specifically targeted while maintaining non acne-associated strains alive and in good shape.

The present invention meets this need.

SUMMARY OF THE INVENTION

The present invention arises from the unexpected finding by the inventors that a set of 218 SNPs across the hem locus (comprising the genes hemA, hemB, hemC, hemD, hemE, hemH, hemL, and hemY) allows to classify the known *C. acnes* strain diversity in 2 groups, herein called α -type and β -type. Importantly, this new typing strictly correlates with the published porphyrin production levels, in contrast to the deoR gene, with α -type containing all *C. acnes* producing high amounts of porphyrins whereas β -type contain all the *C. acnes* producing low or no amounts of porphyrins. Moreover, after developing a method for the estimation of relative abundances of α -type and β -type *C. acnes* strains in shotgun metagenomic sequencing data, and applying it to a public data set associated to a study characterizing the skin microbiome in acne vulgaris (Barnard et al. (2016) Scientific Reports 6:39491), the present inventors demonstrated that α -type *C. acnes* strains formed significantly larger fraction of the *C. acnes* population in patients suffering from acne vulgaris than healthy subjects.

The present invention thus relates to a method for treating or preventing a *Cutibacterium acnes* bacteria-associated disease in a subject, said method comprising modulating, in particular reducing, in said subject, the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908,

A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1;

and

wherein β -type *C. acnes* bacteria are *C. acnes* bacteria which are not α -type *C. acnes* bacteria.

The present invention also relates to a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising specifically reducing the amount of α -type *C. acnes* bacteria in said subject,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432,

5

A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395,

wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention also relates to a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising genetically modifying a DNA sequence in α -type *C. acnes* bacteria in said subject, to generate at least one change, preferably in the hem locus of said α -type *C. acnes* bacteria,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

6

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G 1278, T1295, T1318, A1336, G1343, C1346, G 1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention also relates to a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising increasing the amount of β -type *C. acnes* bacteria in said subject,

wherein β -type *C. acnes* bacteria are *C. acnes* bacteria which are not α -type *C. acnes* bacteria,

α -type *C. acnes* bacteria being *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641,

7

T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8, in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention further concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent inducing the modulation, preferably the reduction, in said subject, of the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926,

8

C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1;

and

wherein β -type *C. acnes* bacteria are *C. acnes* bacteria which are not α -type *C. acnes* bacteria.

The present invention also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which specifically reduces the amount of α -type *C. acnes* bacteria in said subject,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

9

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G 1278, T1295, T1318, A1336, G1343, C1346, G 1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which genetically modifies a DNA sequence in α -type *C. acnes* bacteria in said subject, to generate at least one change, preferably in the hem locus of said α -type *C. acnes* bacteria,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

10

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G 1278, T1295, T1318, A1336, G1343, C1346, G 1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention further concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which increases the amount of β -type *C. acnes* bacteria in said subject,

wherein β -type *C. acnes* bacteria are *C. acnes* bacteria which are not α -type *C. acnes* bacteria,

α -type *C. acnes* bacteria being *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199,

11

C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention also concerns a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising specifically reducing the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

12

and/or

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

The present invention also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which specifically reduces the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject by genetically modifying a DNA sequence in α -type *C. acnes* bacteria in said subject, to generate at least one change, preferably in the hem locus of said α -type *C. acnes* bacteria, and

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

13

and/or

- (ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

Another object of the invention concerns a pharmaceutical composition comprising a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a programmable nuclease, or a gene editing enzyme or system, designed to specifically target a α -type *C. acnes* bacteria,

wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

- (ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

14

Another object of the invention concerns a method for determining if a subject is at risk of developing acne, in particular acne vulgaris, said method comprising the steps of:

- a) determining the presence and/or amount of α -type *C. acnes* bacteria in a sample, wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10,

and/or

- (ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1,

and

- b) based on the presence and/or amount determined at step a), determining if said subject is at risk of developing acne, in particular acne vulgaris.

Another object of the invention concerns a method of diagnosing acne, in particular acne vulgaris in a subject, said method comprising the steps of:

- a) determining the presence and/or amount of α -type *C. acnes* bacteria in a sample, wherein α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90% of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G 1278, T1295, T1318, A1336, G1343, C1346, G 1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10, and/or

- (ii) in which the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1, and

- b) based on the presence and/or amount determined at step a), diagnosing acne, in particular acne vulgaris, in said subject.

DETAILED DESCRIPTION

α -type *C. acnes* bacteria and β -type *C. acnes* bacteria

Cutibacterium acnes (formerly *Propionibacterium acnes*) is a gram-positive rod-shaped aerotolerant bacteria, first isolated from skin in 1897. It belongs to the order Actinomycetales, is part of the Propionibacteriaceae family, and belongs to the genus *Cutibacterium*. *C. acnes* is one of the most prevalent and abundant bacteria on human skin where it can be found both on the skin surface (stratum corneum) and in the hair follicle. Inside the hair follicle, it is in direct contact with a large diversity of living cells such as keratinocytes, stem cells, sebaceous cells and immune cells, unlike on the stratum corneum where it is mostly in contact with the dead corneocyte. *C. acnes* is a commensal bacterium but has also been associated with several skin diseases such as acne vulgaris or progressive macular hypomelanosis.

C. acnes strains were previously classified into two main types, I and II, on the basis of their cell wall carbohydrate content and serum lectin responses (Johnson et al. (1982) J Bacteriol. 109(3):1047-66). Subclusters of these types were then identified on the basis of the R ecA and tly genes and the use of the QUBPa1 and QUBPa2 antibodies (McDowell et al. (2005) J Clin Microbiol. 43(1):326-334). An additional phylotype, type III, corresponding to strains with filamentous appendages, was then added to the classification (McDowell et al. (2008) J Medical Microbiol. 57:218-224). Multi-locus sequence typing (MLST) methods were also developed to increase typing resolution. The Belfast scheme, based on 7 target genes, differentiates type I into clades IA1, IA2, IB and IC (McDowell et al. (2011) Microbiology 157:1990-2003), while the Aarhus scheme, based on 9 target genes, differentiates type I into clades I-1a, I-1b and I-2 (Lomholt et al. (2010) PLoS ONE 5(8):e12277). SLST and whole-genome sequencing (WGS) techniques were then used to develop new methods differentiating *C. acnes* strains into SLST types (Scholz et al. (2014) P LoS ONE 9(8): e104199).

Mass spectrometry-based methods were also used to characterize *C. acnes* strains, and, when associated with profiling of ribosomal subunit proteins, in MALDI-MS prototyping, enabled the discrimination of all phylotypes mentioned above (Teramoto et al. (2019) Proc. Jpn. Acad. Ser. B 95:612-623).

Based on 16S rRNA gene analysis, *C. acnes* strains were also divided in ribotypes defined and referenced in Fitz-Gibbon et al. (2013) J. Invest. Dermatol. 133:2152-2160.

However, as underlined by Mayslich et al. (Mayslich et al. (2021) Microorganisms 9:303), there is an important need of standardization among all the above nomenclatures, and, as highlighted above, none of them enables a robust and efficient distinction between acne strains and non acne-associated strains.

The inventors of the present invention identified a set of 218 SNPs across the hem locus (comprising the genes hemA, hemB, hemC, hemD, hemE, hemH, hemL, and hemY) which allows to classify the known *C. acnes* strain diversity in 2 groups, herein called α -type and β -type. Importantly, this new typing strictly correlates with the published porphyrin production levels, in contrast to the deoR gene, with α -type containing all *C. acnes* producing high amounts of porphyrins whereas β -type contains all the *C. acnes* producing low or no amounts of porphyrins. Moreover, after developing a method for the estimation of relative abundances of α -type and β -type *C. acnes* strains in shotgun metagenomic sequencing data, and applying it to a

public data set associated to a study characterizing the skin microbiome in acne vulgaris (Barnard et al. (2016) Scientific Reports 6:39491), the present inventors demonstrated that α -type *C. acnes* strains formed significantly larger fraction of the *C. acnes* population in patients suffering from acne vulgaris than healthy subjects.

The inventors thereby identified α -type *C. acnes* bacteria as being *C. acnes* bacteria in which at least 90% of the marker positions defined in Table 1 below have the nucleotide variant defined as “alpha variant”, the positions being defined with respect to reference gene sequence SEQ ID NO: 3 for hemA gene, SEQ ID NO:4 for hemB gene, SEQ ID NO:5 for hemC gene, SEQ ID NO: 6 for hemD gene, SEQ ID NO: 7 for hemE gene, SEQ ID NO: 8 for hemH gene, SEQ ID NO: 9 for hemL gene and SEQ ID NO: 10 for hemY gene.

Values indicated in the column entitled “reference position (0-based)” are 0-based which means that the first nucleotide in the gene sequence is indexed as 0. As will be easily understood by the skilled person, a position can alternatively be defined in “1-based” way, meaning that the first nucleotide in the gene sequence is indexed as 1. In that case, all the “0-based” defined positions would be incremented by 1. For example, the “0-based” defined position T173 in the hemA gene is identical to the “1-based” defined position T174 in the hemA gene. Values indicated in the column entitled “reference position (1-based)” are 1-based. In the present document, unless specifically indicated, values of the reference positions are 0-based.

TABLE 1

Set of reference position for alpha/beta types of the hem locus				
gene	alpha variant	beta variant	reference position (0-based)	reference position (1-based)
hemA	T	G	173	174
hemA	A	G	479	480
hemA	T	C	506	507
hemA	A	G	614	615
hemA	T	G	629	630
hemA	C	A	671	672
hemA	T	C	971	972
hemA	A	G	989	990
hemA	T	C	1085	1086
hemA	T	C	1094	1095
hemB	C	T	142	143
hemB	A	G	155	156
hemB	A	G	176	177
hemB	A	G	183	184
hemB	C	T	224	225
hemB	C	T	245	246
hemB	G	A	248	249
hemB	T	C	281	282
hemB	G	A	311	312
hemB	C	T	314	315
hemB	T	C	346	347
hemB	C	T	374	375
hemB	G	T	377	378
hemB	C	T	452	453
hemB	T	C	626	627
hemB	C	G	683	684
hemB	A	G	693	694
hemB	A	G	720	721
hemB	C	T	738	739
hemB	C	T	794	795
hemB	A	C	920	921
hemB	T	G	944	945
hemB	T	C	992	993
hemB	G	C	1022	1023
hemB	T	C	1031	1032
hemC	G	A	77	78
hemC	A	G	96	97

TABLE 1-continued

Set of reference position for alpha/beta types of the hem locus				
gene	alpha variant	beta variant	reference position (0-based)	reference position (1-based)
hemC	C	G	149	150
hemC	A	G	233	234
hemC	T	C	236	237
hemC	T	C	255	256
hemC	A	G	257	258
hemC	A	C	260	261
hemC	C	T	317	318
hemC	A	G	347	348
hemC	T	C	425	426
hemC	T	C	437	438
hemC	A	G	464	465
hemC	A	G	486	487
hemC	A	G	487	488
hemC	C	T	488	489
hemC	A	G	524	525
hemC	A	G	545	546
hemC	A	G	590	591
hemC	T	C	605	606
hemC	A	G	644	645
hemC	A	G	665	666
hemC	C	T	710	711
hemC	C	G	713	714
hemC	A	G	737	738
hemC	A	G	770	771
hemC	A	G	776	777
hemC	T	A	785	786
hemC	T	C	802	803
hemC	G	A	830	831
hemC	C	T	842	843
hemC	T	C	845	846
hemC	G	A	855	856
hemC	C	G	893	894
hemC	A	T	896	897
hemC	C	T	897	898
hemC	G	A	902	903
hemC	G	T	908	909
hemC	A	G	941	942
hemC	A	G	947	948
hemD	T	C	11	12
hemD	T	C	41	42
hemD	T	C	50	51
hemD	A	G	150	151
hemD	T	C	155	156
hemD	G	A	159	160
hemD	G	T	176	177
hemD	T	C	257	258
hemD	C	T	266	267
hemD	T	C	278	279
hemD	G	C	323	324
hemD	A	G	324	325
hemD	T	C	362	363
hemD	T	C	367	368
hemD	T	C	374	375
hemD	G	A	387	388
hemD	A	G	398	399
hemD	A	G	399	400
hemD	T	C	407	408
hemD	G	A	422	423
hemD	G	A	425	426
hemD	G	A	432	433
hemD	A	G	446	447
hemD	A	G	459	460
hemD	T	C	464	465
hemD	G	A	595	596
hemD	T	C	617	618
hemD	T	A	643	644
hemD	A	G	644	645
hemD	C	T	680	681
hemE	T	C	241	242
hemE	T	C	476	477
hemE	A	G	530	531
hemE	T	C	704	705
hemE	A	G	728	729
hemE	G	C	767	768
hemE	T	A	834	835

19

TABLE 1-continued

Set of reference position for alpha/beta types of the hem locus				
gene	alpha variant	beta variant	reference position (0-based)	reference position (1-based)
hemE	A	G	867	868
hemE	A	G	872	873
hemE	A	C	882	883
hemE	A	G	957	958
hemH	T	C	11	12
hemH	T	C	23	24
hemH	A	G	48	49
hemH	T	C	89	90
hemH	C	T	188	189
hemH	G	A	338	339
hemH	T	C	464	465
hemH	T	C	470	47
hemH	G	A	536	537
hemH	T	C	548	549
hemH	T	C	608	609
hemH	G	A	626	627
hemH	T	C	632	633
hemH	T	C	641	642
hemH	T	C	67	672
hemH	T	C	674	675
hemH	C	T	683	684
hemH	C	T	689	690
hemH	A	T	740	741
hemH	G	A	745	746
hemH	G	A	787	788
hemH	A	G	797	798
hemH	T	C	893	894
hemH	A	T	898	899
hemH	T	C	1034	1035
hemH	T	C	1039	1040
hemH	T	C	1046	1047
hemH	C	G	1071	1072
hemH	G	A	1095	1096
hemH	C	T	1109	1110
hemH	T	C	1153	1154
hemH	T	C	1358	1359
hemH	G	C	1496	1497
hemH	C	T	1511	1512
hemH	C	T	1527	1528
hemH	G	A	1532	1533
hemH	T	C	1598	1599
hemH	T	C	1712	1713
hemH	A	G	1850	1851
hemH	G	C	1901	1902
hemH	A	G	1926	1927
hemH	C	T	1931	1932
hemH	C	T	1997	1998
hemL	T	C	107	108
hemL	C	T	197	198
hemL	T	A	229	230
hemL	A	G	242	243
hemL	C	A	310	311
hemL	A	G	338	339
hemL	A	G	374	375
hemL	T	C	389	390
hemL	G	T	392	393
hemL	T	C	442	443
hemL	C	T	467	468
hemL	G	A	470	471
hemL	C	T	482	483
hemL	G	T	518	519
hemL	C	G	641	642
hemY	A	G	89	90
hemY	A	G	236	237
hemY	A	G	354	355
hemY	C	T	383	384
hemY	C	T	392	393
hemY	A	G	395	396
hemY	T	C	398	399
hemY	T	C	465	466
hemY	T	C	474	475
hemY	C	T	521	522
hemY	T	C	538	539
hemY	C	T	566	567
hemY	A	C	593	594

20

TABLE 1-continued

Set of reference position for alpha/beta types of the hem locus				
gene	alpha variant	beta variant	reference position (0-based)	reference position (1-based)
hemY	A	C	767	768
hemY	T	A	782	783
hemY	G	A	863	864
hemY	G	A	920	921
hemY	T	A	927	928
hemY	A	G	932	933
hemY	G	A	935	936
hemY	C	A	979	980
hemY	G	C	989	990
hemY	C	G	1019	1020
hemY	T	C	1055	1056
hemY	G	A	1091	1092
hemY	T	C	1136	1137
hemY	G	A	1143	1144
hemY	G	T	1152	1153
hemY	T	C	1166	1167
hemY	T	G	1172	1173
hemY	A	G	1176	1177
hemY	T	C	1184	1185
hemY	C	T	1199	1200
hemY	C	T	1205	1206
hemY	G	A	1268	1269
hemY	G	A	1278	1279
hemY	T	C	1295	1296
hemY	T	C	1318	1319
hemY	A	G	1336	1337
hemY	G	A	1343	1344
hemY	C	T	1346	1347
hemY	G	C	1355	1356
hemY	T	G	1359	1360
hemY	C	G	1395	1396

In the context of the invention, α -type *C. acnes* bacteria are thus preferably *C. acnes* bacteria which comprise at least 90%, in particular at least 90.5%, at least 91%, at least 91.5%, at least 92%, at least 92.5%, at least 93%, at least 93.5%, at least 94%, at least 94.5%, at least 95%, at least 95.5%, at least 96%, at least 96.5%, at least 97%, at least 97.5%, at least 98%, at least 98.5%, at least 99%, at least 99.5% or 100% of the following nucleotide variants in the hem locus:

- in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,
- in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,
- in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,
- in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G 387, A398, A399, T407, G 422, G 425, G 432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,
- in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

21

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

Alternatively, α -type *C. acnes* bacteria may be *C. acnes* bacteria which comprise at least 197, at least 198, at least 199, at least 200, at least 201, at least 202, at least 203, at least 204, at least 205, at least 206, at least 207, at least 208, at least 209, at least 210, at least 211, at least 212, at least 213, at least 214, at least 215, at least 216, at least 217 or all of the alpha nucleotide variants as defined in Table 1.

Alternatively, using "1-based" defined positions, α -type *C. acnes* bacteria may be *C. acnes* bacteria which comprise at least 90%, in particular at least 90.5%, at least 91%, at least 91.5%, at least 92%, at least 92.5%, at least 93%, at least 93.5%, at least 94%, at least 94.5%, at least 95%, at least 95.5%, at least 96%, at least 96.5%, at least 97%, at least 97.5%, at least 98%, at least 98.5%, at least 99%, at least 99.5% or 100% of the following nucleotide variants in the hem locus:

in the hemA gene: T174, A480, T507, A615, T630, C672, T972, A990, T1086, and T1095, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C143, A156, A177, A184, C225, C246, G249, T282, G312, C315, T347, C375, G378, C453, T627, C684, A694, A721, C739, C795, A921, T945, T993, G1023, and T1032, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G78, A97, C150, A234, T237, T256, A258, A261, C318, A348, T426, T438, A465, A487, A488, C489, A525, A546, A591, T606, A645, A666, C711, C714, A738, A771, A777, T786, T803, G831, C843, T846, G856, C894, A897, C898, G903, G909, A942, and A948, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T12, T42, T51, A151, T156, G160, G177, T258, C267, T279, G324, A325, T363, T368, T375, G388, A399, A400, T408, G423, G426, G433, A447, A460, T465, G596, T618, T644, A645, and C681, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T242, T477, A531, T705, A729, G768, T835, A868, A873, A883, and A958, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T12, T24, A49, T90, C189, G339, T465, T471, G537, T549, T609, G627, T633, T642, T672, T675, C684, C690, A741, G746, G788, A798, T894, A899, T1035, T1040, T1047, C1072, G1096, C1110, T1154, T1359, G1497, C1512, C1528, G1533, T1599, T1713, A1851, G1902, A1927, C1932, and

22

C1998, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T108, C198, T230, A243, C311, A339, A375, T390, G393, T443, C468, G471, C483, G519, and C642, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A90, A237, A355, C384, C393, A396, T399, T466, T475, C522, T539, C567, A594, A768, T783, G864, G921, T928, A933, G936, C980, G990, C1020, T1056, G1092, T1137, G1144, G1153, T1167, T1173, A1177, T1185, C1200, C1206, G1269, G1279, T1296, T1319, A1337, G1344, C1347, G1356, T1360 and C1396, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

By "hem locus" is meant herein a locus included in the porphyrin biosynthesis loci of *C. acnes* bacteria and comprising 8 hem genes, namely the hemA, hemB, hemC, hemD, hemE, hemH, hemL and hemY genes.

Alternatively or in addition, in the context of the invention α -type *C. acnes* bacteria are *C. acnes* bacteria in which the nucleic acid sequence of the hem locus (which comprises the genes hemA, hemB, hemC, hemD, hemE, hemH, hemL and hemY) is at least 97% identical, in particular at least 97.1% identical, at least 97.2% identical, at least 97.3% identical, at least 97.4% identical, at least 97.5% identical, at least 97.6% identical, at least 97.7% identical, at least 97.8% identical, at least 97.9% identical, at least 98% identical, at least 98.1% identical, at least 98.2% identical, at least 98.3% identical, at least 98.4% identical, at least 98.5% identical, at least 98.6% identical, at least 98.7% identical, at least 98.8% identical, at least 98.9% identical, at least 99% identical, at least 99.1% identical, at least 99.2% identical, at least 99.3% identical, at least 99.4% identical, at least 99.5% identical, at least 99.6% identical, at least 99.7% identical, at least 99.8% identical, at least 99.9% identical, or 100% identical, to the sequence SEQ ID NO: 1.

Preferably, α -type *C. acnes* bacteria are *C. acnes* bacteria (i) which comprise at least 90%, in particular at least 90.5%, at least 91%, at least 91.5%, at least 92%, at least 92.5%, at least 93%, at least 93.5%, at least 94%, at least 94.5%, at least 95%, at least 95.5%, at least 96%, at least 96.5%, at least 97%, at least 97.5%, at least 98%, at least 98.5%, at least 99%, at least 99.5% or 100% of the alpha nucleotide variants as defined in Table 1, or which comprise at least 197, at least 198, at least 199, at least 200, at least 201, at least 202, at least 203, at least 204, at least 205, at least 206, at least 207, at least 208, at least 209, at least 210, at least 211, at least 212, at least 213, at least 214, at least 215, at least 216, at least 217 or all of the alpha nucleotide variants as defined in Table 1, and

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical, in particular at least 97.1% identical, at least 97.2% identical, at least 97.3% identical, at least 97.4% identical, at least 97.5% identical, at least 97.6% identical, at least 97.7% identical, at least 97.8% identical, at least 97.9% identical, at least 98% identical, at least 98.1% identical, at least 98.2% identical, at least 98.3% identical, at least 98.4% identical, at least 98.5% identical, at least 98.6% identical, at least 98.7% identical, at least 98.8% identical, at least 98.9% identical, at least 99% identical, at least 99.1% identical, at least 99.2% identical, at least 99.3% identical, at least 99.4% identical, at least 99.5% identical, at least 99.6% identical, at least 99.7% identical, at least 99.8% identical, at least 99.9% identical, or 100% identical, to the sequence SEQ ID NO: 1.

23

In the context of the invention, β -type *C. acnes* bacteria are *C. acnes* bacteria which are not α -type *C. acnes* bacteria as defined above.

Preferably, β -type *C. acnes* bacteria are *C. acnes* bacteria which comprise at least 90%, in particular at least 90.5%, at least 91%, at least 91.5%, at least 92%, at least 92.5%, at least 93%, at least 93.5%, at least 94%, at least 94.5%, at least 95%, at least 95.5%, at least 96%, at least 96.5%, at least 97%, at least 97.5%, at least 98%, at least 98.5%, at least 99%, at least 99.5% or 100% of the following nucleotide variants in the hem locus:

in the hemA gene: G173, G479, C506, G614, G629, A671, C971, G989, C1085, and C1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: T142, G155, G176, G183, T224, T245, A248, C281, A311, T314, C346, T374, T377, T452, C626, G683, G693, G720, T738, T794, C920, G944, C992, C1022, and C1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: A77, G96, G149, G233, C236, C255, G257, C260, T317, G347, C425, C437, G464, G486, G487, T488, G524, G545, G590, C605, G644, G665, T710, G713, G737, G770, G776, A785, C802, A830, T842, C845, A855, G893, T896, T897, A902, T908, G941, and G947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: C11, C41, C50, G150, C155, A159, T176, C257, T266, C278, C323, G324, C362, C367, C374, A387, G398, G399, C407, A422, A425, A432, G446, G459, C464, A595, C617, A643, G644, and T680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: C241, C476, G530, C704, G728, C767, A834, G867, G872, C882, and G957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: C11, C23, G48, C89, T188, A338, C464, C470, A536, C548, C608, A626, C632, C641, C671, C674, T683, T689, T740, A745, A787, G797, C893, T898, C1034, C1039, C1046, G1071, A1095, T1109, C1153, C1358, C1496, T1511, T1527, A1532, C1598, C1712, G1850, C1901, G1926, T1931, and T1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: C107, T197, A229, G242, A310, G338, G374, C389, T392, C442, T467, A470, T482, T518, and G 641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: G89, G236, G354, T383, T392, G395, C398, C465, C474, T521, C538, T566, C593, C767, A782, A863, A920, A927, G932, A935, A979, C989, G1019, C1055, A1091, C1136, A1143, T1152, C1166, G1172, G1176, C1184, T1199, T1205, A1268, A1278, C1295, C1318, G1336, A1343, T1346, C1355, G1359 and G1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

Alternatively, using "1-based" defined positions, β -type *C. acnes* bacteria may be *C. acnes* bacteria which comprise at least 90%, in particular at least 90.5%, at least 91%, at least 91.5%, at least 92%, at least 92.5%, at least 93%, at least 93.5%, at least 94%, at least 94.5%, at least 95%, at least 95.5%, at least 96%, at least 96.5%, at least 97%, at least 97.5%, at least 98%, at least 98.5%, at least 99%, at least 99.5% or 100% of the following nucleotide variants in the hem locus

24

in the hemA gene: G174, G480, C507, G615, G630, A672, C972, G990, C1086, and C1095, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: T143, G156, G177, G 184, T225, T246, A249, C282, A312, T315, C347, T375, T378, T453, C627, G684, G694, G721, T739, T795, C921, G945, C993, C1023, and C1032, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: A78, G97, G150, G234, C237, C256, G258, C261, T318, G348, C426, C438, G465, G487, G488, T489, G525, G546, G591, C606, G645, G666, T711, G714, G738, G771, G777, A786, C803, A831, T843, C846, A856, G894, T897, T898, A903, T909, G942, and G948, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: C12, C42, C51, G151, C156, A160, T177, C258, T267, C279, C324, G325, C363, C368, C375, A388, G399, G400, C408, A423, A426, A433, G447, G460, C465, A596, C618, A644, G645, and T681, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: C242, C477, G531, C705, G729, C768, A835, G868, G873, C883, and G958, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: C12, C24, G49, C90, T189, A339, C465, C471, A537, C549, C609, A627, C633, C642, C672, C675, T684, T690, T741, A746, A788, G798, C894, T899, C1035, C1040, C1047, G1072, A1096, T1110, C1154, C1359, C1497, T1512, T1528, A1533, C1599, C1713, G1851, C1902, G1927, T1932, and T1998, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: C108, T198, A230, G 243, A311, G 339, G375, C390, T393, C443, T468, A471, T483, T519, and G642, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: G90, G237, G355, T384, T393, G396, C399, C466, C475, T522, C539, T567, C594, C768, A783, A864, A921, A928, G933, A936, A980, C990, G1020, C1056, A1092, C1137, A1144, T1153, C1167, G1173, G1177, C1185, T1200, T1206, A1269, A1279, C1296, C1319, G1337, A1344, T1347, C1356, G1360 and G1396, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

Alternatively, β -type *C. acnes* bacteria may be *C. acnes* bacteria which comprise at least 197, at least 198, at least 199, at least 200, at least 201, at least 202, at least 203, at least 204, at least 205, at least 206, at least 207, at least 208, at least 209, at least 210, at least 211, at least 212, at least 213, at least 214, at least 215, at least 216, at least 217 or all of the beta nucleotide variants as defined in Table 1.

Alternatively or in addition, β -type *C. acnes* bacteria are preferably *C. acnes* bacteria in which the nucleic acid sequence of the hem locus (which comprises the genes hemA, hemB, hemC, hemD, hemE, hemH, hemL, and hemY) is at least 97% identical, in particular at least 97.1% identical, at least 97.2% identical, at least 97.3% identical, at least 97.4% identical, at least 97.5% identical, at least 97.6% identical, at least 97.7% identical, at least 97.8% identical, at least 97.9% identical, at least 98% identical, at least 98.1% identical, at least 98.2% identical, at least 98.3% identical, at least 98.4% identical, at least 98.5% identical, at least 98.6% identical, at least 98.7% identical, at least 98.8% identical, at least 98.9% identical, at least 99% identical, at least 99.1% identical, at least 99.2% identical,

at least 99.3% identical, at least 99.4% identical, at least 99.5% identical, at least 99.6% identical, at least 99.7% identical, at least 99.8% identical, at least 99.9% identical, or 100% identical, to the sequence SEQ ID NO: 2.

Preferably, β -type *C. acnes* bacteria are *C. acnes* bacteria

(i) which comprise at least 90%, in particular at least 90.5%, at least 91%, at least 91.5%, at least 92%, at least 92.5%, at least 93%, at least 93.5%, at least 94%, at least 94.5%, at least 95%, at least 95.5%, at least 96%, at least 96.5%, at least 97%, at least 97.5%, at least 98%, at least 98.5%, at least 99%, at least 99.5% or 100% of the beta nucleotide variants as defined in Table 1, or which comprise at least 197, at least 198, at least 199, at least 200, at least 201, at least 202, at least 203, at least 204, at least 205, at least 206, at least 207, at least 208, at least 209, at least 210, at least 211, at least 212, at least 213, at least 214, at least 215, at least 216, at least 217 or all of the beta nucleotide variants as defined in Table 1, and

(ii) in which the nucleic acid sequence of the hem locus is at least 97% identical, in particular at least 97.1% identical, at least 97.2% identical, at least 97.3% identical, at least 97.4% identical, at least 97.5% identical, at least 97.6% identical, at least 97.7% identical, at least 97.8% identical, at least 97.9% identical, at least 98% identical, at least 98.1% identical, at least 98.2% identical, at least 98.3% identical, at least 98.4% identical, at least 98.5% identical, at least 98.6% identical, at least 98.7% identical, at least 98.8% identical, at least 98.9% identical, at least 99% identical, at least 99.1% identical, at least 99.2% identical, at least 99.3% identical, at least 99.4% identical, at least 99.5% identical, at least 99.6% identical, at least 99.7% identical, at least 99.8% identical, at least 99.9% identical, or 100% identical, to the sequence SEQ ID NO: 2.

As used herein, "percent identity" between two sequences, means the percentage of identical nucleotides or amino acids between the two sequences to be compared, obtained with the best alignment of said sequences. As used herein, "best alignment" or "optimal alignment", means the alignment for which the determined percentage of identity (see below) is the highest. Sequence comparison between two sequences is usually realized by comparing these sequences that have been previously aligned according to the best alignment; this comparison is realized on segments of comparison in order to identify and compare the local regions of similarity. The best sequence alignment to perform comparison can be realized, besides manually, by using the global homology algorithm developed by Smith and Waterman (Smith and Waterman (1981) Ad. App. Math. 2:482), by using the local homology algorithm developed by Needleman and Wunsch (Needleman and Wunsch (1970) J. Mol. Biol 48:443), by using the method of similarities developed by Pearson and Lipman (Pearson and Lipman (1988) Proc. Natl. Acad. Sci. USA 85:2444), by using computer software using such algorithms (GAP, BESTFIT, BLAST P, BLAST N, FASTA, TFASTA, etc.), by using the MUSCLE multiple alignment algorithms (Edgar (2004) Nucleic Acids Research 32:1792). To get the best local alignment, one can preferably use Needleman-Wunsch algorithm. The identity percentage between two sequences is determined by comparing these two sequences optimally aligned, the sequences being able to comprise additions or deletions in respect to the reference sequence in order to get the optimal alignment between these two sequences. The percentage of identity may be calculated by determining the number of identical positions between these two sequences,

and dividing this number by the total number of compared positions, and by multiplying the result obtained by 100 to get the percentage of identity between these two sequences. Preferably, for purposes herein, percent identity values refers to values generated using the pairwise sequence alignment program EMBOSS Needle that creates an optimal global alignment of two sequences using the Needleman-Wunsch algorithm, wherein all search parameters are set to default values, i.e. Scoring matrix=BLOSUM62 for protein sequences and DNAsfull for DNA sequences, Gap open=10, Gap extend=0.5, End gap penalty=false, End gap open=10 and End gap extend=0.5.

Method of Treatment or Prevention of *C. acnes* Bacteria-Associated Disease

The present invention concerns a method for treating or preventing a *Cutibacterium acnes* bacteria-associated disease in a subject, said method comprising modulating, in said subject, the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria,

wherein

α -type *C. acnes* bacteria are as defined in the section "α-type *C. acnes* bacteria and β-type *C. acnes* bacteria" above, and

β -type *C. acnes* bacteria are as defined in the section "α-type *C. acnes* bacteria and β-type *C. acnes* bacteria" above.

The present invention also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent inducing the modulation, in said subject, of the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria,

wherein

α -type *C. acnes* bacteria are as defined in the section "α-type *C. acnes* bacteria and β-type *C. acnes* bacteria" above, and

β -type *C. acnes* bacteria are as defined in the section "α-type *C. acnes* bacteria and β-type *C. acnes* bacteria" above.

The present invention further concerns the use of a therapeutic or prophylactic agent inducing the modulation of the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria, for the manufacture of a medicament for the treatment or prevention of a *C. acnes* bacteria-associated disease,

wherein

α -type *C. acnes* bacteria are as defined in the section "α-type *C. acnes* bacteria and β-type *C. acnes* bacteria" above, and

β -type *C. acnes* bacteria are as defined in the section "α-type *C. acnes* bacteria and β-type *C. acnes* bacteria" above.

By "*C. acnes* bacteria-associated disease" is meant herein a disease, in particular a skin disease, which is caused by *C. acnes* bacteria, or is associated with *C. acnes* bacteria presence, or in which *C. acnes* bacteria play a negative role.

Examples of *C. acnes* bacteria-associated disease include, but are not limited to, acne (such as acne vulgaris, acne inversa, acne conglobata, or acne fulminans), keratitis, synovitis acne pustulosis hyperostosis osteitis (SAPHO) syndrome, endocarditis, medical implant biofilm infection, prosthetic joint infections, surgical wound infections, vascular graft infections, anaerobic arthritis, cardiovascular device-related infections (such as prosthetic valve endocarditis), ocular implant infections, breast implant illness, sciatica, conjunctivitis, shunt-associated and/or spinal hardware central nervous system infections, shunt-associated

central nervous system infections, sarcoidosis, endophthalmitis, osteomyelitis, allergic alveolitis, rheumatoid arthritis, infectious arthritis, chronic juvenile arthritis, chronic destructive oligoarthritis, degenerative disc disease, dental infections, ulcerative colitis hyperpyrexia, cerebral abscess, subdural empyema, peritonitis, periodontitis, endodontic infections, chronic rhinosinusitis, folliculitis, corneal ulcer, prostate inflammation, chronic prostatitis, primary biliary cirrhosis, hidradenitis suppurativa, pulmonary angitis, atherosclerosis, prostatic cancer, progressive macular hypomelanosis, eczema, psoriasis, seborrheic dermatitis, rosacea, lichen sclerosus, lichen planus, prurigo nodularis and lichen simplex chronicus.

In a particular embodiment, said *C. acnes* bacteria-associated disease is a *C. acnes* bacteria-associated skin disease. Examples of *C. acnes* bacteria-associated skin diseases include, but are not limited to, acne (such as acne vulgaris, acne inversa, acne conglobata or acne fulminans), surgical skin wound infections, folliculitis, eczema, psoriasis, seborrheic dermatitis, rosacea, lichen sclerosus, lichen planus, prurigo nodularis and lichen simplex chronicus.

In a particular embodiment, said *C. acnes* bacteria-associated disease is a *C. acnes* bacteria-associated inflammatory disease. Examples of *C. acnes* bacteria-associated inflammatory diseases are well-known from the skilled person, and include without limitation acne (such as acne vulgaris, acne inversa, acne conglobata, or acne fulminans), sarcoidosis, SAPHO syndrome, eczema, psoriasis, seborrheic dermatitis, rosacea, lichen sclerosus, lichen planus, prurigo nodularis and lichen simplex chronicus.

Therefore, in a preferred embodiment, said *C. acnes* bacteria-associated disease is selected from the group consisting of acne (such as acne vulgaris, acne inversa, acne conglobata, or acne fulminans), sarcoidosis, SAPHO syndrome, eczema, psoriasis, seborrheic dermatitis, rosacea, lichen sclerosus, lichen planus, prurigo nodularis and lichen simplex chronicus.

In a more preferred embodiment, said *C. acnes* bacteria-associated disease is acne, more particularly acne vulgaris.

As used in the context of the invention, acne vulgaris is a chronic, inflammatory disorder of the pilosebaceous gland. It affects almost all humans at some point of their lives with 15 to 20% suffering from moderate to severe forms of acne. Acne may be classified as mild, moderate or severe. Mild acne typically corresponds to cases wherein there are less than 20 comedones and/or less than 15 inflammatory lesions and/or a total lesion count strictly inferior to 30. Moderate acne typically corresponds to cases wherein there are between 20 and 100 comedones and/or between 15 and 50 inflammatory lesions and/or a total lesion count between 30 and 125. Severe acne typically corresponds to cases wherein there are more than 5 pseudocysts and/or more than 100 comedones and/or more than 50 inflammatory lesions.

In a particular embodiment, said *C. acnes* bacteria-associated disease is severe acne.

As used herein, the terms “treatment”, “treating,” and the like, refer to obtaining a desired pharmacologic or physiologic effect. The effect may be therapeutic in terms of a partial or complete cure of a disease or disorder, or a symptom of a disease or disorder, or an adverse effect attributable to a disease or disorder. “Treatment” as used herein, includes inhibiting the disease, disorder or condition, i.e., arresting its development; and relieving the disease, disorder or condition i.e., causing regression of the disease, disorder or condition.

When applied to acne, the term “treatment” for example includes the reduction of the number of acne lesions, sebum

excretion, follicular keratinization, comedo formation, bacterial colonization of the follicle, *C. acnes* proliferation and/or inflammation. It further encompasses for example the decrease of the severity of the disease, such as the decrease of severe acne to moderate or mild acne, or the decrease of moderate acne to mild acne.

As used herein, the term “prevention”, “preventing”, and the like, includes preventing or decreasing the likelihood or severity of the onset of a disease. This includes prophylactic treatment of those having an enhanced risk of developing such disease. An elevated risk represents an above-average risk that a subject would develop a disease, which can be determined, for example, through family history, detection of genes causing a predisposition to developing said disease, or treatment with antibiotics.

When applied to acne, the term “prevention” for example encompasses the fact of avoiding reaching a given state of severity of acne, for example avoiding reaching severe acne, or moderate acne, or mild acne.

In a particular embodiment of the invention, the treatment and/or prevention of the *C. acnes* bacteria-associated disease is obtained by modulating the ratio of the amount of α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above, to the amount of β -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

By “amount” is meant herein a relative (for example with respect to the whole bacterial population, in particular skin bacterial population, of the subject, or with respect to the whole *C. acnes* population, in particular skin *C. acnes* population, of the subject) or absolute amount (or concentration), that can be determined by any suitable technique well-known from the skilled person. Examples of techniques suitable to determine the amount of a given bacterial type(s) include direct cell counting, indirect cell counting, P C R (including real-time and quantitative PCR), and shotgun metagenomic sequencing.

By “modulating of the ratio” is meant herein an increase or reduction of the ratio after the implementation of the method and/or after the administration of the therapeutic or prophylactic agent compared to the ratio in the absence of or before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

In a particular embodiment, the modulation of the ratio is a statistically significant modulation.

In preferred embodiments, the treatment and/or prevention of the *C. acnes* bacteria-associated disease, in particular of the *C. acnes* bacteria-associated inflammatory disease, is obtained by reducing the ratio of the amount of α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above, to the amount of β -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

By “reduction of the ratio” is meant herein a reduction of the ratio after the implementation of the method and/or after the administration of the therapeutic or prophylactic agent compared to the ratio in the absence of or before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

Preferably, the reduction of the ratio is a statistically significant reduction.

In a particular embodiment, said modulation, preferably reduction, of the ratio is observed 1 day after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent, in particular 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 2 weeks, 3 weeks, 1 month, 2 months, 3 months, 4 months, 5 months,

6 months or 1 year after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent.

In a particular embodiment, said modulation, preferably reduction, of the ratio is maintained (not necessarily at the same level) for 1 day after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent, in particular 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 2 weeks, 3 weeks, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months or 1 year after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent.

In a particular embodiment, said modulation, preferably reduction, of the ratio is maintained (not necessarily at the same level) for the whole period during which the method is implemented and/or the therapeutic or prophylactic agent is administered.

Preferably, said ratio is the ratio of the amount of α -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, on the skin of said subject to the amount of β -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, on said skin.

According to the present invention, the skin refers to the skin of the arms, especially the hands, the skin of the legs, especially the feet, the skin of the armpits, the skin of the neck, the skin of the chest, the skin of the back, the skin of the scalp, and/or the skin of the face. In a preferred embodiment, the skin is the skin of the face, of the neck, of the chest and/or of the back. In a still preferred embodiment, the skin is the skin of the face.

Specifically Reducing the Amount of α -Type *C. acnes* Bacteria

In a particular embodiment, the treating or preventing method of the invention comprises specifically reducing the amount of α -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, in the subject, in particular on the skin of the subject.

The present invention therefore concerns a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising specifically reducing the amount of α -type *C. acnes* bacteria in said subject, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above. It also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which specifically reduces the amount of α -type *C. acnes* bacteria in said subject, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above. It further concerns the use of a therapeutic or prophylactic agent which specifically reduces the amount of α -type *C. acnes* bacteria in a subject for the manufacture of a medicament for the treatment or prevention of a *C. acnes* bacteria-associated disease, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

As used herein, the term "amount" has the same meaning as disclosed in the section "Method of treatment or prevention of *C. acnes* bacteria-associated disease" above.

By "specifically reducing the amount of α -type *C. acnes* bacteria" is meant herein that the amount, in said subject, in particular on the skin of said subject, of *C. acnes* bacteria of another type than α -type, in particular of β -type *C. acnes* bacteria, is not significantly reduced, and/or the amount, in said subject, in particular on the skin of said subject, of other

bacterial species is not significantly reduced. Preferably, by "specifically reducing the amount of α -type *C. acnes* bacteria" is meant herein that the amount, in said subject, in particular on the skin of said subject, of *C. acnes* bacteria of another type than α -type, in particular of β -type *C. acnes* bacteria, is not significantly reduced.

By "reduction of the amount" is meant herein a reduction of the amount after the implementation of the method and/or after the administration of the therapeutic or prophylactic agent compared to the amount in the absence of or before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

Preferably, the reduction of the amount is a statistically significant reduction.

In a particular embodiment, said reduction of the amount is observed 1 day after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent, in particular 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 2 weeks, 3 weeks, or 1 month after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent.

In a particular embodiment, said reduction of the amount is maintained (not necessarily at the same level) for 1 day after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent, in particular 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 2 weeks, 3 weeks, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months or 1 year after the first implementation of the method and/or after the first administration of the therapeutic or prophylactic agent.

In a particular embodiment, said reduction of the amount is maintained (not necessarily at the same level) for the whole period during which the method is implemented and/or the therapeutic or prophylactic agent is administered.

In a particular embodiment, the treating or preventing method of the invention comprises specifically killing or inhibiting the growth of α -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, in the subject, in particular on the skin of the subject.

The present invention therefore concerns a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising specifically killing or inhibiting the growth of α -type *C. acnes* bacteria in said subject, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above. It also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which specifically kills or inhibits the growth of α -type *C. acnes* bacteria in said subject, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above. It further concerns the use of a therapeutic or prophylactic agent which specifically kills or inhibits the growth of α -type *C. acnes* bacteria for the manufacture of a medicament for the treatment or prevention of a *C. acnes* bacteria-associated disease, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

In a particular embodiment, said α -type *C. acnes* bacteria are killed at a level corresponding to a reduction of at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80% or at least 90%, preferably a reduction of at least 50%, at least 60%, at least 70%, at least 80% or at least 90%, of the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin

population, in the subject, compared to a non-treated population or compared to the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

In a preferred embodiment, said α -type *C. acnes* bacteria are killed at a level corresponding to an at least 0.1 log, at least 0.2 log, at least 0.5 log, at least 1 log, preferably at least 2 log, more preferably at least 3 log, at least 4 log or at least 5 log reduction of the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject, compared to a non-treated population or compared to the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

As used herein, the term “log” refers to the common logarithm which is the base-10 logarithm (also indicated by log₁₀).

By “specifically killing α -type *C. acnes* bacteria” is meant herein that *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, present in said subject, in particular on the skin of said subject, are not significantly killed, in particular are killed at a level at least 1 log, at least 2 log, at least 3 log or at least 4 log inferior to α -type *C. acnes* bacteria. Preferably, by “specifically killing α -type *C. acnes* bacteria” is meant herein that *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, present in said subject, in particular on the skin of said subject, are not significantly killed, in particular are killed at a level at least 1 log, at least 2 log, at least 3 log or at least 4 log inferior to α -type *C. acnes* bacteria. In some particular embodiments, by “specifically killing α -type *C. acnes* bacteria” is meant herein that *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, preferably β -type *C. acnes* bacteria, present in said subject, in particular on the skin of said subject, are killed at a level corresponding to a reduction of less than 20%, preferably less than 10%, of the population of *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, preferably β -type *C. acnes* bacteria, in the subject, compared to a non-treated population or compared to the population of *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, preferably β -type *C. acnes* bacteria, in the subject, preferably in the skin of the subject, before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

In a particular embodiment, the growth of said α -type *C. acnes* bacteria is inhibited at a level corresponding to an at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80% or at least 90%, preferably an at least 50%, at least 60%, at least 70%, at least 80% or at least 90%, reduction of the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject, compared to a non-treated population or compared to the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

In a preferred embodiment, the growth of said α -type *C. acnes* bacteria is inhibited at a level corresponding to an at least 0.1 log, at least 0.2 log, at least 0.5 log, at least 1 log,

preferably at least 2 log, more preferably at least 3 log, at least 4 log or at least 5 log reduction of the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject, compared to a non-treated population or compared to the α -type *C. acnes* bacteria population, in particular α -type *C. acnes* bacteria skin population, in the subject before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

By “specifically inhibiting the growth of α -type *C. acnes* bacteria” is meant herein that the growth of *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, present in said subject, in particular on the skin of said subject, is not significantly inhibited, in particular is inhibited at a level at least 1 log, at least 2 log, at least 3 log or at least 4 log inferior to α -type *C. acnes* bacteria. Preferably, by “specifically inhibiting the growth of α -type *C. acnes* bacteria” is meant herein that the growth of *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, present in said subject, in particular on the skin of said subject, is not significantly inhibited, in particular is inhibited at a level at least 1 log, at least 2 log, at least 3 log or at least 4 log inferior to α -type *C. acnes* bacteria. In some particular embodiments, by “specifically inhibiting the growth of α -type *C. acnes* bacteria” is meant herein that the growth of *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, preferably β -type *C. acnes* bacteria, present in said subject, in particular on the skin of said subject, is inhibited at a level corresponding to a reduction of less than 20%, preferably less than 10%, compared to a non-treated population or compared to the population of *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, preferably β -type *C. acnes* bacteria, in the subject, preferably in the skin of the subject, before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

In a particular embodiment, the treating or preventing method of the invention comprises administering to said subject a therapeutically or prophylactically effective amount of an antibacterial, phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which specifically targets said α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

Similarly, in a particular embodiment of the composition for use according to the present invention or the use of a therapeutic or prophylactic agent according to the present invention, the therapeutic or prophylactic agent is an antibacterial, phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which specifically targets said α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

As used herein, the term “therapeutically effective amount” refers to a nontoxic but sufficient amount of an active agent to provide the desired therapeutic effect.

As used herein, the term “prophylactically effective amount” refers to a nontoxic but sufficient amount of an active agent to provide the desired preventive effect.

By “targets said α -type *C. acnes* bacteria” is meant herein that said therapeutic or prophylactic agent is able to recognize, preferably specifically recognizes, said α -type *C. acnes*

bacteria, and/or to exert, preferably specifically exert, its activity on said α -type *C. acnes* bacteria.

By “specifically targets said α -type *C. acnes* bacteria” is meant herein that said therapeutic or prophylactic agent recognizes and/or exerts its activity on said α -type *C. acnes* bacteria but does not significantly recognize and/or exert its activity on *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, and/or other bacterial species, present in said subject, in particular on the skin of said subject. Preferably, by “specifically targets said α -type *C. acnes* bacteria” is meant herein that said therapeutic or prophylactic agent recognizes and/or exerts its activity on said α -type *C. acnes* bacteria but does not significantly recognize and/or exert its activity on *C. acnes* bacteria of another type than α -type, in particular β -type *C. acnes* bacteria, present in said subject, in particular on the skin of said subject.

By “antibacterial” is meant herein a compound inhibiting or preventing bacterial growth, killing bacteria, or reducing the number of bacteria. Said antibacterial can typically be a small molecule, a protein, peptide or polypeptide, or a nucleic acid such as a DNA or RNA. Examples of antibacterials include antibiotics, bacteriocins and endolysins. Therefore, in a particular embodiment, said antibacterial is an antibiotic, a bacteriocin and/or an endolysin, which specifically targets said α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

By “bacteriocin” is meant herein a proteinaceous toxin produced by bacteria to kill or inhibit growth of other bacteria. Bacteriocins are categorized in several ways, including producing strain, common resistance mechanisms, and mechanism of killing. Such bacteriocins have been described from gram negative bacteria (e.g. microcins, colicin-like bacteriocins and tailocins) and from gram positive bacteria (e.g. Class I, Class II, Class III or Class IV bacteriocins).

In the context of the invention, said bacteriocin may be a wild-type bacteriocin or an engineered bacteriocin, in particular a bacteriocin mutant, variant or chimera, typically comprising modifications and/or alterations of the amino acid sequence. Such alterations and/or modifications may comprise mutations such as deletions, insertions and additions, substitutions or combinations thereof and/or chemical changes of the amino acid residues, e.g. biotinylation, acetylation, pegylation, chemical changes of the amino-, thiol- or carboxyl-groups. Such modified bacteriocin typically exhibits the lytic activity of the wild-type bacteriocin. However, said activity can be the same, higher or lower than the activity of the wild-type bacteriocin. Said activity can be about 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190 or about 200% of the activity of the wild-type bacteriocin or even more. The activity can be measured by assays well known in the art by a person skilled in the art as e.g. the plate lysis assay or the liquid lysis assay which are e.g. described in Briers et al. (2007) *J. Biochem. Biophys Methods* 70:531-533, or Donovan et al. (2006) *FEMS Microbiol Lett.* 265:133-139. Alternatively, such modified bacteriocin can exhibit a different lytic activity compared to the wild-type bacteriocin, such as a lytic activity with a different specificity compared to the wild-type bacteriocin.

By “endolysin” or “lysin” is meant herein enzymes used by bacteriophages at the end of their replication cycle to degrade the peptidoglycan of the bacterial host from within, resulting in cell lysis and release of progeny virions. They

are typically either β (1,4)-glycosylases (lysozymes), transglycosylases, amidases or endopeptidases.

In the context of the invention, said endolysin may be a wild-type endolysin or an engineered endolysin, in particular a endolysin mutant, variant or chimera, typically comprising modifications and/or alterations of the amino acid sequence. Such alterations and/or modifications may comprise mutations such as deletions, insertions and additions, substitutions or combinations thereof and/or chemical changes of the amino acid residues, e.g. biotinylation, acetylation, pegylation, chemical changes of the amino-, SH- or carboxyl-groups. Such modified endolysin typically exhibit the lytic activity of the wild-type endolysin. However, said activity can be the same, higher or lower than the activity of the wild-type endolysin. Said activity can be about 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190 or about 200% of the activity of the wild-type endolysin or even more. The activity can be measured by assays well known in the art by a person skilled in the art as e.g. the plate lysis assay or the liquid lysis assay which are e.g. described in Briers et al. (2007) *J. Biochem. Biophys Methods* 70:531-533, or Donovan et al. (2006) *FEMS Microbiol Lett.* 265:133-139. Alternatively, such modified endolysin can exhibit a different lytic activity compared to the wild-type endolysin, such as a lytic activity with a different specificity compared to the wild-type endolysin.

By “phage” or “bacteriophage” is meant herein a bacterial virus. Bacteriophages are obligate intracellular parasites that multiply inside bacteria by co-opting some or all of the host biosynthetic machinery. Bacteriophages can be found inside bacteria as a prophage whose genome is integrated in the bacterial chromosome or as a phage-plasmid whose genome is part of an extrachromosomal plasmid (such phage-plasmids are for example disclosed in Ravin et al. (1999) *Molecular Microbiology* 34(5):980-994). Phages contain nucleic acid (i.e., genome) and proteins, and may be enveloped by a lipid membrane. Depending upon the phage, the nucleic acid genome can be either DNA or RNA, and can exist in either circular or linear forms. The size of the phage genome varies depending upon the phage. The simplest phages have genomes that are only a few thousand nucleotides in size, while the more complex phages may contain more than 100,000 nucleotides in their genome, and in rare instances more than 1,000,000. The number and amount of individual types of protein in phage particles will vary depending upon the phage.

In the context of the invention, the terms “recombinant” and “engineered” are used interchangeably. By “recombinant” is meant herein a nucleic acid, protein or microorganism containing genetic materials derived from multiple different sources and/or obtained using molecular biology and/or synthetic biology tools, or being encoded by genetic materials derived from multiple different sources and/or obtained using molecular biology and/or synthetic biology tools. Therefore, in the context of the invention, a “recombinant” nucleic acid, protein or microorganism is not a wild-type nucleic acid, protein or microorganism.

As used herein, a “recombinant phage” refers to a phage, as defined above, that has been modified at the protein and/or nucleic acid level. Therefore, in the context of the invention a “recombinant phage” is not a wild-type phage or a phage obtained after natural evolution, but a phage which has been voluntarily modified, for example using molecular biology and/or synthetic biology tools.

As used herein, the term “phagemid” or “phasmid” are equivalent and refer to a vector that derives from both a plasmid and a bacteriophage genome. A phagemid typically

comprises a phage packaging site and optionally an origin of replication (ori), in particular a bacterial and/or phage origin of replication. In one embodiment, the phagemid does not comprise a functional bacterial origin of replication and thus cannot replicate by itself once injected into a bacterium. Alternatively, the phagemid can comprise a plasmid origin of replication, in particular a bacterial and/or phage origin of replication.

As used herein, the term “packaged phagemid” refers to a phagemid which is encapsidated in a bacteriophage scaffold, phage-derived delivery particle or capsid. Particularly, it refers to a bacteriophage scaffold, phage delivery particle or capsid devoid of a bacteriophage genome. The packaged phagemid may be produced with a helper phage strategy, well known from the man skilled in the art. The helper phage typically comprises all the genes coding for the structural and functional proteins that are indispensable for the phagemid according to the invention to be encapsidated.

As used herein, the term “plasmid” refers to a circular nucleic acid molecule capable of autonomous replication in a host cell. Said plasmid can be a conjugative plasmid. In a particular embodiment, said plasmid is a recombinant plasmid.

As used herein, the term “DNA- or RNA-containing vesicle” refers to an artificial delivery vehicle enclosing at least one nucleic acid molecule (which can be DNA, RNA or a mixture thereof) within a core surrounded by a shell. Said DNA- or RNA-containing vesicle can be chemically-based vesicles, in particular polymer-based vesicles such as polymersomes or nonionic surfactants such as niosomes, lipid-based vesicles such as liposomes, protein-based or peptide-based vesicles, or nanoparticle-based vesicles.

As used herein, the term “extracellular vesicle” or “cell-derived nanovesicle” refers to a nanosized structure in the form of a particle, released or secreted from cells, which enclose biomolecules, such as proteins, nucleic acid molecules, lipids or organelles, in a cell membrane of a lipid layer, typically identical to the cell membrane of the cell from which the biomolecules are derived. Examples of extracellular vesicles include microvesicles, exosomes, oncosomes and apoptotic bodies. Extracellular vesicles typically have an average diameter of 30 nm to 1 μ m, preferably of 100 nm to 1 μ m. Extracellular vesicles can be isolated from naturally formed extracellular vesicles (for example by differential centrifugation culminating in ultrafiltration, density gradient/cushion centrifugation, and immunoaffinity-based capture) or can be obtained by artificial methods (for example by an extrusion process where cells are passed through microfluidic channels, a multi-stage filtration process, etc.).

The bacteria used in the context of the invention can be a wild-type bacteria, such as a wild-type bacteria naturally able to specifically target α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above, for example by naturally producing antibacterials, as defined above, that specifically target α -type *C. acnes* bacteria, or phages, as defined above, that specifically target α -type *C. acnes* bacteria. Alternatively, the bacteria used in the context of the invention are engineered (also called recombinant bacteria), for example bacteria which have been engineered to produce antibacterials as defined above, that specifically target α -type *C. acnes* bacteria, or to produce phages or engineered phages, as defined above, that specifically target α -type *C. acnes* bacteria, or to comprise phagemids or plasmids, as defined above, that specifically target α -type *C. acnes* bacteria.

As used herein, “engineered bacteria” refers to bacteria that have been modified at the protein and/or nucleic acid level. Therefore, in the context of the invention “engineered bacteria” are not a wild-type bacteria or bacteria obtained after natural evolution, but bacteria which have been voluntarily modified, for example using molecular biology and/or synthetic biology tools.

In a particular embodiment, the treating or preventing method of the invention comprises administering to said subject a therapeutically or prophylactically effective amount of a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, as defined above, which specifically targets said α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

Similarly, in a particular embodiment of the composition for use according to the present invention or the use of a therapeutic or prophylactic agent according to the present invention, the therapeutic or prophylactic agent is a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, as defined above, which specifically targets said α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

In a preferred embodiment, the phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, encodes (or comprises a nucleic acid which encodes) a programmable nuclease designed to specifically target said α -type *C. acnes* bacteria.

By “programmable nuclease” is meant herein an enzyme designed to recognize and cleave a particular nucleic acid sequence. Programmable nucleases are well-known from the skilled person and include C R IS P R-Cas systems (including C R IS P R-Cas nucleases), TALE Ns and variants thereof, zinc finger nucleases (ZFN) and variants thereof, natural, evolved or engineered meganuclease or recombinase variants, and any combination and hybrid thereof.

Therefore, in a particular embodiment, said programmable nuclease is selected from CRISPR-Cas nucleases, TALENs and variants thereof, ZFNs and variants thereof, natural, evolved or engineered meganuclease or recombinase variants, and any combination and hybrid thereof.

As used herein, the term “TALEN” or “transcription activator-like effector nuclease” refers to a recombinant protein comprising a “Transcription Activator-like effector (TALE) binding domain” and an endonuclease domain. “Transcription Activator-Like effector (TALE) binding domain” refers to a protein that contains a “repeat region” (also referred to as “repeat domain”), which contains a plurality of repeated highly conserved 33-34 amino acid sequence (referred to as a “repetitive module” and “repeat module”), with the last repetitive module at the C-terminal end being a partial or half (approximately 20 amino acids) length. The combination of repetitive modules in the repeat region binds to a specific nucleotide sequence in the target DNA and/or RNA sequence, such that the TALE N modifies the target DNA and/or RNA within or adjacent to the specific nucleotide sequence in the cell or progeny thereof.

As used herein, the term “zinc finger nuclease” or “ZFN” refers to a chimeric protein comprising a zinc finger DNA-binding domain fused to a nuclease domain from an endonuclease or exonuclease.

As used herein, the term “meganuclease” refers to an endonuclease that binds double-stranded DNA at a recog-

nitition sequence that is greater than 12 base pairs. Naturally-occurring meganucleases can be monomeric (e.g., I-SceI) ordimeric (e.g., I-CreI). The term meganuclease, as used herein, can be used to refer to monomeric meganucleases, dimeric meganucleases, or to the monomers which associate to form a dimeric meganuclease. Due to the large recognition site of meganucleases, this site generally occurs only once in any given genome.

As used herein, the term “recombinase” refers to a DNA modifying enzyme that binds, cleaves, strand exchanges, and rejoins DNA at its respective recombination sites (i.e., an enzyme capable of performing DNA recombination).

“C R IS P R-Cas system” refers to Clustered regularly interspaced short palindromic repeats and their C R IS P R-associated (Cas) proteins. These systems comprise a plurality of diverse RNA guided prokaryotic adaptive immune systems employed by these organisms to defend against foreign parasitic nucleic acids.

The CRISPR-Cas system contains two distinct elements, i.e. i) an endonuclease, in this case the C R IS P R associated nuclease (Cas or “C R IS P R associated protein”) and ii) a guide RNA.

Depending on the type of C R IS P R-Cas system, the guide RNA may be in the form of a chimeric RNA which consists of the combination of a C R IS P R (crRNA) bacterial RNA and a tracrRNA (trans-activating RNA C R IS P R) (Jinek et al. (2012) Science 337(6096):816-21). The guide RNA combines the targeting specificity of the crRNA corresponding to the “spacing sequences” that serve as guides to the Cas proteins, and the conformational properties of the tracrRNA in a single transcript. When the guide RNA and the Cas protein are expressed simultaneously in the cell, the target genomic sequence can be permanently interrupted (and causing disappearance of the targeted and surrounding sequences and/or cell death, depending on the location) or modified. The modification may be guided by a repair matrix.

The CRISPR-Cas system includes two main classes depending on the nuclease mechanism of action:

Class 1 is made of multi-subunit effector complexes and includes type 1, III and IV;

Class 2 is made of single-unit effector modules, like Cas9 nuclease, and includes type II (II-A, II-B, II-C variant), V (V-A, V-B, V-C, V-D, V-E, V-U1, V-U2, V-U3, V-U4, V-U5) and VI (VI-A, VI-B1, VI-B2, VI-C, VI-D).

In a particular embodiment, the programmable nuclease is a CRISPR-Cas nuclease (also referred to herein as Cas nuclease or CRISPR enzyme). A variety of CRISPR enzymes are available for use in the context of the present invention. In some embodiments, the C R IS P R enzyme is a Type II CRISPR enzyme, a Type II-A or Type II-B CRISPR enzyme. In another embodiment, the CRISPR enzyme is a Type I CRISPR enzyme or a Type III CRISPR enzyme. In some embodiments, the C R IS P R enzyme catalyzes DNA cleavage. In some other embodiments, the C R IS P R enzyme catalyzes RNA cleavage.

Non-limiting examples of Cas proteins as part of a multi-subunit effector or as a single-unit effector include Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas10, Cas11 (SS), Cas12a (Cpf1), Cas12b (C2c1), Cas12c (C2c3), Cas12d (CasY), Cas12e (CasX), C2c4, C2c8, C2c5, C2c10, C2c9, Cas13a (C2c2), Cas13b (C2c6), Cas13c (C2c7), Cas13d, Csa5, Csc1, Csc2, Cse1, Cse2, Csy1, Csy2, Csy3, Csf1, Csf2, Csf3, Csf4, Csm1, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csn2, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx13, Csx1, Csx15,

SdCpf1, CmtCpf1, TsCpf1, CmaCpf1, PcCpf1, ErCpf1, FbCpf1, UbcCpf1, AsCpf1, LbCpf1, Mad4, Mad7, Cms1, homologues thereof, orthologues thereof, variants thereof, or modified versions thereof. In some embodiments, the CRISPR enzyme cleaves both strands of the target nucleic acid at the Protospacer Adjacent Motif (PAM) site.

In a particular embodiment, the CRISPR enzyme is any Cas9 protein, for instance any naturally-occurring bacterial Cas9 as well as any variants, homologues or orthologues thereof.

By “Cas9” is meant a protein Cas9 (also called Csn1 or Csx12) or a functional protein, peptide or polypeptide fragment thereof, i.e. capable of interacting with the guide RNA(s) and of exerting the enzymatic activity (nuclease) which allows it to perform the double-strand cleavage of the DNA of the target genome. “Cas9” can thus denote a modified protein, for example truncated to remove domains of the protein that are not essential for the predefined functions of the protein, in particular the domains that are not necessary for interaction with the gRNA(s).

The sequence encoding Cas9 (the entire protein or a fragment thereof) as used in the context of the invention can be obtained from any known Cas9 protein (Fonfara et al. (2014) Nucleic Acids Res. 42(4):2577-90; Koonin et al. (2017) Curr Opin Microbiol. 37:67-78). Examples of Cas9 proteins useful in the present invention include, but are not limited to, Cas9 proteins of *Streptococcus pyogenes* (Sp-Cas9), *Streptococcus thermophiles* (St1Cas9, St3Cas9), *Streptococcus mutans*, *Staphylococcus aureus* (SaCas9), *Campylobacter jejuni* (CjCas9), *Francisella novicida* (Fn-Cas9) and *Neisseria meningitidis* (NmCas9).

In a particular embodiment, the C R IS P R enzyme is dCas9.

The sequence encoding Cpf1 (Cas12a) (the entire protein or a fragment thereof) as used in the context of the invention can be obtained from any known Cpf1 (Cas12a) protein (Koonin et al. (2017) Curr Opin Microbiol. 37:67-78). Examples of Cpf1(Cas12a) proteins useful in the present invention include, but are not limited to, Cpf1(Cas12a) proteins of *Acidaminococcus* sp, Lachnospiraceae bacterium and *Francisella novicida*.

The sequence encoding Cas13a (the entire protein or a fragment thereof) as used in the context of the invention can be obtained from any known Cas13a (C2c2) protein (Abudayyeh et al. (2017) Nature 550(7675):280-284). Examples of Cas13a (C2c2) proteins useful in the present invention include, but are not limited to, Cas13a (C2c2) proteins of *Leptotrichia wadei* (LwaCas13a).

The sequence encoding Cas13d (the entire protein or a fragment thereof) as used in the context of the invention can be obtained from any known Cas13d protein (Yan et al. (2018) Mol Cell. 70(2):327-339.e5.). Examples of Cas13d proteins useful in the present invention include, but are not limited to, Cas13d proteins of *Eubacterium siraeum* and *Ruminococcus* sp.

The sequence encoding Mad4 (the entire protein or a fragment thereof) as used in the context of the invention is disclosed in international application WO2018/236548.

The sequence encoding Mad7 (the entire protein or a fragment thereof) as used in the context of the invention is disclosed in international application WO2018/236548.

The sequence encoding Cms1 (the entire protein or a fragment thereof) as used in the context of the invention is disclosed in international patent application WO2017/141173.

In one embodiment, the C R IS P R enzymes may be coupled to a guide RNA or single guide RNA (sgRNA).

Therefore, in a preferred embodiment, said phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria engineered bacteria, further encodes a guide RNA designed to specifically target said α -type *C. acnes* bacteria.

As used herein, the term “guide RNA” refers to either a crRNA (also known as CRISPR RNA), or the combination of a crRNA and a trRNA (also known as tracrRNA). The crRNA and trRNA may be associated as a single RNA molecule (single guide RNA, sgRNA) or in two separate RNA molecules (dual guide RNA, dgRNA). The trRNA may be a naturally-occurring sequence, or a trRNA sequence with modifications or variations compared to naturally occurring sequences. For clarity, the term “guide RNA” as used herein, and unless specifically stated otherwise, may refer to an RNA molecule (comprising A, C, G, and U nucleotides) or to a DNA molecule encoding such an RNA molecule (comprising A, C, G, and T nucleotides) or complementary sequences thereof. As well-known from the skilled person, a “guide RNA” in the CRISPR-Cas system is utilized for the detection of a target nucleic acid and is responsible for recognizing a portion of the sequence possessed by the target nucleic acid. A guide RNA contains a sequence for recognizing a portion of the sequence carried by the target nucleic acid and is responsible for the specificity of detection of the target nucleic acid in the CRISPR-Cas system. The guide RNA is typically designed to match the Cas protein with which it is administered. Methods for designing and producing a guide RNA are well known.

In a particular embodiment, said guide RNA is designed to specifically target an α -type *C. acnes* bacteria-specific sequence, gene or locus.

In a particular embodiment, said guide RNA is designed to specifically target the hem locus of said α -type *C. acnes* bacteria.

In a more preferred embodiment, said guide RNA is designed to specifically target at least one of the nucleotide variants in the hem locus associated with α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above. Preferably, said guides RNA are selected to have at least one mismatch, preferably two mismatches, between α -type *C. acnes* bacteria and β -type *C. acnes* bacteria and/or a mismatch in the PAM site.

Therefore, in a still more particular embodiment, said guide RNA is designed to specifically target at least one of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G 387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

Preferably, said phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, encodes a plurality (at least 2, 3, 4, 5 or more) guide RNAs specifically targeting a plurality (at least 2, 3, 4, 5 or more) of the nucleotide variants in the hem locus associated with α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above.

In a particular embodiment, said phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, comprises a sequence encoding a guide RNA selected from the group consisting of SEQ ID NOs: 13 to 16. Genetically Modifying a DNA Sequence in the α -Type *C. acnes* Bacteria

In a particular embodiment, the treating or preventing method of the invention comprises genetically modifying a DNA sequence in the α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above, in said subject, in particular on the skin of said subject, to generate at least one change in said α -type *C. acnes* bacteria, preferably in the hem locus of said α -type *C. acnes* bacteria.

Similarly, in a particular embodiment of the composition for use according to the present invention or the use of a therapeutic or prophylactic agent according to the present invention, the therapeutic or prophylactic agent induces the genetic modification of a DNA sequence in the α -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above, in said subject, in particular on the skin of said subject, to generate at least one change in said α -type *C. acnes* bacteria, preferably in the hem locus of said α -type *C. acnes* bacteria.

The present invention therefore concerns a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising genetically modifying a DNA sequence in the α -type *C. acnes* bacteria in said subject, in particular on the skin of said subject, to generate at least one change in said α -type *C. acnes* bacteria, preferably in the hem locus of said α -type *C. acnes* bacteria. It also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or pro-

phylactic agent which genetically modifies a DNA sequence in the α -type *C. acnes* bacteria in said subject, in particular on the skin of said subject, to generate at least one change in said α -type *C. acnes* bacteria, preferably in the hem locus of said α -type *C. acnes* bacteria. It further concerns the use of a therapeutic or prophylactic agent which genetically modifies a DNA sequence in the α -type *C. acnes* bacteria in a subject, in particular on the skin of said subject, to generate at least one change in said α -type *C. acnes* bacteria, preferably in the hem locus of said α -type *C. acnes* bacteria, for the manufacture of a medicament for the treatment or prevention of a *C. acnes* bacteria-associated disease.

By “genetically modifying a DNA sequence” or “genetic modification of a DNA sequence” is meant herein that at least one nucleotide of a DNA sequence is modified. Said modification can be a substitution of at least one nucleotide, an insertion of at least one nucleotide, a deletion of at least one nucleotide, or a modification of the nucleobase of at least one nucleotide (such as methylation).

In the context of the invention, said genetic modification preferably occurs in the hem locus of said α -type *C. acnes* bacteria, in other words, said genetic modification preferably generates at least one change in the hem locus of said α -type *C. acnes* bacteria.

In a particular embodiment, said genetic modification preferably generates at least one change in at least one of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G 387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989,

C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

In a particular embodiment, said genetic modification generates at least one change in at least one of the nucleotide variants in the hem locus associated with α -type *C. acnes* bacteria into a nucleotide variant in the hem locus associated with β -type *C. acnes* bacteria, as defined in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria”.

Therefore, in a particular embodiment, said genetic modification generates at least one, more particularly at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 218 or all of the following changes in the hem locus:

in the hemA gene: T173G, A479G, T506C, A614G, T629G, C671A, T971C, A989G, T1085C, and T1094C, wherein the positions are defined with reference to sequence SEQ ID NO: 3,

in the hemB gene: C142T, A155G, A176G, A183G, C224T, C245T, G248A, T281C, G311A, C314T, T346C, C374T, G377T, C452T, T626C, C683G, A693G, A720G, C738T, C794T, A920C, T944G, T992C, G1022C, and T1031C, wherein the positions are defined with reference to sequence SEQ ID NO: 4,

in the hemC gene: G77A, A96G, C149G, A233G, T236C, T255C, A257G, A260C, C317T, A347G, T425C, T437C, A464G, A486G, A487G, C488T, A524G, A545G, A590G, T605C, A644G, A665G, C710T, C713G, A737G, A770G, A776G, T785A, T802C, G830A, C842T, T845C, G855A, C893G, A896T, C897T, G902A, G908T, A941G, and A947G, wherein the positions are defined with reference to sequence SEQ ID NO: 5,

in the hemD gene: T11C, T41C, T50C, A150G, T155C, G159A, G176T, T257C, C266T, T278C, G323C, A324G, T362C, T367C, T374C, G387A, A398G, A399G, T407C, G422A, G425A, G432A, A446G, A459G, T464C, G595A, T617C, T643A, A644G, and C680T, wherein the positions are defined with reference to sequence SEQ ID NO: 6,

in the hemE gene: T241C, T476C, A530G, T704C, A728G, G767C, T834A, A867G, A872G, A882C, and A957G, wherein the positions are defined with reference to sequence SEQ ID NO: 7,

in the hemH gene: T11C, T23C, A48G, T89C, C188T, G338A, T464C, T470C, G536A, T548C, T608C, G626A, T632C, T641C, T671C, T674C, C683T, C689T, A740T, G745A, G787A, A797G, T893C, A898T, T1034C, T1039C, T1046C, C1071G, G1095A, C1109T, T1153C, T1358C, G1496C, C1511T, C1527T, G1532A, T1598C, T1712C, A1850G, G1901C, A1926G, C1931T, and C1997T, wherein the positions are defined with reference to sequence SEQ ID NO: 8,

in the hemL gene: T107C, C197T, T229A, A242G, C310A, A338G, A374G, T389C, G392T, T442C, C467T, G470A, C482T, G518T, and C641G, wherein the positions are defined with reference to sequence SEQ ID NO: 9, and

in the hemY gene: A89G, A236G, A354G, C383T, C392T, A395G, T398C, T465C, T474C, C521T, T538C, C566T, A593C, A767C, T782A, G863A, G920A, T927A, A932G, G935A, C979A, G989C, C1019G, T1055C, G1091A, T1136C, G1143A, G1152T, T1166C, T1172G, A1176G, T1184C, C1199T, C1205T, G1268A, G1278A, T1295C, T1318C, A1336G,

G1343A, C1346T, G1355C, T1359G and C1395G, wherein the positions are defined with reference to sequence SEQ ID NO: 10.

The genetic modification can be a point mutation(s), a deletion(s), insertion(s) or any combination thereof. Preferably, the genetic modification is a point modification.

The genetic modification preferably eliminates, reduces, or increases the expression of a gene. The genetic modification can be in the translated or untranslated regions of a gene. The genetic modification can be in the promoter region of a gene or within any other region involved in gene regulation.

In a particular embodiment, the genetic modification eliminates and/or reduces the expression of one or several genes of the hem locus, i.e. hemA, hemB, hemC, hemD, hemE, hemH, hemL, and/or hemY genes in the *C. acnes* bacteria, preferably in α -type *C. acnes* bacteria. In a more particular embodiment, the genetic modification(s) eliminate(s) or reduce(s) porphyrin production by the *C. acnes* bacteria, in particular by α -type *C. acnes* bacteria.

In some embodiments, the genetic modification results in the change in at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20, 25, 30, 35, 40, 45, 50, 100, 200, 500, etc. amino acids to a different amino acid. In some embodiments, the genetic modification introduces a stop codon. In some embodiments, the genetic modification is outside protein coding sequences, within RNA, or within regulatory sequences. In some embodiments, the genetic modification introduces one or more rare codons so the expression of the protein is affected, preferably reduced. In some embodiments, the start codon is modified so the normal start is changed or abrogated.

In a particular embodiment, said genetic modification results in the fact that the modified *C. acnes* bacteria does not belong anymore to the group of α -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above. In a preferred embodiment, said genetic modification results in the fact that the modified *C. acnes* bacteria becomes a β -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

The therapeutic or prophylactic agent which genetically modifies a DNA sequence in the α -type *C. acnes* bacteria to generate at least one change in said α -type *C. acnes* bacteria, preferably in the hem locus of said α -type *C. acnes* bacteria may be a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a gene editing enzyme or system targeting said hem locus, in particular at least one of the nucleotide variants as defined above, in said α -type *C. acnes* bacteria.

In a particular embodiment, the treating or preventing method of the invention comprises administering to said subject a therapeutically or prophylactically effective amount of a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a gene editing enzyme or system targeting said hem locus, in particular at least one of the nucleotide variants as defined above, in said α -type *C. acnes* bacteria.

Similarly, in a particular embodiment of the composition for use according to the present invention or the use of a therapeutic or prophylactic agent according to the present invention, the therapeutic or prophylactic agent is a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a gene editing enzyme

or system targeting and genetically modifying said hem locus, in particular at least one of the nucleotide variants as defined above, in said α -type *C. acnes* bacteria.

As used herein, the term "gene editing" refers to a type of genetic engineering in which DNA is inserted, deleted, modified or replaced in the genome of a living organism.

By "gene editing system" is meant herein a combination of components required for gene editing of a genome in a cell. The various components of the system, such as polypeptides, gRNA, etc., may exist independently of each other, or may exist in any combination thereof.

By "gene editing enzyme" is meant herein an enzyme enabling the gene editing of a genome in a cell. Said gene editing enzyme can typically be part of a gene editing system as defined above, and be used with another component of the system.

In a particular embodiment, the gene editing enzyme or system is a base editing enzyme or system.

Base editing (BE) refers to the ability to substitute a specific nucleotide base pair on a DNA or RNA molecule by another. In base editing technology, there is no exchange of DNA, instead an enzymatic reaction converts a nucleotide to another one leading to a mismatch at the level of dsDNA that is then corrected by the cell machinery.

In some embodiments, the base editing system comprises one or more of the following enzymes and systems:

Cytosine base editors (CBE) and Adenosine base editors (ABE), as described in Rees and Liu (2018) NatRev Genet 19:770-788.

Examples of DNA Base Editors Include

Cytosine Base Editor (CBE) that converts C:G into T:A (Komor et al. (2016) Nature 533:420-4);

Adenine Base Editor (ABE) that converts A:T into G:C (Gaudelli et al. (2017) Nature 551(7681):464-471);

Cytosine Guanine Base Editor (CGBE) that converts C:G into G:C (Chen et al. (2020) Precise and programmable C:G to G:C base editing in genomic DNA. Biorxiv; Kurt et al. (2020) CRISPR C-to-G base editors for inducing targeted DNA transversions in human cells. Nature Biotechnology);

Cytosine Adenine Base Editor (CABE) that converts C:G into A:T (Zhao et al. (2020) New base editors change C to A in bacteria and C to G in mammalian cells. Nature Biotechnology);

Adenine Cytosine Base Editor (ACBE) that converts A:T into C:G (WO2020181180);

Adenine Thymine Base Editor (ATBE) that converts A:T into T:A (WO2020181202);

Thymine Adenine Base Editor (TABE) that converts T:A into A:T (WO2020181193; WO2020181178; WO2020181195).

Base editors differ in the base modification enzymes. CBE rely on ssDNA cytidine deaminase among which: APOBEC1, rAPOBEC1, APOBEC1 mutant or evolved version (evoAPOBEC1), and APOBEC homologs (APOBEC3A (eA3A), Anc689), Cytidine deaminase 1 (CDA1), evoCDA1, FERNY, evoFERNY.

ABE rely on deoxyadenosine deaminase activity of a tandem fusion TadA-TadA* where TadA* is an evolved version of TadA, an *E. coli* tRNA adenosine deaminase enzyme, able to convert adenosine into Inosine on ssDNA. TadA* include TadA-8a-e and TadA-7.10.

Except from base modification enzyme there has been also modifications implemented to base editor to increase editing efficacy, precision and modularity:

the addition of one or two uracil DNA glycosylase inhibitor domain (UGI) to prevent base excision repair mechanism to revert base edition;
 the addition of Mu-GAM that decrease insertion-deletion rate by inhibiting Non-homologous end joining mechanism in the cell (NHEJ);
 the use of nickase active Cas9 (nCas9 D10A) that, by creating nicks on the non-edited strand favors its repair and consequently the fixation of the edited base;
 the use of diverse Cas proteins from for example different organisms, mutants with different PAM motifs or different fidelity or different family (e.g. Cas12a).

Non-limiting examples of DNA-based editor proteins include BE1, BE2, BE3, BE4, BE4-GAM, HF-BE3, Sniper-BE3, Target-AID, Target-AID-NG, ABE, EE-BE3, YE1-BE3, YE2-BE3, YEE-BE3, BE-PLUS, SaBE3, SaBE4, SaBE4-GAM, Sa(KKH)-BE3, VQR-BE3, VRER-BE3, EQR-BE3, xBE3, Cas12a-BE, Ea3A-BE3, A3A-BE3, TAM, CRISPR-X, ABE7.9, ABE7.10, ABE7.10*, xABE, ABESa, VQR-ABE, VRER-ABE, Sa(KKH)-ABE, ABE8e, SpRY-ABE, SpRY-CBE, SpG-CBE4, SpG-ABE, SpRY-CBE4, SpCas9-NG-ABE, SpCas9-NG-CBE4, enAsBE1.1, enAsBE1.2, enAsBE1.3, enAsBE1.4, AsBE1.1, AsBE1.4, CRISPR-Abest, CRISPR-Cbest, eA3A-BE3, AncBE4.

Cytosine Guanine Base Editors (C G B E) consist of a nickase CRISPR fused to:

A cytosine deaminase (rAPOBEC) and base excision repair proteins (e.g. rXRCC1) (Chen et al. Precise and programmable C:G to G:C base editing in genomic DNA. Biorxiv (2020); Chen et al. (2021) Nature Communications 12:1384);

A ratAPOBEC1 variant (R33A) protein and an *E. coli*-derived uracil DNA N-glycosylase (eUNG) (Kurt et al. (2020) CRISPR C-to-G base editors for inducing targeted DNA transversions in human cells. Nature Biotechnology).

Cytosine Adenine Base Editors (CABE) consist of a Cas9 nickase, a cytidine deaminase (e.g. AID), and a uracil-DNA glycosylase (Ung) (Zhao et al. (2020) New base editors change C to A in bacteria and C to G in mammalian cells. Nature Biotechnology).

ACBE include a nucleic acid programmable DNA-binding protein and an adenine oxidase (WO2020181180).

ATBE consist of a Cas9 nickase and one or more adenosine deaminase or an oxidase domain (WO2020181202).

TABE consist of a Cas9 nickase and an adenosine methyltransferase, a thymine alkyltransferase, or an adenosine deaminase domain (WO2020181193; WO2020181178; WO2020181195).

Base editor molecules can also consist of two or more of the above listed editor enzymes fused to a Cas protein (e.g. combination of an ABE and CBE). These biomolecules are named dual base editors and enable the editing of two different bases (Grunewald et al. (2020) A dual-deaminase CRISPR base editor enables concurrent adenine and cytosine editing, Nature Biotechnology; Li et al. (2020) Targeted, random mutagenesis of plant genes with dual cytosine and adenine base editors, Nature Biotechnology).

In a particular embodiment, the base editing enzyme is a fusion protein comprising a Cas9 (e.g., a Cas9 nickase) domain and a deaminase domain. In some embodiments, the base editing enzyme comprises Cas9 and a cytosine deaminase enzyme, such as APOBEC enzymes, or adenosine deaminase enzymes, such as ADAT enzymes, for example as disclosed in U.S. Patent Publ. 2015/0166980. In one embodiment, the deaminase is an ACF1/ASE deaminase. In various embodiments, the APOBEC enzyme is selected

from the group consisting of APOBEC1 deaminase, APOBEC2 deaminase, APOBEC3A deaminase, APOBEC3B deaminase, APOBEC3C deaminase, APOBEC3D deaminase, APOBEC3F deaminase, APOBEC3G deaminase, and APOBEC3H deaminase. In various embodiments, the base editing enzyme comprises a Cas9 domain, a cytosine deaminase domain, and a uracil glycosylase inhibitor (UGI) domain. In one embodiment, the deaminase is an adenosine deaminase that deaminates adenosine in DNA, for example as disclosed in U.S. Pat. No. 10,113,163. In some embodiments, the base editing system further comprises an inhibitor of base repair, such as a nuclease dead inosine specific nuclease (dISN), for example as disclosed in U.S. Pat. No. 10,113,163.

In a particular embodiment, the base editing system comprises a Cytosine base editor (CBE) and/or an Adenosine base editor (ABE) as defined above.

In another particular embodiment, the gene editing enzyme or system is a prime editing enzyme or system.

Prime Editing (PE) allows introduction of insertions, deletions (indels), and multiple base-to-base conversions. Prime editing typically relies on the ability of a reverse transcriptase (RT), fused to a Cas nickase variant, to convert RNA sequence brought by a prime editing guide RNA (pegRNA) into DNA at the nick site generated by the Cas protein. The DNA flap generated from this process is then included or not in the targeted DNA sequence.

Prime editors, as described in Anzalone et al. (2019) Nature 576:149-157, typically consist of a nCas9 fused to a reverse transcriptase used in combination with a prime editing RNA (pegRNA; a guide RNA that includes a template region for reverse transcription).

Prime editing systems typically include:

a Cas nickase variant such as Cas9-H840A fused to a reverse transcriptase domain such as M-MLV RT or its mutant version (M-MLV RT(D200N), M-MLV RT(D200N/L603W), M-MLV RT(D200N/L603W/T330P/T306K/W313F), and

a prime editing guide RNA (pegRNA)

To favor editing, the prime editing system can include the expression of an additional sgRNA targeting the Cas nickase activity towards the non-edited DNA strand ideally only after the resolution of the edited strand flap by designing the sgRNA to anneal with the edited strand but not with the original strand.

Non-limiting examples of prime editing systems include PE 1, PE 1-M1, PE 1-M 2, PE 1-M 3, PE1-M6, PE1-M15, PE1-M3inv, PE 2, PE 3, PE 3b.

In various embodiments, the prime editing system comprises a fusion protein comprising a catalytically impaired Cas9 endonuclease fused to an engineered reverse transcriptase, programmed with a prime editing guide RNA (pegRNA) that both specifies the target site and encodes the desired edit, for example as described in (2019) Nature 576:149-157.

Other suitable gene editing systems or enzymes include: Cas9 Retron precISe Parallel Editing via homologyY ('CRISPEY'), a retron RNA fused to the sgRNA and expressed together with Cas9 and the retron proteins including at least the reverse transcriptase (Sharon et al. (2018) Cell 175:544-557.e16).

The SCRIBE strategy: a retron system expressed in combination with a recombinase promoting the recombination of single stranded DNA, also known as single stranded annealing proteins (SSAPs) (Farzadfard and Lu (2014) Science 346:1256272). Such recombinases include but are not limited to phage recombinases such

as lambda red, recET, Sak, Sak4, and SSAPs described in Wannier et al. Improved bacterial recombineering by parallelized protein discovery. Biorxiv 2020.01.14.906594 (2020) doi:10.1101/2020.01.14.906594.

The targetron system based on group II introns described in Karberg et al. Nat Biotechnol (2001) 19:1162-7 which has been adapted to many bacterial species.

Other retron based gene targeting approaches as described in Simon et al. (2019) Nucleic Acids Res 47:11007-11019.

A bridge system based on IS1111 and IS110 insertion sequence (IS) family members and using a non-coding RNA (ncRNA) with two distinct binding loops that separately recognize the IS DNA donor and its genomic insertion target site. By bridging the donor and target DNA molecules through direct base-pairing interactions, the bispecific bridge RNA facilitates DNA recombination by the IS recombinase (Durrant et al. Nature. 2024 Jun; 630(8018):984-993; Siddiquee et al. Nat Commun 15, 5235 (2024)).

In a particular embodiment, said gene editing system or enzyme is a base editor or a prime editor, as defined above.

The present invention also concerns a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising specifically reducing the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject, wherein α -type *C. acnes* bacteria are *C. acnes* bacteria as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

It also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which specifically reduces the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject by genetically modifying a DNA sequence in α -type *C. acnes* bacteria in said subject, to generate at least one change, preferably in the hem locus of said α -type *C. acnes* bacteria, and wherein α -type *C. acnes* bacteria are *C. acnes* bacteria as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

It further concerns the use of a therapeutic or prophylactic agent for the manufacture of a medicament for the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, wherein said therapeutic or prophylactic agent specifically reduces the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject by genetically modifying a DNA sequence in α -type *C. acnes* bacteria in said subject, to generate at least one change, preferably in the hem locus of said α -type *C. acnes* bacteria, and wherein α -type *C. acnes* bacteria are *C. acnes* bacteria as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

The therapeutic or prophylactic agent may be as defined above.

In a particular embodiment, the therapeutic or prophylactic agent which specifically reduces the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject by genetically modifying a DNA sequence in α -type *C. acnes* bacteria in said subject is a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a gene editing enzyme or system targeting said hem locus, in particular at least one of the nucleotide variants as defined above, in said α -type *C. acnes* bacteria.

In a particular embodiment, the treating or preventing method of the invention comprises administering to said subject a therapeutically or prophylactically effective amount of a therapeutic or prophylactic agent which specifically reduces the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject by genetically modifying a DNA sequence in α -type *C. acnes* bacteria in said subject. Preferably, said therapeutic or prophylactic agent is a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a gene editing enzyme or system targeting said hem locus, in particular at least one of the nucleotide variants as defined above.

The genetic modification may be as defined above and may be carried out as described above.

In a particular embodiment, the genetic modification eliminates and/or reduces the expression of one or several genes of the hem locus, i.e. hemA, hemB, hemC, hemD, hemE, hemH, hemL, and/or hemY genes in α -type *C. acnes* bacteria.

In a more particular embodiment, the genetic modification(s) eliminate(s) or reduce(s) porphyrin production by α -type *C. acnes* bacteria.

By "specifically reducing the expression of at least one hem locus protein in α -type *C. acnes*" is meant herein that the expression of said at least one hem locus protein, i.e. hemA, hemB, hemC, hemD, hemE, hemH, hemL, and/or hemY proteins, in said subject, in particular on the skin of said subject, of *C. acnes* bacteria of another type than α -type, in particular of β -type *C. acnes* bacteria, and/or of other bacterial species, is not significantly reduced, preferably that the expression of said at least one hem locus protein, in said subject, in particular on the skin of said subject, of *C. acnes* bacteria of another type than α -type, in particular of β -type *C. acnes* bacteria is not significantly reduced. Preferably, the expression of said at least one hem locus protein of *C. acnes* bacteria of another type than α -type, in particular of β -type *C. acnes* bacteria, and/or of other bacterial species, is reduced by less than 20%, preferably less than 10%, compared to a non-treated population or compared to the population of *C. acnes* bacteria of another type than α -type, in particular of β -type *C. acnes* bacteria, and/or of other bacterial species, in the subject before the implementation of the method and/or the administration of the therapeutic or prophylactic agent.

The genetic modification(s) inducing the reduction of expression of at least one hem locus protein may affect or not the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria, i.e. the genetic modification(s) may result or not in the fact that the modified *C. acnes* bacteria does not belong anymore to the group of α -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above. In some particular embodiments, the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria is not affected by the genetic modification(s). Increasing the Amount of β -Type *C. acnes* Bacteria

In a particular embodiment, the treating or preventing method of the invention comprises increasing the amount of β -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, in said subject, in particular on the skin of said subject.

The present invention therefore concerns a method for treating or preventing a *C. acnes* bacteria-associated disease in a subject, said method comprising increasing the amount of β -type *C. acnes* bacteria in said subject, in particular on

the skin of said subject. It also concerns a composition for use in the treatment or prevention of a *C. acnes* bacteria-associated disease in a subject, said composition comprising a therapeutic or prophylactic agent which increases the amount of β -type *C. acnes* bacteria in said subject, in particular on the skin of said subject. It further concerns the use of a therapeutic or prophylactic agent which increases the amount of β -type *C. acnes* bacteria in a subject, in particular on the skin of said subject, for the manufacture of a medicament for the treatment or prevention of a *C. acnes* bacteria-associated disease in said subject.

As used herein, the term "amount" has the same meaning as disclosed in the section "Method of treatment or prevention of *C. acnes* bacteria-associated disease" above.

In a particular embodiment, the amount of β -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, is specifically increased, in said subject, in particular on the skin of said subject.

By "specifically increase the amount of β -type *C. acnes* bacteria" is meant herein that the amount, in said subject, in particular on the skin of said subject, of *C. acnes* bacteria of another type than β -type, in particular of α -type *C. acnes* bacteria, is not significantly increased, and/or the amount, in said subject, in particular on the skin of said subject, of other bacterial species is not significantly increased.

By "increase of the amount" is meant herein an increase of the amount after the implementation of the method compared to the amount in the absence of or before the implementation of the method.

In a particular embodiment, the increase of the amount is a statistically significant increase.

In a particular embodiment, said increase of the amount is observed 1 day after the first implementation of the method, in particular 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 2 weeks, 3 weeks, or 1 month after the first implementation of the method.

In a particular embodiment, said increase of the amount is maintained (not necessarily at the same level) for 1 day after the first implementation of the method, in particular 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 2 weeks, 3 weeks, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months or 1 year after the first implementation of the method.

In a particular embodiment, said increase of the amount is maintained (not necessarily at the same level) for the whole period during which the method is implemented.

In a particular embodiment, the treating or preventing method of the invention comprises administering β -type *C. acnes* bacteria, as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, in said subject, in particular on the skin of said subject.

In particular, the therapeutic or prophylactic agent which increases the amount of β -type *C. acnes* bacteria may be a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, which encodes a gene editing enzyme or system targeting the hem locus of α -type *C. acnes* bacteria, in particular the alpha nucleotide variants as defined in Table 1, to generate changes in said α -type *C. acnes* bacteria, said changes resulting in the fact that the modified α -type *C. acnes* bacteria becomes β -type *C. acnes* bacteria.

Subject, Regimen and Administration

The subject according to the invention is an animal, preferably a mammal, even more preferably a human. However, the term "subject" can also refer to non-human animals, in particular mammals.

The human subject according to the invention may be a new-born, a child, an infant, an adolescent or an adult at any age. In a particular embodiment, the human subject is an adolescent.

In a preferred embodiment, the subject has been diagnosed with, or is at risk of developing acne, in particular acne vulgaris. In a preferred embodiment, the subject has been diagnosed with acne, in particular acne vulgaris, using the diagnostic method disclosed in the section "Risk of developing acne and diagnosis" below. In a preferred embodiment, the subject has been identified as being at risk of developing acne, in particular acne vulgaris, using the method disclosed in the section "Risk of developing acne and diagnosis" below.

In a particular embodiment, the subject has never received any treatment before the treating or preventing method of the invention.

In a particular embodiment, the subject has already received at least one line of treatment, preferably several lines of treatment, prior to the treating or preventing method of the invention.

Preferably, the treatment is administered regularly, preferably between every day and every month, more preferably between every day and every two weeks, more preferably between every day and every week, even more preferably the treatment is administered every day. In a particular embodiment, the treatment is administered several times a day, preferably 2 times a day.

The duration of treatment according to the invention is preferably between 1 day and 1 year, more preferably between 1 week and 6 months, still more preferably between 2 weeks and 3 months, even more preferably between 3 weeks and 2 months. In a particular embodiment, the duration of the treatment is about 1 month. Alternatively, the treatment may last as long as the disease persists.

The form of the pharmaceutical compositions, the route of administration and the dose of administration of therapeutic or prophylactic agents according to the invention can be adjusted by the man skilled in the art according to the type and severity of the disease, and to the patient or subject, in particular its age, weight, sex, and general physical condition.

Particularly, the amount of therapeutic or prophylactic agent to be administered has to be determined by standard procedure well known by those of ordinary skills in the art. Physiological data of the patient or subject (e.g. age, size, and weight) and the routes of administration have to be taken into account to determine the appropriate dosage, so as a therapeutically or prophylactically effective amount will be administered to the patient or subject.

For example, the total amount of phages, engineered phages or packaged phagemids according to the invention, for each administration is typically comprised between 10^4 and 10^{15} delivery vehicles.

The preferred route of administration used in the context of the invention is the topical administration.

It is envisioned that the treating or preventing method of the invention can be used in combination with other standard treatments. For example, in some embodiments, the treating or preventing method of the invention is implemented before, after, or simultaneously with administration of an additional therapeutic agent or with implementation of an additional treating or preventing method. In some embodiments, said additional therapeutic agent may be a topical antibiotic. Non-limiting examples of topical antibiotics include clindamycin, doxycycline, erythromycin, and tetracycline. In some embodiments, said additional therapeutic

agent may be an oral antibiotic. Non-limiting examples of oral antibiotics include erythromycin; or a tetracycline, such as doxycycline or minocycline. Other additional therapeutic agents may comprise anti-inflammatory agents, antioxidants, acids, or a combination thereof.

In some embodiments, said additional treating or preventing method is light therapy. In some embodiments, said additional treating or preventing method is a laser treatment. In some embodiments, said additional treating or preventing method comprises treating the subject with blue light. In some embodiments, said additional treating or preventing method comprises treating the subject with red light. In some embodiments, said additional treating or preventing method comprises treating the subject with a targeted laser device. In some embodiments, said additional treating or preventing method comprises treating the subject with a laser that targets porphyrins.

A combination of additional therapeutic agent(s) and of additional treating or preventing method(s), as disclosed above, may also be used.

Pharmaceutical Composition

The present invention also concerns a pharmaceutical composition comprising a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, extracellular vesicle, bacteria or engineered bacteria, as defined in the section "Specifically reducing the amount of α -type *C. acnes* bacteria" above, which encodes a programmable nuclease as defined in the section "Specifically reducing the amount of α -type *C. acnes* bacteria" above, or a gene editing enzyme or system as defined in the section "Genetically modifying a DNA sequence in the α -type *C. acnes* bacteria" above, designed to specifically target a α -type *C. acnes* bacteria, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above.

In a particular embodiment, said pharmaceutical composition further comprises a pharmaceutically acceptable vehicle.

By "pharmaceutically acceptable vehicle" is meant herein pharmaceutical ingredients known to be useful for preparing pharmaceutical compositions to be administered to patients in need. Such ingredients are safe and non-sensitizing under the conditions used.

Pharmaceutically acceptable vehicles are known to one of ordinary skill in the art, and include common excipients, diluents, or carriers.

In a particular embodiment, said pharmaceutical composition further comprises at least one pharmaceutical acceptable excipient and/or adjuvant.

By "pharmaceutically acceptable excipient" is meant herein a non-pharmaceutically active additive used in the manufacture of a pharmaceutical composition, which allows the pharmaceutically active ingredient to be manufactured into a pharmaceutical composition or a galenic formulation providing the necessary bioavailability of the medicament to the patient upon the administration of the pharmaceutical composition. The excipient is preferably compatible with the other ingredients of the composition and produces no adverse effect, allergic reaction or other undesirable reaction when it is administered to a human or an animal.

By "pharmaceutically acceptable adjuvant" is meant herein a substance that is non-toxic, biologically tolerable, and otherwise biologically suitable for administration to a subject, added to a pharmaceutical composition to strengthen the role of the excipient or of a therapeutically active ingredient.

In a particular embodiment, said pharmaceutical composition further comprises an additional therapeutically active agent.

By "additional therapeutically active agent" is meant herein any chemical, biochemical, organic, inorganic compound, composition, element or substance that is designated or can be used for the treatment of diseases, disorders, malfunctions, etc. of a living being or of biological material in general, and which is different from the therapeutic or prophylactic agents defined in the sections "Method of treatment or prevention of *C. acnes* bacteria-associated disease", "Specifically reducing the amount of α -type *C. acnes* bacteria", "Genetically modifying a DNA sequence in the α -type *C. acnes* bacteria" and/or "Increasing the amount of β -type *C. acnes* bacteria" above.

Said additional therapeutically active agent can be selected from the group consisting of nonsteroidal anti-inflammatory drugs (NSAIDs) analgesics; opiate agonist analgesics; salicylate analgesics; H1-blocker antihistamines; H2-blocker antihistamines; anti-infective agents; anti-aerobic anti-infectives; antifungal antibiotic anti-infectives; macrolide antibiotic anti-infectives; miscellaneous beta-lactam antibiotic anti-infectives; penicillin antibiotic anti-infectives; quinolone antibiotic anti-infectives; tetracycline antibiotic anti-infectives; antituberculosis antimycobacterial anti-infectives; antiprotozoal anti-infectives; antimalarial antiprotozoal anti-infectives; anti-retroviral anti-infectives; antiviral anti-infective agents; alkylating antineoplastic agents; nitrosourea alkylating antineoplastic agents; antimetabolite antineoplastic agents; pyrimidine analog antimetabolite antineoplastic agents; hormonal antineoplastics; natural antineoplastics; antibiotic natural antineoplastics; vinca alkaloid natural antineoplastics; autonomic agents; anticholinergic autonomic agents; antimuscarinic anticholinergic autonomic agents; ergot alkaloid autonomic agents; cholinergic agonist parasympathomimetics; cholinesterase inhibitor parasympathomimetics; alpha-blocker sympatholytics; beta-blocker sympatholytics; adrenergic agonist sympathomimetics; cardiovascular agents; beta-blocker antianginals; calcium-channel blocker antianginals; nitrate antianginals; cardiac glycoside antiarrhythmics; class I antiarrhythmics; class II antiarrhythmics; class III antiarrhythmics; class IV antiarrhythmics; alpha-blocker antihypertensives; angiotensin-converting enzyme inhibitor (ACE inhibitor) antihypertensives; beta blocker antihypertensives; calcium-channel blocker antihypertensive agents; central-acting adrenergic antihypertensives; diuretic antihypertensive agents; peripheral vasodilator antihypertensives; antilipemics; bile acid sequestrant antilipemics; HMG-CoA reductase inhibitor antilipemics; inotropes; cardiac glycoside inotropes; thrombolytic agents or enzymes; dermatological agents; dermatological corticosteroid anti-inflammatory agents; antifungal topical anti-infectives; antiviral topical anti-infectives; topical antineoplastics; electrolytic and renal agents; loop diuretics; potassium-sparing diuretics; thiazide diuretics; uricosuric agents; enzymes such as RNase and DNase; immunosuppressive agents; antiemetics; salicylate gastrointestinal anti-inflammatory agents; gastric acid-pump inhibitor anti-ulcer agents; H2-blocker anti-ulcer agents; digestants; prokinetic agents; opiate agonist intravenous anesthetics; hematopoietic anti anemia agents; coagulation agents; anticoagulants; growth receptor inhibitors; abortifacients; antidiabetic agents; oral contraceptives; progestin contraceptives; estrogens; fertility agents; parathyroid agents; pituitary hormones; progestins; thyroid hormones; immunobiologic agents; immunoglobulins; amide local anesthetics; ester local anesthetics; musculoskeletal corti-

costeroid anti-inflammatory agents; musculoskeletal anti-inflammatory immunosuppressives; musculoskeletal non-steroidal anti-inflammatory drugs (NSAIDs); skeletal muscle relaxants; reverse neuromuscular blocker skeletal muscle relaxants; neurological agents; anticonvulsants; barbiturate anticonvulsants; benzodiazepine anticonvulsants; anti-Parkinson's agents; antivertigo agents; opiate agonists; opiate antagonists; beta-blocker anti-glaucoma agents; miotic anti-glaucoma agents; ophthalmic aminoglycoside anti-infectives; ophthalmic quinolone anti-infectives; ophthalmic corticosteroid anti-inflammatory agents; ophthalmic nonsteroidal anti-inflammatory drugs (NSAIDs); antipsychotics; benzodiazepine anxiolytics, sedatives and hypnotics; psychostimulants; antitussives; bronchodilators; adrenergic agonist bronchodilators; respiratory corticosteroid anti-inflammatory agents; antidotes; heavy metal antagonists/chelating agents; deterrent substance abuse agents; withdrawal substance abuse agents; minerals, such as iron, calcium, and magnesium; vitamin B compounds, such as cyanocobalamin (vitamin B12) and niacin (vitamin B3); vitamin C compounds; vitamin D compounds, such as calcitriol; vitamin A, vitamin E, and vitamin K compounds; anti-bleeding agents; anthelmintic anti-infectives; sclerosants; anabolic agents; antacids; anti-asthmatic agents; anticholesterolemic and anti-lipid agents; anti-diarrheals; anti-manic agents; anti-nauseants; anti-obesity agents; antipyretic and analgesic agents; antispasmodic agents; antithrombotic agents; anti-uricemic agents; anti-tussives; appetite suppressants; cerebral dilators; coronary dilators; decongestants; diagnostic agents; erythropoietic agents; expectorants; gastrointestinal sedatives; hyperglycemic agents; hypoglycemic agents; ion exchange resins; laxatives; mucolytic agents; neuromuscular drugs; peripheral vasodilators; psychotropics, stimulants; thyroid and antithyroid agents; and uterine relaxants.

In a particular embodiment, said pharmaceutical composition may further comprise a skin care agent.

By "skin care agent" is meant herein an agent that has one or more beneficial effects on the care and/or hygiene of the skin. The skin care agent can be selected from the group consisting of antioxidants, free-radical scavengers, skin protecting agents, skin conditioning agents, skin soothing agents, exfoliators, moisturizers, emollients and humectants.

By "skin conditioning agent" is meant herein an agent that can maintain the skin in a good condition. Examples of skin conditioning agents include urea, guanidine, aloe vera, glycolic acid and glycolate salts such as ammonium and quaternary alkyl ammonium, lactic acid and lactate salts such as sodium lactate, ammonium lactate and quaternary alkyl ammonium lactate, polyhydroxy alcohols such as sorbitol, glycerol, hexanetriol, tocopherol, propylene glycol, butylene glycol, hexylene glycol, polyethylene glycol, carbohydrates such as alkoxyated glucose, starches, starch derivatives, glycerin, pyrrolidone carboxylic acid (PCA), lactamide monoethanolamine, acetamide monoethanolamine, volatile silicone oils, nonvolatile silicone oils, *Helianthus annuus* seed oil, phospholipids, *Salix alba* (willow) bark extract, glycine soja seed extract, and mixtures thereof.

Examples of skin soothing agents include bisabolol.

As used herein, "skin protecting agents" are agents that protect the skin against chemical irritants and/or physical irritants, such as UV light, including sunscreens, anti-wrinkle and anti-skin atrophy agents.

Examples of UV blocking agents include 2-ethylhexyl p-methoxycinnamate, 2-ethylhexyl N,N-dimethyl-p-aminobenzoate, p-aminobenzoic acid, 2-phenylbenzimidazole-5-sulfonic acid, octocrylene, oxybenzone, homomenthyl

salicylate, octyl salicylate, 4,4'-methoxy-t-butyl dibenzoylmethane, 4-isopropyl dibenzoylmethane, 3-benzylidene camphor, 3-(4-methylbenzylidene) camphor, anthanilates, ultrafine titanium dioxide, zinc oxide, iron oxide, silica, 4-N,N-(2-ethylhexyl)methylaminobenzoic acid ester of 2,4-dihydroxybenzophenone, 4-N,N-(2-ethylhexyl)-methylaminobenzoic acid ester with 4-hydroxydibenzoylmethane, 4-N,N-(2-ethylhexyl)-methylaminobenzoic acid ester of 2-hydroxy-4-(2-hydroxyethoxy)benzophenone and 4-N,N-(2-ethylhexyl)-methylaminobenzoic acid ester of 4-(2-hydroxyethoxy)dibenzoylmethane.

Examples of anti-wrinkle and anti-skin atrophy agents include retinoic acid and its derivatives, retinol, retinyl esters, salicylic acid and its derivatives, sulfur-containing D and L amino acids except cysteine, alpha-hydroxy acids (e.g., glycolic acid and lactic acid), phytic acid, lipoic acid and lysophosphatidic acid.

Examples of antioxidants and/or free-radical scavengers include ascorbic acid, salts of ascorbic acid such as ascorbyl palmitate and sodium ascorbate, ascorbyl glucosamine, vitamin E (i.e., tocopherols such as α -tocopherol), derivatives of vitamin E (e.g., tocopheryl acetate), retinoids such as retinoic acid, retinol, trans-retinol, cis-retinol, mixtures of trans-retinol and cis-retinol, 3-dehydroretinol and derivatives of vitamin A (e.g., retinyl acetate, retinal and retinyl palmitate, also known as tetinyl palmitate), lipoic acid, sodium citrate, sodium sulfite, lycopene, anthocyanids, bioflavonoids (e.g., hesperitin, naringen, rutin and quercetin), superoxide dismutase, glutathione peroxidase, butylated hydroxytoluene (BHT), butylated hydroxyanisole (BHA), indole-3-carbinol, pycnogenol, melatonin, sulfuraphane, pregnenolone, lipoic acid and 4-hydroxy-5-methyl-3[2H]-furanone.

Examples of exfoliants include hydroxy carboxylic acids such as alpha hydroxy acids or beta hydroxy acids, keto acids and hydroxybenzoic acids.

Examples of moisturizers include lactic acid and other hydroxy acids and their salts, glycerol, propylene glycol, butylene glycol, sodium PCA, sodium hyaluronate, Carbowax 200, Carbowax 400 and Carbowax 800.

Examples of emollients or humectants include panthenol, cetyl palmitate, glycerol (glycerin), PPG-15 stearyl ether, lanolin alcohol, lanolin, lanolin derivatives, cholesterol, petrolatum, isostearyl neopentanoate, octyl stearate, mineral oil, isocetyl stearate, myristyl myristate, octyl dodecanol, 2-ethylhexyl palmitate (octyl palmitate), dimethicone, phenyl trimethicone, cyclomethicone, C₁₂-C₁₅ alkyl benzoates, dimethiconol, propylene glycol, *Theobroma grandiflorum* seed butter, ceramides (e.g. ceramide 2 or ceramide 3), hydroxypropyl bispalmitamide MEA, hydroxypropyl bislauramide MEA, hydroxypropyl bisisostearamide MEA, 1,3-bis(N-2-(hydroxyethyl)stearylaminol)-2-hydroxy propane, bis-hydroxyethyl tocopheryl succinoylamido hydroxypropane, urea, aloe, allantoin, glycyrrhetic acid, safflower oil, oleyl alcohol, oleic acid, stearic acid, dicaprylate/dicaprate, diethyl sebacate, isostearyl alcohol, pentylene glycol, isononyl isononanoate, and 1,3-bis(N-2-(hydroxyethyl)palmitoylamino)-2-hydroxypropane.

The pharmaceutical composition of the invention can be formulated under any suitable form well-known from the skilled person.

In a particular embodiment, said composition is in the form of aqueous, hydroalcoholic or oily solutions, of dispersions in the form of solutions or dispersions of the lotion or serum type, of emulsions, in particular with liquid or semi-liquid consistency of the milk type, typically obtained by dispersion of a fatty phase in an aqueous phase (O/W) or

55

conversely (W/O), or suspensions or emulsions of soft semi-solid or solid consistency of the cream type, of cream, of aqueous or anhydrous gel, of microemulsions, of nanoe-mulsions, of microcapsules, of microparticles, of ionic and/or nonionic type vesicular dispersions, of stick, of aerosol 5 spray, of pump spray, or of foam. In a particular embodiment, said composition is in the form of an emulsion, of a microemulsion or of a nanoemulsion.

The pharmaceutical composition of the invention can be in any form well-known from the skilled person, typically 10 depending on its administration route.

For oral administration, the pharmaceutical composition can be in the form of tablets, capsules, sugar-coated pills, syrups, suspensions, solutions, powders, granules, emul- 15 sions, suspensions of microspheres or nanospheres or lipid or polymer vesicles providing controlled release.

For parenteral administration, the pharmaceutical composition can be in the form of solutions or suspensions for infusion or for injection.

For topical application, the pharmaceutical composition can be in the form of unguents, creams, milks, ointments, powders, impregnated tampons, solutions, gels, sprays, lotions or suspensions. It can also be in the form of suspen- 20 sions of microspheres or nanospheres or lipid or polymer vesicles or polymer patches or hydrogels providing controlled release. This formulation for topical application can be in anhydrous form, in aqueous form or in the form of an emulsion.

Preferably, the pharmaceutical composition of the invention is for topical application.

The pharmaceutical composition according to the present invention can be prepared by mixing the essential ingredient (s), and optional ingredient(s), if necessary.

The method and means to mix the above essential and optional ingredients are not limited. Any conventional 35 method and means can be used to mix the above essential and optional ingredients to prepare the formulation according to the present invention.

Risk of Developing Acne and Diagnosis

The present invention also relates to a method for determining if a subject is at risk of developing acne, in particular acne vulgaris, said method comprising the steps of: 40

- a) determining the presence and/or amount of α -type *C. acnes* bacteria in a sample, in particular a skin sample, of said subject, wherein α -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, and
- b) based on the presence and/or amount determined at step a), determining if said subject is at risk of developing acne, in particular acne vulgaris. 45

The present invention also relates to a method for determining if a subject is at risk of developing acne, in particular acne vulgaris, said method comprising the steps of:

- a) determining the presence and/or amount of β -type *C. acnes* bacteria in a sample, in particular a skin sample, of said subject, wherein β -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, and
- b) based on the presence and/or amount determined at step a), determining if said subject is at risk of developing acne, in particular acne vulgaris. 50

The present invention also relates to a method of diagnosing acne, in particular acne vulgaris, in a subject, said method comprising the steps of:

- a) determining the presence and/or amount of α -type *C. acnes* bacteria in a sample, in particular a skin sample, of said subject, wherein α -type *C. acnes* bacteria are as 65

56

defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, and

- b) based on the presence and/or amount determined at step a), diagnosing acne, in particular acne vulgaris, in said subject.

The present invention also relates to a method of diagnosing acne, in particular acne vulgaris, in a subject, said method comprising the steps of:

- a) determining the presence and/or amount of β -type *C. acnes* bacteria in a sample, in particular a skin sample, of said subject, wherein β -type *C. acnes* bacteria are as defined in the section " α -type *C. acnes* bacteria and β -type *C. acnes* bacteria" above, and
- b) based on the presence and/or amount determined at step a), diagnosing acne, in particular acne vulgaris, in said subject.

As used herein, the term "amount" has the same meaning as disclosed in the section "Method of treatment or prevention of *C. acnes* bacteria-associated disease" above.

In a preferred embodiment of said methods, the amount of α -type *C. acnes* bacteria or β -type *C. acnes* bacteria, is the relative abundance of α -type *C. acnes* bacteria or β -type *C. acnes* bacteria.

As used herein, the term "presence" means that the amount of said α -type *C. acnes* bacteria or β -type *C. acnes* bacteria in said sample is within the detectable range of the techniques used to detect bacteria, more particularly said α -type *C. acnes* bacteria or β -type *C. acnes* bacteria. In other words, when it is determined that said α -type *C. acnes* bacteria or β -type *C. acnes* bacteria are not present in said sample, it means that they are below the detection threshold of the technique used for this determination step. 30

Preferably, said sample is a skin sample.

The skin sample may be any sample taken from the skin of a subject. A skin sample may be obtained by any means known in the art including, but not limited to, swabbing the skin with a tool able to collect skin cells (e.g., a Q-tip or cotton swab), placing an adhesive or tape on the surface of the skin and removing the adhesive or tape, thereby yielding a skin sample on the adhesive or tape, or through a biopsy (e.g., a shave biopsy, a punch biopsy, an incisional biopsy, a saucerization biopsy, or an excisional biopsy). As will be understood by the skilled person, the skin sample needs to be obtained by any means which enables collecting the bacteria present on the skin of the subject, preferably in the skin follicle of the subject. Therefore, the skin sample comprises at least a part of the skin microbiome.

In some embodiments, the skin sample further includes epithelial cells, epidermal cells, dermal cells, hair/skin follicles, adipose tissue (i.e., subcutaneous fat), and/or connective tissue.

As underlined above and in the examples below, the present inventors showed that a set of 218 SNPs across the hem locus (comprising the genes hemA, hemB, hemC, hemD, hemE, hemF, hemG, hemH, and hemY) can separate the known *C. acnes* strain diversity in 2 groups, herein called α -type and β -type. Importantly, this new typing correlates with the published porphyrin production levels, more precisely than the deoR gene. Moreover, after developing a method for the estimation of relative abundances of α -type and β -type *C. acnes* strains in shotgun metagenomic sequencing data, and applying it to a public data set associated to a study characterizing the skin microbiome in acne vulgaris (Barnard et al. (2016) Scientific Reports 6:39491), the present inventors demonstrated that α -type *C. acnes* strains were significantly more abundant in patients suffering from acne vulgaris than healthy subjects.

Therefore, in the methods for determining if a subject is at risk of developing acne, in particular acne vulgaris, of the invention, the subject is determined as being at risk of developing acne, in particular acne vulgaris, when (i) the α -type *C. acnes* bacteria are determined to be present in said sample and/or when the amount, in particular the relative abundance (for example with respect to the whole bacterial population, in particular skin bacterial population, of the subject, or with respect to the whole *C. acnes* population, in particular skin *C. acnes* population, of the subject), of α -type *C. acnes* bacteria is determined to be higher than a reference value or (ii) when the β -type *C. acnes* bacteria are determined to be present in said sample and/or when the amount, in particular the relative abundance (for example with respect to the whole bacterial population, in particular skin bacterial population, of the subject, or with respect to the whole *C. acnes* population, in particular skin *C. acnes* population, of the subject), of β -type *C. acnes* bacteria is determined to be lower than a reference value. Similarly, in the diagnosis method of the invention, the subject is diagnosed as suffering from acne, in particular acne vulgaris, when (i) the α -type *C. acnes* bacteria are determined to be present in said sample and/or when the amount, in particular the relative abundance (for example with respect to the whole bacterial population, in particular skin bacterial population, of the subject, or with respect to the whole *C. acnes* population, in particular skin *C. acnes* population, of the subject), of α -type *C. acnes* bacteria is determined to be higher than a reference value or (ii) when the β -type *C. acnes* bacteria are determined to be present in said sample and/or when the amount, in particular the relative abundance (for example with respect to the whole bacterial population, in particular skin bacterial population, of the subject, or with respect to the whole *C. acnes* population, in particular skin *C. acnes* population, of the subject), of β -type *C. acnes* bacteria is determined to be lower than a reference value.

In a particular embodiment, said reference value is the amount, in particular the relative abundance (for example with respect to the whole bacterial population, in particular skin bacterial population, of the subject, or with respect to the whole *C. acnes* population, in particular skin *C. acnes* population, of the subject), of α -type *C. acnes* bacteria, respectively β -type *C. acnes* bacteria, in a corresponding sample, in particular skin sample, of a healthy subject.

By "healthy subject" is meant herein a subject who does not suffer from acne, in particular acne vulgaris, and preferably who does not suffer from a *C. acnes*-associated disease, in particular skin disease and/or from an inflammatory disease and/or from a skin disease and more preferably who further does not present any sign of skin discomfort such as signs associated with sensitive skin, sensitized skin, fragile skin or weakened skin, or any uncomfortable and unaesthetic manifestations of sensitized, fragile and/or weakened skin.

By "sensitive skin" is meant herein a skin which, by nature, does not tolerate aggressive agents well, especially environmental agents such as pollutants, climate factors (wind, cold, heat), emotional factors, especially stress and/or chemical agents (heavy metals, detergents, compounds contained in cosmetic compositions such as fragrances, preservatives, alcohols, pH, AHA or dermatological treatments, such as vitamin A acid) and/or aggressive conditions, including perspiration and mechanical aggression such as waxing, shaving, rubbing and even water, especially hard water. Sensitive skin is not pathological skin, unlike allergic skin. Nevertheless, it may react to aggressive agents and/or conditions by unaesthetic and/or uncomfortable cutaneous

manifestations such as stinging, feeling of heat or warmth, tension, tingling, tightness and redness. Thus the "sensitive skin" character may be estimated by the subject themselves with subjective cutaneous sensations or by the dermatologist with objective cutaneous reactions.

By "sensitized skin" is meant herein a skin and/or mucosa momentarily made sensitive, as defined above, therefore nonpathological as such.

By "fragile" and/or "weakened skin" (i.e. skin made momentarily fragile), is meant herein a skin whose barrier function is weakened. This may be linked to the status of the individual; elderly people and infants have more fragile skin, for example. This state may result from chemical or physical aggression (abrasion, rubbing for example).

The uncomfortable and unaesthetic manifestations of sensitized, fragile and/or weakened skin are the same as for sensitive skin, without these manifestations and/or skin conditions being considered to involve the prevention and/or treatment of a pathology.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skills in the art to which this invention belongs.

It must be noted that as used herein, the singular forms "a", "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a cell" includes a plurality of such cells (e.g., a population of such cells). Similarly, reference to "a nucleic acid" includes one or more of such nucleic acids.

Although the invention has been described in conjunction with specific embodiments thereof, many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations to fall within the spirit and broad scope of the appended claims.

All publications, patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if such individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present invention. To the extent that section headings are used, they should not be construed as necessarily limiting.

The following examples are given for the purpose of illustrating various embodiments of the invention and are not meant to limit the present invention in any fashion.

BRIEF DESCRIPTION OF THE SEQUENCES

SEQ ID NO:	Type	Organism	Description
1	Genomic DNA	<i>C. acnes</i>	hem locus sequence reference for α -type <i>C. acnes</i> bacteria
2	Genomic DNA	<i>C. acnes</i>	hem locus sequence reference for β -type <i>C. acnes</i> bacteria
3	Genomic DNA	<i>C. acnes</i>	hemA gene sequence reference
4	Genomic DNA	<i>C. acnes</i>	hemB gene sequence reference
5	Genomic DNA	<i>C. acnes</i>	hemC gene sequence reference
6	Genomic DNA	<i>C. acnes</i>	hemD gene sequence reference
7	Genomic DNA	<i>C. acnes</i>	hemE gene sequence reference
8	Genomic DNA	<i>C. acnes</i>	hemH gene sequence reference
9	Genomic DNA	<i>C. acnes</i>	hemL gene sequence reference
10	Genomic DNA	<i>C. acnes</i>	hemY gene sequence reference

-continued

BRIEF DESCRIPTION OF THE SEQUENCES			
SEQ ID NO:	Type	Organism	Description
11	Artificial sequence		Mad4 nuclease
12	Artificial sequence		non-targeting gRNA
13	Artificial sequence		IA1 gRNA
14	Artificial sequence		HemB_1 gRNA
15	Artificial sequence		HemL_2 gRNA
16	Artificial sequence		HemY_2 gRNA

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1: Phylogenetic tree of the hem locus with annotated SLST clusters and the a/R3 typing division herein disclosed.

FIG. 2: Published porphyrin production levels along with their assigned a/R3 type and the deoR repressor presence (deoR+) or absence (deoR-). FIG. 2A: Adapted from FIG. 1 of Barnard et al. (2020) mSphere 5:e00793-19; FIG. 2B: Adapted from FIG. 1 of Johnson et al. (2016) mSphere 1(1):e00023-15; FIG. 2C: Adapted from FIG. S4 of Johnson et al. (2016) mSphere 1(1):e00023-15.

FIG. 3: Extract of the alignment of the glpR/deoR reference DNA sequence (SEQ ID NO:17 encoding SEQ ID NO:18) with the consensus glpR/deoR DNA sequence for most strains of the SLST D cluster (SEQ ID NO:19 encoding SEQ ID NO:20). Insertion of poly-G in the SLST D variant DNA sequence leads to a frameshift and the generation of a premature stop codon at position 105 whereas GlpR/DeoR reference protein is 310 AA long.

FIG. 4: Difference between ratio of *C. acnes* population classified as the α -type in healthy skin metagenomes and the skin metagenomes of patients with acne vulgaris

FIG. 5: The hem locus that is conserved across *C. acnes* strains and which comprises 8 hem genes: hemA, hemB, hemC, hemD, hemE, hemH, hemL and hemY. A representative from each SLST cluster is shown.

FIG. 6: Specific nuclease-mediated killing of alpha-type *C. acnes* strains. A) Specific nuclease-mediated killing of *C. acnes* phylotypes and alpha vs beta strains. Each diluted population of four *C. acnes* strains (one alpha-type strain: SLST A5; three beta-type strains: SLST D1, D2, K9) was transduced with three different packaged phagemids, each packaged with a DNA payload carrying a chloramphenicol cassette, a nuclease (SEQ ID NO: 11) expression cassette and either a non-targeting gRNA (SEQ ID NO: 12, black bars), or a gRNA that targets the strains A5, D1, D2 (SEQ ID NO: 13, grey bars), or a HemB gRNA that targets only the strain SLST A5 (SEQ ID NO: 14, white bars). Y axis: Colony-forming units (cfu) per μ L of packaged phagemid on BHI+3 μ g/mL chloramphenicol plates. B) Specific nuclease-mediated killing of *C. acnes* alpha strains vs beta strains. Each diluted population of four *C. acnes* strains (three alpha-type strains: 2 $\times$ SLST A1, SLST A5; one beta-type strain: SLST K9) was transduced with six different packaged phagemids, each packaged with a DNA payload carrying a chloramphenicol cassette, a nuclease (SEQ ID NO: 11) expression cassette as well as a non-targeting gRNA (SEQ ID NO: 12) or a set of Hem gRNAs that target the three alpha-type strains at different positions of the hem locus:

HemB_1 gRNA (SEQ ID NO: 14) targets HemB alpha nucleotide variant A155; HemL_2 gRNA (SEQ ID No.: 15) targets HemL alpha nucleotide variants C389 and T392; HemY_2 gRNA (SEQ ID No: 16) targets HemY alpha nucleotide variants T1199 and T1205. Y axis: Colony-forming units per μ L of packaged phagemids on BHI+3 μ g/mL chloramphenicol plates.

EXAMPLES

Example 1

Heme molecule, a porphyrin ring complexed with iron, is an enzymatic cofactor and a source of iron essential for survival of bacterial pathogens (Choby et al. (2016) J. Mol. Biol. 428:3408-3428). *Cutibacterium acnes* (hereinafter *C. acnes*) is the most abundant bacteria inhabiting human skin (Fitz-Gibbon et al. (2013) J. Invest. Dermatol. 133:2152-2160, Oh et al (2014) Nature 515:59-64). It is also known to be involved in acne vulgaris, an inflammatory skin disease affecting the vast majority of adolescents and young adults (Bhate et al. (2013) British Journal of Dermatology 168: 474-485). *C. acnes* has been shown to produce porphyrins that trigger inflammation (Schaller et al. (2005) British J Dermatol. 153:66-71, Spittaels et al. (2021) iScience 24:102575). Interestingly, *C. acnes* strains from different lineages vary in their porphyrin production levels (Johnson et al. (2016) mSphere 1:e00023-15, Barnard et al. (2020) mSphere 5:e00793-19), and the presence of a putative repressor gene from the deoR family has previously been hypothesized to be responsible for low levels of porphyrin production (Johnson et al. (2016) mSphere 1:e00023-15). However, such explanation is not complete, as high levels of porphyrin production were observed also for some strains which encode for deoR in their genomes (Barnard et al. (2020) mSphere 5:e00793-19). Moreover, while the lineage of the strains is usually indicative of their low/high porphyrin production, exceptions have been described in the literature (Johnson et al. (2016) mSphere 1:e00023-15).

Key enzymes involved in the porphyrin biosynthesis pathway in *C. acnes* are encoded by 8 genes (hemA, hemB, hemC, hemD, hemE, hemH, hemL, hemY). Those genes are clustered in a locus (hereafter referred to as the hem locus) that is conserved across all *C. acnes* strains.

The inventors found that a set of 218 SNPs across the hem locus genes can separate the known *C. acnes* strain diversity in two groups, α -type and β -type. While the inventors can unambiguously assign types from classical *C. acnes* typing schemes (SLST, recA) to either the α - or β -type, α and β -types are not monophyletic based on the core genome phylogeny (i.e. overall similarity of the strain genomes does not predict whether the strain will be a or β). Importantly, a/p typing strictly correlates with presence/absence with the published porphyrin production levels, with α -type strains being high producers and β -type strains low producers.

Furthermore, the inventors have developed a method for the estimation of relative abundances of α and β -type *C. acnes* strains in shotgun metagenomic sequencing data, and applied it to a public dataset associated to a study characterizing the skin microbiome in acne vulgaris (Barnard et al. (2016) Scientific Reports 6:39491). They found α -type strains to form a significantly larger fraction of the *C. acnes* population in acne vulgaris patients than in healthy volunteers. Therefore a treatment modality to specifically kill α -type *C. acnes* strains, inhibit specifically the growth of

α -type *C. acnes* strains or inactivate porphyrin production in *C. acnes* α -type strains has the potential to prevent and/or cure acne vulgaris.

SLST Cluster Definition

The genomic assembly of 475 *C. acnes* strains were either sequenced by Eligo Bioscience or downloaded from public repositories. For each assembly, the single-locus sequence type (SLST) was predicted using in silico PCR amplification and sequence alignment with the primers and reference sequences described by Scholz et al. (Scholz et al. (2014) PloS ONE 9:e104199). A pangenomic analysis of the whole collection was also performed using roary (Page et al. (2015) Bioinformatics 31:3691-3693) and panX (Ding et al. (2018) Nucleic Acids Res. 46:e5), and a phylogenetic tree based on the core genome was built using RAxML (Stamatakis (2014) Bioinformatics 30:1312-1313). Based on the branch lengths of the core genome phylogenetic tree, the inventors defined a coarser typing scheme, SLST cluster, consisting in the grouping of SLST types starting with the same letter (e.g. SLST cluster A includes SLST A1, A2, etc.).

α and β -type definition

The hem locus genes were extracted from all available assemblies and a phylogenetic tree was built based on their concatenated amino acid sequences using IQ-TREE (Minh et al. (2020) Mol. Biol. Evol. 37:1530-1534) with edge-equal partition model (Chernomor et al. (2016) Syst. Biol. 65:997-1008) and revised Dayhoff matrix substitution model (Kosiol et al. (2005) Mol Biol Evol 22:193-199). In the obtained tree (FIG. 1) the inventors observed two distinct groups that do not match the core genome phylogeny—one comprising SLST clusters A, B, C, E, F, G, and H (named the α -type) and the other comprising SLST clusters D, K and L (named the β -type). Importantly, no SLST cluster was split between the types.

In order to obtain a set of marker SNPs for those types, the inventors computed a multiple sequence alignment on the nucleotide level for each gene with MUSCLE (Edgar (2004) Nucleic Acids Res 32:1792-1797) and identified positions that discriminate between the α - and the β -type, while being conserved in more than 90% of the genomes from each type. This procedure identified 218 marker positions (see Table 1 above). Based on this result, the inventors defined the α -type (and the β -type) as strains that in at least 90% of said positions have the α -characteristic (or the β -characteristic) nucleotide variants.

Analysis of Published Porphyrin Levels

Available scientific literature was searched for reported levels of porphyrin production by *C. acnes* strains. Data from two articles (Johnson et al. (2016) mSphere 1:e00023-15, Barnard et al. (2020) mSphere 5:e00793-19) were aggregated, and the strains used were typed using the available genome sequences and the SLST scheme. In cases when genomic sequences were not available, SLST clusters were inferred from the lineages reported by the authors.

Based on the SLST clusters, the strains were assigned to either α - or β -type. The inventors observed that the strains classified as high or low porphyrin producers by the authors exactly correspond to α - and β -type strains (FIG. 2), respectively, offering a more reliable explanation of the production levels than the presence of the deoR repressor. Moreover, by multiple sequence alignment of the deoR genes the inventors found that the SLST cluster D strains bear a poly-G insertion in this gene, which results in a shifted reading frame and a nonsense mutation in the majority of the cases, HL025PA1 included (FIG. 3). It is therefore expected that deoR in SLST cluster D is not functional and cannot explain the low porphyrin production measured. Therefore, the inventors

considered the SNPs in Table 1 above to be biomarkers for high porphyrin production (α -type). Inversely, β -type is associated with low porphyrin production.

Metagenomic Analysis

Metagenomic shotgun sequencing data from Barnard et al. (2016) Scientific Reports 6:39491, including follicular plugs samples from 42 acne vulgaris patients and 40 healthy volunteers, were obtained from the NCBI dbGaP resource (accession number phs001655). Raw reads were pre-processed using fastp (Chen et al. (2018) Bioinformatics 34:i884-i890) and aligned against the human reference genome and the PhiX genome using bowtie2 (Langmead et al. (2012) Nature Methods 9:357-359). The reads aligning to either the host or PhiX were removed, while those passing QC were mapped to the reference hem locus using bwa mem (Li (2013) arXiv:1303.3997) and only properly paired mappings were kept for downstream analysis.

Since the locus is always single-copy in the available *C. acnes* genomes and does not have close orthologues in other species (the closest one, in *Cutibacterium modestum*, has 86% identity at the nucleotide level), the inventors did not include other reference sequences in the alignment database. However, in order to validate their approach, they also used Kraken2 (Wood et al. (2019) Genome Biology 20:257) to remove reads assigned to other species than *C. acnes* from the dataset, observing no difference in our results.

For each of the 218 discriminative positions, the fraction of α -type and β -type variants was computed from the alignment pileup. The overall relative abundance of each type within the *C. acnes* population was calculated using the median frequency of its signature SNPs across the 218 positions. Type α was found to be significantly more abundant in the *acnes* vulgaris patients than in the healthy controls (p-value 0.0112; Mann-Whitney U test) (FIG. 4).

Example 2

Specific Nuclease-Mediated Killing of *C. acnes* Phylotypes and Alpha Vs Beta Strains

This example demonstrates the nuclease-mediated specific killing of *Cutibacterium acnes* phylotypes as well as alpha strains vs. beta strains. The nuclease can be programmed to target and kill specific phylotypes (in this example, phylotypes IA1 and II) that encompass both alpha and beta strains according to the definition in the section “ α -type *C. acnes* bacteria and β -type *C. acnes* bacteria” above (FIG. 6A). Similarly, the nuclease can be programmed to target and kill only alpha strains while leaving beta strains intact (FIG. 6B).

Four *C. acnes* strains (one alpha-type strain: SLST A5; three beta-type strains: SLST D1, D2, K9) were grown anaerobically overnight at 37° C. and diluted the following day to an OD600 of 0.05. Strains SLST A5, SLST D1 and SLST D2 belong to phylotype IA1. SLST K9 belongs to phylotype II. Each diluted *C. acnes* population was transduced with three different packaged phagemids, each packaged with a DNA payload carrying a chloramphenicol cassette, a Mad4 nuclease (SEQ ID NO: 11) expression cassette and either a non-targeting gRNA (SEQ ID NO: 12), or a gRNA that targets phylotypes IA1, i.e. the strains A5, D1, D2 (SEQ ID NO: 13), or a HemB gRNA (HemB_1 gRNA) that targets HemB alpha nucleotide variant A155 and thus only alpha-type *C. acnes* strains such as the strain SLST A5 (SEQ ID NO: 14) Transduced cells were serially diluted and plated on BHI agar with 3 μ g ml⁻¹ chloramphenicol post transduction and incubated in an anaerobic chamber at 37° C.

63

As shown in FIG. 6A, packaged phagemids programmed with a HemB_1 gRNA that targets HemB alpha nucleotide variant A155 selectively reduced the number of the alpha-type strain (SLST A5) by more than 4 logs while not affecting the growth of any of the beta-type strains (SLST D1, D2 and K9). Packaged phagemids programmed with a IA1 gRNA selectively reduced the number of colonies from the strains SLST A5, D1, D2 by more than 104 fold while not affecting the number of colonies from the strain SLST K9, as anticipated.

Specific Nuclease-Mediated Killing of *C. acnes* Alpha Strains Vs Beta Strains

Four *C. acnes* strains (three alpha-type strains: 2x SLST A1, SLST A5; one beta-type strain: SLST K9) were grown anaerobically overnight at 37° C. and diluted the following day to an OD600 of 0.05. Each diluted *C. acnes* population was transduced with six different packaged phagemids, each

64

packaged with a DNA payload carrying a chloramphenicol cassette, a Mad4 nuclease (SEQ ID NO: 11) expression cassette as well as a non-targeting gRNA (SEQ ID NO: 12) or a set of Hem gRNAs that target the three alpha-type strains at different positions of the hem locus: HemB_1 gRNA (SEQ ID NO: 14) targets HemB alpha nucleotide variant A155; HemL_2 gRNA (SEQ ID No.: 15) targets HemL alpha nucleotide variants C389 and T392; HemY_2 gRNA (SEQ ID No: 16) targets HemY alpha nucleotide variants T1199 and T1205. Transduced cells were serially diluted and plated on BHI agar with 3 µg ml⁻¹ chloramphenicol post transduction. Plates were incubated in an anaerobic chamber at 37° C.

As shown in FIG. 6B, packaged phagemids programmed with a Hem gRNA that targets an alpha nucleotide variant, selectively reduced the number of the three alpha-type strains by more than 99.9% while not affecting the number of colonies from the beta-type strain.

SEQUENCE LISTING

```

Sequence total quantity: 20
SEQ ID NO: 1          moltype = DNA length = 10225
FEATURE               Location/Qualifiers
source                1..10225
                     mol_type = genomic DNA
                     note = Alpha type locus reference
                     organism = Cutibacterium acnes

SEQUENCE: 1
tcaggcaactg atcacctggg aagccgctg ggcagccttc ggcagggcgg cgcggaaaac 60
ctcgaaagct tcgtcgctcg gagcgctcga gaggaaccag gcctcaaagc acgatggcgg 120
caggttgacc ccggcatcta gcatcgctg gaagaacgcg gtgtaagcct cggcattctg 180
agactttggc ttgtcgtaat tgccgacgg agagtcgacg ccgaggaata cgtgaagag 240
atttcggcg ttctgcaggc ggtacgtgac gccggctgcg tcaagcgcg atacgagctc 300
actgcgccat ttctcagagc gcgcatcgag gcgctggtac accccggcgt cggccagctg 360
cagggttttg acaccgcgca cgggtggccag cgggttacgg gacaaaagtgc cggcctggta 420
gacggggcgg gtaggagcca gcaaatccat gacctcggcg cggccaccta gggcagccag 480
cgccatacca ccgccgatga ccttgccgaa cgtgaaaata tcgggagcga agtcctcgcc 540
ggcctgctgt tgcaaacccc agaattccga cgggctgcac cggaaatccgg tgagtacttc 600
gtcgggtgatc atgagggcac cgtgctcggt tgtgaggcga cgaatcgcg cattgaagcc 660
gggttcgggg ggaacaactc ccatattggc cgggcaggac tccacgatca ctgcagcaat 720
ctggtcgcca cgtcgggaga acatcgctgc gaggcgggcg acgtcgttgt aatccaccac 780
aatagtgctg gcgctcgagg cggctggcac accggccgaa ccgggcaggc cagccgtcgc 840
cacgcgggaa ccgcgccggc cagcaggac gtcagaatgg ccgtggtagt taccggcaaa 900
cttgacgatg acgtcacgcc cagtgggtgc accgggctaaa cggaccgcag tcattgtgcg 960
ctcagtgctt gttgagagca aacgcacctt ctgcaccgga gccatccgct cgatgatggc 1020
atccgctagt tcggtttcgg tgccgggtcgg ggcgcccagg gacagccctt tcgacgcggc 1080
cacctgaacc gcagagacga cctcggggag ggcgtgcccc agcaaagccg gacccaggtt 1140
gcacaccaga tcgacgtagc gcgtgccctc ggctgtccag atccacggcg cttgggctctt 1200
ctcaatgaag cgaaggaaac ccgccaccga gccgtaaagg ccgaccgggg aatcaacccc 1260
gccggggaat acggcctggg cggcgagaaa cagttcgggc ttggaggtca tttcttctt 1320
ccggtgtggg ggtttgtggt gagcatgatc acgacgggtc agaggtcctc gcgaaacatc 1380
tgccgcgcct cgattgccca gtaggccaag acaagatcgg caccggcacg cagcatcgac 1440
gtcagggact ctccgatgca acgtcggcga tcgatccaac cgtttgcggc agcggcctcg 1500
agcatggcgt attctccaga gacctggtaa gcggcaaccg gtacctgcga caccctcgcg 1560
acatcggaac acacgtcgag gtatggggag gcgggcttca ccatgaccag gtcggctccc 1620
tcggcgaggt cgagaagggg ttccgcgagg ccttcccggc ggttcggtgg atcctgctgg 1680
taagtcctgc gatcacccct aagggaaacag ttgacggcgt ccggaaggcg gccaaagaag 1740
gctgaagcgt acttagctga ataggccatg atggacacat ccgtgaatcc ctccgcatcc 1800
agagcatcgc gaatgacagc gatctgacgc tccatcatcc ccgacgggga gaccatgtgg 1860
gctccggcgc gcgcttggga gaccgccatc gcggcatata gcggcacagt ggagtcatta 1920
tcgacgcgac catcaggggt cagagctccg cagtgctcat gatcgggtga ctcgctaga 1980
caggtgtcgg cacagatcac cagatcgctc ccgacggcct cgcgcaccgc ggagatgccg 2040
gcgttgagga tgcttctcag gtcccaggcc tgagaaccga tctcgtctt gtgctccggc 2100
ataccaaaac gttcaataac accaataacc gcctcggcag cttgggtggc gagctcctt 2160
atcgactcgg tggagtgtcg gaattggcgg ggcaggctgg tgatctcgac gggggaatcg 2220
atccctctct tgacgaagc ccgacgcat aactggcgcg gggcgacgcg ggtttcggcg 2280
accatccggc gcacgcgagg agtgggtcgt agtcgacgcg gacgagaaac aggcgcgacg 2340
ccctcgggca atggcttgac cccagccatg cggggatcac tggcatcagg caggtacggg 2400
ttcatgatca acttctctcg cgaacggcgg tcggtcaggc gaccaaaggc tcaaccgcct 2460
caaggacggt ttcaggagtg ggggaggagg ccaccgatg cgtgggtgta ccgagtactt 2520
cgagtgttct tgcagtgggc ttaccgaggg cgacgagcct ctgacgacgc catcccagcg 2580
gctctaccag ctcgacgagg gcacgtccca ccgaggagcg agtgacgacg acgaccccg 2640
cagacggcca cgactggcg atgacctcag gcagtttctc gacggcagcg gtggtataga 2700
tctccaatcg gtttaccgtg cagccgacct tcgtcagccc cagcggtacg gtatcgcgcg 2760
ttaactggga cgtggttagg aggatactgg cgactccacc tggccactcg gcgaccagac 2820

```

-continued

cagcgccgct	gtgattctcg	gggacgaagt	cgacgtcaat	cccacgtgtg	cgcagcgcag	2880
cagaggtagc	ggcgccgacc	gcagctactt	tgatgttggg	attcggcaac	ggggtagggg	2940
gcagtcggac	ggtggtcttc	gaggtgacga	ccagccagtc	ggcctgccac	aggtcagggg	3000
cagtaagcct	ctccgcatcc	ggcaagggtga	cgggtcgcgct	cagcgctgtg	tggatcactc	3060
cataaaccggc	ctgcttcaaa	gtcgtcacga	atthgtcgtg	ggagtcgtca	cggggaaacaa	3120
caactggggg	cttgacgaga	tccgacggat	ttgtggtcat	cgagtgcttc	ctgaagcttt	3180
gagatctgca	acgtcgccgg	caccatcgte	cagtaactct	tccgcagccg	tcacccccag	3240
cgtatgcgcc	ggcgccgacc	ctgggtcgac	ctccgcggtt	aattcgacgc	tgacctgttc	3300
ccgaacgggt	cgggaaccgt	cgacagcagc	cactacaaca	tccagggtta	acactttag	3360
ggattttccc	gaaacaggct	tgagttgagc	taatgctccg	atgggggcag	cgcaccccg	3420
ctctagatgg	gagagcaccg	cacgttcggc	ggtggcgccc	agcctggttc	tgcggtcgte	3480
aatccgtgcc	agcgtcttgg	cgagcttgcc	atgacgtgaa	tcagcgggtc	ggcattcgac	3540
ggctagcgct	ccctgagcag	gggcaggcag	tactaccgac	gggtcgagga	actcagtgac	3600
ggcgccgcta	gagctagtc	gagctagtc	ggaggtcgcg	agcacgcagg	cgtcgaggtc	3660
ctccttgcca	cgttttttgg	cgtagcggcc	gagtcctttg	acgcgaccca	ggcgggtgtc	3720
gacattttcca	cggatatega	cgatcgtcag	gtccggtcta	accgcccaaca	actgtgcggc	3780
ccgacgtggc	gagccgtacc	cgacgttagc	ttcttgcggg	aggctcgtaa	gggtgagccc	3840
atcgccggcc	accaaagcgt	cacgcgggtc	cgtcgagct	ggacccggac	cgatcgccag	3900
tcctaaccga	cgtcctgtcg	gtagatcttt	gaaggagtgg	acggcaatgt	cggcttctcc	3960
ctcgaggagg	ggcgacggga	tcggcgaggc	gaacaccccg	attcctccca	tgcccgctaa	4020
cgaggccgct	gaggtgtccc	cgtgagtagt	gatcgctcgc	agattgacgt	ccagccctc	4080
cgactgagc	agcccccga	ccatctctga	ttgagtagtg	gccagagtgc	tgccacgggt	4140
tcccagacgg	atcgacagaa	tattgcctat	tgtggcactc	agtaccgacg	ccgattccag	4200
ctcgcaatca	gacaagttag	cggacagttc	ggcgggcccg	cggcaccaga	ggcgcaatgc	4260
ccgaacccga	tgcccccgac	cccactaagc	cgatccgtcc	ctgtccagac	tggttcaata	4320
ccctagctcg	ggcggtgcaga	ccatcgccgc	atgctcgggc	ctgtggagtg	gtccgtgcga	4380
gggaacctgt	ccagtgaaatg	atctgggagt	ctacgacctg	ctccgcggag	aggctccacg	4440
cgagtaggac	ggaagcgtcc	gtagagcct	gatcgacgt	caaagcgaca	ttcgcttcac	4500
cagatcgacc	ataggaaacc	cgtagcaagt	ggagaccttc	gggacacgat	tctctcaagc	4560
tgggcccactt	cgcgcagatc	tgagtaagt	ccttggcagt	aacggtgggtc	gttccggggg	4620
ccaccagcat	ggcggttccc	cgtgggctcg	agtcacgccc	gggtgctcgc	aaggcgagat	4680
tgacgtggcc	gacggcgccg	cccggtgcca	actcggggcc	aacgggtgct	atgacattgc	4740
tgagcagttt	tgtggcaggt	ccagccgaac	aggccacgac	gacccgaggg	gtcagggaag	4800
aacgaatctc	ggcgctgggt	acggggggga	aagatggatc	cggattgccc	gaggtaccag	4860
cgggtctcac	tacccactga	tctccgtggc	gggttagccc	gcgagctcct	actcgggtca	4920
tgggtcgctcg	ggcgccctgt	tctgcctgac	gacgcagctc	ggccggcatg	atgaacatgc	4980
caaccggtac	cgacgcgact	gcaggccccc	ctggcatccg	tgagcggcac	agcgcactc	5040
ccgctcgctag	cgatccgggt	gcccgtgggt	cggccctcag	accgggggag	acgacgtcga	5100
ctgacagtag	atgtgggtcg	gcggcggtga	ttcccccgcc	gatcgggggt	acgagacgcc	5160
gcaatacctc	tgaccccata	cgggaggcga	cgagcgtacc	caggctcgca	gggagatcct	5220
caggcatggg	tgcgcgatca	agtttcggcg	cgcgggcagc	ctcttccgct	cccaggggca	5280
tgatgacgac	tggtgctcag	aagtttcggc	ggatgccaag	aatggcatta	tccgggcatc	5340
cgaagcaggc	gtcctcgagg	tacacccaag	acggggcgat	ggtcggcttc	tcaaccgtca	5400
atcccaactc	gggtggcaggt	tcagcgggtg	ctggtagacg	gaccgcccac	ccctcggccc	5460
cgatgtcgta	gctgattccc	ggcagcgtcg	acctggcaat	cagcccgccg	gtgtacccgc	5520
gggattcgac	aagtaccggg	tcgacacctt	cctgggtgaa	ctgcccaggc	ggccagaggc	5580
cagccatgcc	cccgcacatg	acgacggcgt	cgacgtgctg	cggatcagaa	atgttcatcg	5640
tcgctactcg	gctcacaggg	agtgacccag	ctccaccagc	cgtgtcaaga	cgttgggggt	5700
ggtgttgagg	ggcacaccgt	gaccagattt	gacgacgtgt	ccaggggcct	gacgtcctgc	5760
cgcgacgata	tcgcccgttg	gggactccag	cgggtcccat	ccggcaaaaga	gcaaggcggg	5820
atcgatgttg	ccctgaaccg	gcacttccat	cccgagagcc	ataagagtgg	cgcaggccct	5880
atcaaggctga	gtccggtggt	cgacaccaag	tactgggtac	cgtgacgcag	tagctgccac	5940
ctgcgcaaac	tcagatagca	gatgggtggc	attgatgcca	aagtgaacca	ccgggatggt	6000
cggatcgacg	gtagccagag	cccgcgtgga	gaaagggtcg	acgctggctc	tataatcgct	6060
tgcgtcagc	gacccagccc	aggaatcaaa	gagctggaag	gcacgcgcgc	cgcgcgcag	6120
ctgagcagcc	aggaaggcag	catccaagtc	ggcaaccac	tccatcagcc	ttgccagggt	6180
ctgtgggtct	gagtgcatca	tcgctcgggc	agacagatgg	tcgcgggaag	ggccgcccct	6240
gacgaggtag	gcggcgatag	taaacggggc	acccgcgaaa	ccgatgagcg	gtgtggcgct	6300
gcctaaactcg	gtgatcgcca	ggcggtcgagc	ctcctcgata	accgagggggt	cctcgacatg	6360
acgtcgagta	atgcgggtcaa	cgtcggaggc	agtgccggagc	gggtcgtcga	gaacggggcc	6420
gactcctggc	tcaatgcgca	cctctgcccc	gcacagagct	agcggcacc	tgatgtcgga	6480
gtagagaaag	gcccccgctga	cgtgggtggc	gcgcacaggt	tgccaggtaa	tctcggcgcc	6540
gagctcaggg	ttgaggcaag	actcgagcat	cgtgggtccc	tctcgggctt	cgcggtaact	6600
cggtagcgaa	cggcctgcct	ggcgcataaa	ccacacccgg	gtgacggtag	ggcggtgccc	6660
agtcattggct	tgacagaagc	gtgaatctgc	catgaaggta	gtgggtctcag	tcatcaccac	6720
gattgtgccc	taccacgtgc	agatgcggcc	cacaggacgg	ttattggggc	agttatgagc	6780
ttagtatgta	ctgggggttcg	tcgttacagc	atgggtgaat	aaggcttcgt	gggattgtgt	6840
gtcttgaccg	tggtatcatg	cgagcagggg	cttggcgctg	tttccagggc	ggccgcgcag	6900
gttgacggac	ttggcctggc	attgaccgat	catccccaga	tcggggggcg	ccttgtgtcg	6960
tcaacgtgca	accgggtgtg	cctgatcggt	gaaacctccc	ctgaagccgt	cgcgcagggg	7020
ttcgacaggg	ccgcgttacg	caggtgcatt	gccgatcacg	gagcaaacgt	gttagctgaa	7080
tctgcccagc	tcgtctgtga	gaacgacgcc	gtttggcggt	tggtccggtg	ggctgcgggc	7140
atggagtcga	tggttttttg	ggaacgtgag	gtcgccggac	agatgaagcg	tgctctcagt	7200
gaggcgccgc	gtgaacagac	gtctcctac	accattggtc	acgtcgttga	ggaagctctc	7260
aagacgtcac	gtcatgtcgc	taccgagact	gcgttggtcg	ctgaggggag	caccgttgtc	7320
gcggtcgggc	ttgacctcgt	tgcccagcgg	atggacttgg	atggcgctcg	ggtgctcgtt	7380
atggggagctg	gctcctatgc	cgggtcgctg	tgccccagc	tgagttctcg	gggagtcgcc	7440
gagattcaag	tccattcagc	ttccggagct	gcgcgcggat	tcgctcgacg	ccaccgggtc	7500
agtgaggctc	tcgacatcga	cgcgcgcgtg	gcacaagctg	atcttgtcgt	cacctgocgt	7560

-continued

```

gggtcgggag ttcttgcgtt gttctgtgag gctgtctggc gcgcgggtcga cgtctgaagg 7620
ggcccgcatc tcatgggtgct cgatttagca atcagcgggg acgttgagga gctgtcccc 7680
gctggggtag aggtcatcga ccttgagacg atccggcagg ctgttcccgc ctctgcggaa 7740
gctgagcggg ccgcccgcga acacatcatc gctacgggag tccgtcactt tgccgtagac 7800
ctogaacgtc gtcggatggc cccggctgtc gtcgctctgc gcgacgtcat ttcagatctc 7860
gttactgtcg aactggagcg tctgcccgaa gaaggctcgg tccctgtcta tgaggtagcg 7920
gcgtctcttc gacgactcgc tgcactatg gcgcacatcc cgtcggctcg agccaggatg 7980
gcctctgagc agggcttggg cgatcgctgg cttaactcat tgtcggatgt attggggatt 8040
gacgtggata tcgcccgcac cgtcatcgac atgtccagct tcgccaatgc tgaactgcatg 8100
acgtgtccgg ttaccggcct gcgggtggag gaccttgcca cagatgcggc ccccgagggtg 8160
aggagaggac atcgtgacgg ctaatcccta cgtgtctccc acggaccgtg tggcccotta 8220
tagcgcctgt cttgtcgttt cttttggcgg tcgcggttcc cctgaagagg tcatgccttt 8280
tctgagaagg gttcccccac gtcgcatccc tgaagaacgc ttggtgatg tcgcccgaca 8340
ctatgaccgt tctcgggtg tccagccgat taacgatgcg acggacgttt tctcaacgc 8400
gattgggaat gaactgccc gtcacggagt acgagttcca gtgtgttag ggaaccgcaa 8460
cggcaccccc ttccttgagg aagcgtgac tgacatgac gcgcacgggg tgcgtcgggt 8520
gttagcgggt ttcacctctg cctacgccag ttactccggg tgcgtcagt atcgcgagga 8580
gatcgcgact gccctggctc acgccgggat caccgatatg caggtcgaca aggtacctcc 8640
ctttaacgag gcgcggggat tcattcgcgc taacgcggaa gctctgatg aggcgtttat 8700
gcggatcccg cctactcctc ttgaggctac ccgctcgtt tctgctact actccatccc 8760
tgactctatg caggacgcat ctggcgctgg gcagccgggg acggattaca tttctcagca 8820
caaggcgggt tgtgagaggg tcgctggcca ggtgcgcag gtcttcggga atatgccga 8880
gtgggacttg gccatttgc tccagtcggg gcgacccagc gaccctggg tgaaccgga 8940
cattatcgat cacctccgca gtcttcccga acaggcgctg caatctgttg tctgctctcc 9000
aattggcttc gttgctgacc acatggaggt cgttaatgac cttgattacg aggcgcgca 9060
ggcagctaa gaggtcgggc tgccttcac gcgggcggcg acccctggca cccatcccgc 9120
ctttatcgct gatctccgcg ggtcactctt gtcccaggcc gcggcggtc gtggtgaggg 9180
tggcaatcta acgtcgtggc cggtccttg cgtcgcgggt tgttgcgac gctaccgga 9240
cgctcaggac attccagcgg tctccgggtg tgatgtcgag tccgtagctg ccgggtgcga 9300
tgtatcgat gctgaaccag gagggtgtga tttgtgcca agcggctcag cgaagtgcagt 9360
cgatcgccg gggccagagg cgttcgaact ggagactcct ccttccccct acaaccggtt 9420
gactaaggag gcccccatgt ctgatcattc gagggtgat tccgtgattg agggaccacg 9480
cgacgacgaa gttcccgcg gtcctacac cgcaccgacc gatccccgcg ataccgccgt 9540
cattcccag gaggtaaac cctccagcaa gtggcgcatg tactcggtat tccgcgtcgc 9600
gaccgcgctt cctccgagg atgacgagc tgcgcgctt gttgagggt ccgacgagtg 9660
ggcgggacag tccgggtgctg acacccgagg atggtacgac ctgtcgggct tgcgcgctaa 9720
cgccgacctg ctggtgtggt gggttagcga cgatccggcg gtgctgcagg atgcctacca 9780
tcggttccgc gcatctgggc tggggcgcca cctggaacgg gtgtggtcga acgtaggagt 9840
tcacgtctct gcggagttca acaagtctca cctgccctca tgccttgcgg gcatcgcccc 9900
gaggcggttg tctgcgttct accccttcat tcgctccaag gagtggtaac tcttccggc 9960
ggctgatcgt gcccgcatgc tgcgcgaaca cggcatcgtc ggcccgccca gctccgacgt 10020
caaaagcgtc acgttggcgg cattcgcgct cggggactac gagtggattc tgcggttga 10080
gggtgacgac ctcgcgcgta ttgtcgatgt catgaaggac cttcgttacg ttgaagcccg 10140
tcgttatgtc gacgtggata ttcctctctt caccggcgag cgcgtttctc cgttggtgtg 10200
ggccgaccgt cagatgagag cctga 10224

```

```

SEQ ID NO: 2      moltype = DNA length = 10224
FEATURE          Location/Qualifiers
source           1..10224
                 mol_type = genomic DNA
                 note = Beta type locus reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 2
tcaggctctc atctggcggt cggcccacac caccggagaa acgcgctcgc cggtaaaaga 60
gggggtatcc acgtcgacat aacgacgggc ttcaacgtaa cgaaggctct tcatgacatc 120
aacaacacgg gcgaggtcgt caccctccaa ggcgagaatc cactcgtagt ccccgagcgc 180
gaatgccgcc aacgtcgacg ccttgacgtc ggagctggct gcgccaacga taccgtgttc 240
gcgcagcatg cgggaacgat cagcccgccg aaggaggtag cactccttgg agcgaatgaa 300
ggggtaaaac gcagcccaac gcctcggggc gatgcccggc aagcatgagg gcagggtgaga 360
cttgttgaac tccgcaggag gatgaactcc tacgttcgac cacaccgggt caagggtggcg 420
ccccagccca gatgcgggga aacgatggta ggcgtcctgc agcaccgccc gatcgtcgct 480
aaccaccac accagcaggt cggcgcttagc gcgcaggcct gacaagctgt accatccacg 540
agtgtcgaca ccggagtgct cggcccactc gtccggaaccc tcaacaaggc ggcgacgctc 600
gtcatcctcg gcagggaagc cggtcgcgac gcggaatacc gagtacatcg cccatttgc 660
ggaggcggtg acctcctcgg gaatgaccgg ggtgtcgcgg ggatcggtcg gtgcgggtga 720
ggagccggcg ggaacttctg cgtcgcgtgg tccctcaatc accgaatcag cactcgaatg 780
gtcagacatg ggggtctctc tagtcaacgg gttgtagggg gaaggaggag tctccagttc 840
gaccgctctc ggcccccggc gatcgactgc actcgtgag ccgctgggca caaaaaccgc 900
acctcctggt tcagcatcga ctacatcggc accggcagct acagactcga catcactacc 960
ggagacgggt ggaatgtcct cagcgtcccg gtagcgtcgg caacagccag cggcgccagg 1020
agccggccac gacgttagat tgcacccctc accacgagcc gcccgccctc gggacaagat 1080
gagcccgccg agatcagcga taaaggccgg atgggtgcca cgggtcgccc ccccgctgaa 1140
ggcgagcccg gacaccttgg ctgcctcggc ggcctcgtaa tcaaggctat taacgacctc 1200
catgtgtgca gcaacgaagc caattggagc gacgacaaca gactgcacgc cctgctcggg 1260
aagattgcgg aggtgatcga taatgtccgg ttcaagccac gggtcgttgg gacgccccga 1320
cctcgagcag taggccaagt cccactgcgg catattcccc aaaacctgac gcacctggcc 1380
ggcgaccttc tcacaaaacc ctttgtgtg ggaatgtag tccgttcccc gctgcccagc 1440
cccgatgcg tctgcgcatg actcagggat ggagtgaagt acgaaaacga cgcgggtagc 1500
ttcgagagga gtagggtgga tccgcataaa cgcctgcctc agagcttccc cgttagcgcg 1560

```

-continued

aatgaatccc	ggcgccctcg	tgaagggggg	tacettgtcg	acctgcatat	cggtgatacc	1620
gacgtgagcc	agagcagtcg	cgatctcctc	gcgatactga	cggcaccogg	agtaactggc	1680
gtaggcagag	gtgacaaccg	ctaaccaccg	acgtaccccg	tgcgcgtgca	tgtcagtcag	1740
cgttctctca	aggaaggggg	taccgttacg	gttccctaac	agcaactggaa	ctcgtactcc	1800
gtgacggcgc	agttcattcc	caatcgcggt	gacgaaaacg	tccgtcgcat	cgttaatcgg	1860
gctaaccaccg	ccgaaacggg	catagtgccg	ggcgacatca	gccaaagcgt	cttcagggat	1920
gcgaccgtgg	gagacccttc	tcagaaaagg	catgacctct	tcgggggaac	gcggaccgcc	1980
aaaagaaaacg	acaagcacgg	cgccataaag	ggccaacggg	tccgtggggg	cagcgtaggg	2040
gttagcggtc	acgatgtcct	ctcctcacct	cggggtgccc	catctgcggc	aaggctctcc	2100
acccgcaggc	cggtaaccgg	acacgtcatg	cagtcagcat	tggcgaagct	ggacatgtca	2160
atgacgggtg	ccgcgatgtc	cacgtcgatc	cccaatacat	ccgacaatga	gttaagccag	2220
cgatcgccca	agccctgctc	agaggccatc	ctggctcgag	ccgacgggat	gtgcgccata	2280
gacgcagcga	cccgtcgaag	ggacgcggct	acctcatcga	cagggaccga	gccttcttcg	2340
ggcagacgct	ccagttcagg	agttaacgaga	tctgaaatga	cgtcgcgtag	agcgacgaca	2400
gcccggggcca	tccgacgacg	ttcgagggtc	acggcaaaag	gacggactcc	cgtagcgatg	2460
atgtgttcgg	cggcgggccc	ctcagcttcc	gcagaggcgg	gaacagcctg	ccggatcgte	2520
tcaaggctga	tgcctctac	cccagcgggg	acaggctcct	caacgtcccc	actgattgct	2580
aaatcgagca	ccatgagatc	tcggccctct	cgagcatcga	ccgcgcgcgg	agcagcctca	2640
gcccgaacacg	caggaacccc	gcaccacggg	cagggtgacga	caagatcagc	ttgtgccagc	2700
acggcgctga	tgtcgagagc	ctcactgacc	cgggtggcgtc	gagcaaatcc	ggcgccacgt	2760
ccggaggctg	aatggacttg	aatctcgggc	accccccgag	aactcagctg	ggcgacgac	2820
gcaccggcat	aggagccagt	ccccataaac	agcaccggag	cgccatccaa	gcccattccg	2880
tgggcaacga	gatcaagccc	gaccgcgaca	acgggtgcgtc	cctcagcagc	caacgcagtc	2940
tcggtagcga	catgacgtga	cgtcttgaga	gcttctctcaa	cgacgtgacc	aatggtgtag	3000
gagacgggtc	gttcacggcg	cgctcactg	agagcacgct	tcatctgtcc	ggcgacctca	3060
cgttccccaa	aaacccatcga	ctccatgccc	gcagccaccc	ggaaacaaacg	ccaaacggcg	3120
tcggttctcac	agacgagctg	ggcagattcg	gctaaccagt	ttgtctccgtg	atcggaatg	3180
caactgcgta	acgcggccctc	gtcgaacct	tgcgcgacgg	cttcagggga	ggtttcaacg	3240
atcaggcaca	cccgggtgca	cgttgacagc	acaaggggcg	cccggatctg	gggatgatcg	3300
gtcaatgccca	ggccaagtcc	gtcaacctgc	gcggccgcct	cggaaacgac	gccaagcccc	3360
tgctcggcat	gaccacgggt	caagacacac	aatcccacga	agcgttattc	aacctgctg	3420
taacgacga	gcccacgaa	tacgaagctc	ataactcgcc	caataatcgt	cccatggcg	3480
gcatccgcac	gtggtggggc	acaatcgggg	tgtagactga	gaccactacc	ttcacggcag	3540
attcacgcgt	tctgcaagcc	atgactgggc	accgccttac	cgtcgccccg	gtgtggttta	3600
tgcgccaaacg	aggccgttcg	ctaccggagt	accgcgaagc	ccgagagggc	accacgatgc	3660
tcgagtcttg	cctcaaccct	gagctcgccg	ccgagattac	ctgcccacct	gtgcgccgcc	3720
accacgtcga	cggggccgtt	ctctactccg	acatcatggt	gccgtagcc	ctggcggggg	3780
cagagggtcg	catlgagcca	ggagtccggc	ccgttctcga	cgagcccgtc	cgcacgcct	3840
ccgacgttga	ccgcatfact	cgacgtcatg	tcgaggaccc	ctcggtatc	gaggaggctg	3900
caacgctggc	gatcacccag	ttaggcgacg	tcacaccgct	catcggttcc	gcgggtgccc	3960
cgtttactat	cgcgcctcac	ctcgtcgagg	gcggccctcc	ccgcgaccat	ctgtctgccc	4020
gggcatgat	gcatcagac	ccacagacct	gggcaaggct	catggagtgg	gttgccgact	4080
tggatgctgc	cttcctggct	gctcagctgc	gcggcggggc	gcgtgccatc	cagctctttg	4140
attctctggc	tgggtctcgt	ggcgcgagcg	attatcgagc	cagcgtcgca	ccgttctctc	4200
agcgggctct	ggctaccgtc	gatccgaaca	tcccggtggt	tactttggc	gtcaatgccca	4260
cccatctgct	gtctgagttt	gcgcagggtg	cagctactgc	gtcacggtag	ccagtacttg	4320
gtgtcgacca	ccggattagc	ctcgatgagg	cctgcgccac	tcttatggct	tcgggtagtg	4380
agatgccggt	tcagggcaac	atcgatcccc	ccttgctctt	cgccgtagtg	gaaccgctgg	4440
aaacgcgccac	ccgcgcatac	gtcacggcag	gacgtcaggc	ccctggacac	gtcgtcaatc	4500
tgggtcacgg	cgtgccctcc	aacaccgacc	ccaacgtctt	gacacggctg	gtggagctgg	4560
tgcactccct	gtgaaccgag	tgcgcacaat	gaacatttct	gatccgcagc	acgtcgacgc	4620
cgtcgtcatt	ggcgggggca	tggtcgccct	cgtggccgcc	tggcagttca	cccaggaggg	4680
tctgcaccgg	gtactttgtc	aatcccgcgg	gtacactggt	gggctgattg	ccaggtcgac	4740
gctggccgga	gtcagctacg	acatcggggc	cgagggggtg	gcggtccgtc	taccagacac	4800
cgctgaactg	ggcaaccagt	tggggttgac	gggtgagaag	ccgaccatcg	ccccgtcttg	4860
ggtgtacttc	gatgacgect	gcttcgcgat	gcccgataat	gccattcttg	gcatcccgcc	4920
gaatttctcg	gaccacgcgg	tcgtcatcgc	tctgggagcg	gaaaaggctg	ctcgcccgcc	4980
tgagctcgat	cgcgcgccca	tgccagagga	tcttctctcg	gacctgggta	cgtctgtggc	5040
ctcccggatg	gggtccagaag	tactgcgacg	tctcgtcacc	ccaatcgccg	ggggaaatcca	5100
cgccgcgat	ccacatctac	tgtcagccga	cgtcgtctcc	cccggcttga	gggctgcccg	5160
ccaggccacc	ggatcgctac	ccgcgggagt	tgcgctgtgc	cgctcacgga	tgcacagcgg	5220
gcccgcagtc	gcgtcggtag	ccggtggcat	gttcatcatg	ccggccgagc	tgcgtcgtea	5280
ggcagaacag	gcccggcgca	cgaccatgac	ccgagtagga	gctcgcgagc	taaccgcgca	5340
cgaagatcag	tggctcgtgg	agaccgctgg	aaacctctcg	aatccggatc	catctttgcc	5400
ccccgtcacc	aacggcgaga	gtcgtttctt	cctgacccct	cggtcgctcg	tagcctgttc	5460
ggctggacct	gccacaaaac	tgtcagcaaa	tgtcatcgag	accgttggac	ccgagatggc	5520
gccaggcgcg	ccgatcgggc	acgtcaatct	cgccttgacg	gcaccgcagc	tagactccag	5580
ccacgcggga	aacggcatgc	tgggtgccc	cggaaacgac	accgttactg	ccaaggcact	5640
taccacacatg	tcggcgaaag	ggcccagctt	gagagaatca	tgtcccgaag	gtctccactt	5700
gctacgggtt	tcctatggct	gatccgggtg	aacgaatgct	tctttgagca	tcgaccaggc	5760
gctagcggac	gcctccgctc	tactcgtgtg	cgatctctcg	gcggagcagg	tcgtagactc	5820
ccagatcatt	cactggacag	gttccctcgc	accgacaact	ccacagaccc	gagcatggcg	5880
gacgggtctg	caccgcagc	tagaggcatt	gaaccagctc	ggacgggggc	gaattggctt	5940
agtcggggcg	tgggcatcgg	gttcgggcat	tgcgcgtctg	gtggcgagc	cccgcggaac	6000
tgtccgtcaa	cttgtctgat	ggcagactga	atccgcgtcg	gtactgagtg	ccacaatagg	6060
caatatgtct	gcgatccgtc	tgggaacccg	tgcagcact	ctggctacta	ctcaatcaga	6120
taggtgtgcg	aggctgctca	gatcggaggg	gctggacgtc	gatctgacga	cgatcactac	6180
tacgggggac	acctcgacgg	cctcgttagc	ggcgatggga	ggaatcgggg	tcttcgcctc	6240
ggcgatccgt	gcgacctccc	tcaagggaga	agccgacatt	gccgtccact	cattcaagga	6300

-continued

```

cctaccgaca ggaagtcgce tgggectgce gatcggtgce gttccagctc gacgagacce 6360
gcgtgacgct ttggtggccc gtgatgggct cacccttcac gaactccgc aggaagctag 6420
cgtcggaacg ggctcgccac gtccggccgc acagtgtgtg gcggtcagac cagacctgac 6480
gatcgtcgac atccgtggaa acgtcgacac ccggtcggtt cgtgtcaagg gactcggcca 6540
ctacgccaaa gggtggggca aggaggacct cgacgccgtc gtgctcgcgg cctctggact 6600
agctcgactg ggcatagcgc gcaccgtcac tgagtccctc gaccgctcgg tagtctgtcc 6660
tgccctgccc cagggagcgc tagccgtcga atgcccggacc gctgattcgc gtcattggcaa 6720
gctcgccaaag gcgttggcac ggattgacga ccgcagaacc aggtcgcccg ccactggcga 6780
acgtgcggtg ctctcccatc tggaggcggg gtgcgctgcc cccatcgagg catttggtca 6840
gctcaagcca gtttcgggaa gttcccccaca agtgttaact ctggatgttg tctagctac 6900
tgtcgatggc tcccgcatca ttccggaaaca ggtcagcgctc gaattacccg cggaggtgga 6960
ttcagcactg gctgctgcgc atacgctggg ggtgacggct gccaggaggt tgcgtgacga 7020
tggtgccgccc gacgttgccg atctcaaagc ttcaggaagc actcgatgac cacaaccct 7080
gcccgatctg tcaagccccc ggttgtcgtt ccccgcgacg actccacga caaatctgtg 7140
acgactttga agcaggccgg ttatggagtg atccacacag cgctgacgcg caccgtcacc 7200
ttgcccgaag cagagggggt cactaccctt gaactgtggc atgcccactg gctggtcgtc 7260
acctcgaaag caacgctcgg actgctccct acccgtgtgc cgaatcccaa catcaaagta 7320
gcccgggtcg gtgtcgctac ctccgctgcg ctgcgcacac gtgggatgta cgtcgacttc 7380
gtcccgaagc atcacagcgg cgctggtctg gtcccgaggt ggccaggcgg aaccgcccagc 7440
atctcctcac caacgtccca gttggccgccc gacacgctac cgctgggact aacgaagatc 7500
ggctgcacgg tgaaccgatt ggaggtctac accaccgctg ccgtcgagaa actacctgag 7560
gtcatcgccc agtctggccc gtctgtcggg gtccgctcgc tcaactgcctc ctcggtggga 7620
cgtgcccctg tccagctggt ggagccgctg ggatggcgctc atcagaagct cgtgccctc 7680
ggcaagccca ctgcaagaa acctcgaggag ctccgctcac cagcgatgc ggtggcctcc 7740
tcccctactc ctgaagccgt ccttgaggcg gttgagtcct tggtcgctg accgacggcc 7800
gttcgcgaag gaagttgatc atgaaccctg acctgctga tgcagtgat ccccgcatgg 7860
ctggggctcaa gccattgccc gaggggctgc gtccgtgttc tctgtccgct cgactacgca 7920
ccactcctgc ctgcccgcga atggctgcgc aaaccgctg cgtccccgcc cagttgatac 7980
tgccggcggt cgtcaagggg ggggtcgatt ccccgtcga gatcaccag ctcgcccggc 8040
aatctcagca ctccaccgag tctgattaaag agctcgccac ccaagctgcc gagggccggt 8100
tcggtggtat tgacctgttt ggtatccccg aacataaaga cgagatcggt tctcaggcct 8160
gggacccgaa aggcattctc aaccgcccga tctctgctgt gcgcgagggc gtcggcgacg 8220
atctggtgat ctgtgcccag acctgtctag acgagttcac cgaacatgga cattgcccga 8280
ctctgacccc tgatggtcgc gtcgataatg actccaccgt gccgtatat gccgcgatgg 8340
cgggtctccca agcgcgcgccc ggagcccaca tgggtctccc gtcggggatg atggacggtc 8400
agatcgctgt catctcgcat gctctggatc gcgagggatt caccgacgtg tccatcatgg 8460
cctattcagc taagtacgct tcaagcctct ttggccctct ccgggacgccc gtagactgtt 8520
cccttaaggg tgatcgcaaa gcttaccagc aggatccatc gaaccgcccg gaaggcctac 8580
gcgaaacctc tctcgacctc gccgagggag ccgatctggt catggtgaag cccgcctccc 8640
actacctcga cgtgtgtccc gatgtcgcgc aggtgtcgca ggtaccggtt gccgcttacc 8700
aggtctctgg agaatacgcc atgtctgagg ccgctgccc caacggttgg atcgatcgcc 8760
cagcggtgat cggagagtc ctgacctcga tctgtcgtgc cggtgcccgt ctctgtttga 8820
cctactgggc aatcgaagcg gccagatgt tccgcgagga cctctgaccc gtggtgatca 8880
tgctcaccac aaaccccaca ccggaaggaa gacatgacct ccaacgcgca actgttctct 8940
gcccggcagg ccgtcattcc cggcgggggt gattccccgg tgcgggctta cggctcggta 9000
ggcgggggtc tccgcttcac cgagaaggcc ggaatctggga cgccgaaggc 9060
acgcgctacg tccgatctgt gtgcacctgg ggcccggctc tgcctgggca tgcgcgtccc 9120
gaggtcgtct ctccggttca ggaggccgct tcgaaggggc tgccttttgg cgcgccgacc 9180
cgacccgaaa ccgaactagc ggaatgccatc atcgagcggg tggatccggt cgagaagggtg 9240
cgtttcgtct cgacaggcac tgaggcgaca atgaactcgg tccgtttggc ccgtggcgcc 9300
accggtcgtg acgtcatcgt caagtttgcc ggttaattacc accggccattc tgacgcccgt 9360
ctccgcccgc ccggttccgg tgaagcagc gctggtctac ccggttcggc cgggtgtgcca 9420
gcccgcctcga ctgcccagac tatttggtgt gattacaacg acgtcgcgc cctcgacgcg 9480
gtgttctcgc agcgtggcga ccagattgct gcagtgatcg tggagtccgt cccggccaat 9540
atgggagttg ttccgcccga gcccggtctc aatgcgcgca ttcgtcgctt cacaaccgag 9600
cacggcgccc tcatgatcac cgacgaagta ctcaccgat tccgggtcag cccgtcggga 9660
ttctgggggt tgcaacagca ggccggcgag gacttcgcgc ccgacatttt caccctcgcc 9720
aaggtcgtgg gcggaggcat gccgctgggt gccacgcccga tgcattggac 9780
ttgctggctc ctaccggccc cgtctaccag gccggcactt tgtccggtaa cccgctggcc 9840
accgtcgcgg gtgtcaaaac cctacagctg gccgacgccc aggtgtatca gccctcgtat 9900
gtccgctccg agaaatggcg caatgagctc gaatccgcac ttgacgcagc cggcgtcacg 9960
taccgctcgc agaaccgccc aaatttcttc agcgtattcc tccgctcga ctcgccggtg 10020
cgcaattacg acaacgccaa gtctcagaat gccgaggctt acaccgctt cttccacgcg 10080
atgctagatg ccggggtcaa cctgcccgca tctgtctttg aggcctgggt cctctcgga 10140
gctcacgacg acgaagcttt cgaggttttc cgcgcgccc tgcgaggggc tgcggaggcg 10200
gctgcccagg tgatcagtcg ctga 10224

```

```

SEQ ID NO: 3      moltype = DNA length = 1224
FEATURE          Location/Qualifiers
source           1..1224
                 mol_type = genomic DNA
                 note = hemA reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 3
gtgctgtcaa cgtgcaaccg ggtgtgctg atcgttgaaa cctcccctga agcgtcgcg 60
caagggttcc acgaggccgc gttacgcagg tgcattgccc atcaccgagc aaacgtgtta 120
gccgaatctg accagctcgt ctgtgagAAC gacgccgttt ggctgttgtt cgggtgggct 180
gcgggcattg actcgatggg ttttggggaa cgtgaggtcg ccgacagat gaagcgtgct 240
ctcagtgagg cgcgcgctga acagaccgtc tcctacacca ttggtcacgt cgttgaggaa 300

```

-continued

```

gctctcaaga cgtcacgtca tgcctgtacc gagactgcgt tggtgtgtga gggacgcacc 360
gttgtgcgcg tcggggttga tctcgttgcc cagcggatgg gcttggatgg cgtcgggtg 420
ctcgttatgg ggactggctc ctatgcctgg cgtcgtgctg cccagctgag ttctcggggg 480
gtcgcgcgaga ttcaagtcca ttacgcctcc ggacgtgccc cgggatttgc tcgacgccac 540
cgggtcagtg aggtctctga catcgacgcc gtgctggcac aagctgatct tgcgtcacc 600
tgccgtgggt cggggggttc tgcgtgtctg gctgaggctg ctcggcgccg ggtcgatgct 660
cgaaggggccc gagatctcat ggtgctcgat ttagcaatca gtggggacgt tgaggagcct 720
gtccccgctg gggtagaggt catcgacctt gagacgatcc ggcaggctgt tcccgcctct 780
gcggaagctg agcggggccc cgccgaacac atcatcgcta cgggagtcgg tcaactttgc 840
gtagacctcg aacgtcgtcg gatggccccg gctgtcgtcg ctctgcgcga cgtcatttca 900
gatctcgtaa ctgctgaact ggagcgtctg cccgaagaag gctcgggtccc tgcgatgag 960
gtagcggcgt cccctcgacg gctcgtgctg tctatggcgc acatcccgtc ggctcgagcc 1020
aggatggcct ctgagcaggg cttgggcgat cgttggctta actcattgtc ggatgtattg 1080
gggacgcagc ttgacatcgc ggcaccctgc attgacatgt ccagcttcgc caatgctgac 1140
tgcattgacgt gtcgggttac cggcctgctg gtggaggacc ttgcgcgaga tgcggcacc 1200
cgaggtgagg agaggacatc gtga 1224

```

```

SEQ ID NO: 4      moltype = DNA length = 1047
FEATURE          Location/Qualifiers
source           1..1047
                 mol_type = genomic DNA
                 note = hemB reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 4
atgaaccctg acctgctga tgcacgtgat ccccgcatgg ctgggggtcaa gccattgccc 60
gaggggctgc gtctgtttc tctcgcgtcg cactacgca ccactcctgc gatgcgccga 120
atggtcgccg aaaccctgct cgtccccgcc cagttgatac tgccggcggtt cgtcaagggg 180
ggggtcgatt ccccctgcat gatcaccagc ctgcccggcc aatttcagca ctccaccgag 240
tcgattaaag agcctgcacc ccaagctgcc gagggcgcta tcggtggtat tgacctgttt 300
ggtatccccg aacataaaga cgagatcggt tctcaggcct gggaccgcga aggcacacct 360
aacgcccgca tctctgctgt gcgcgagggc gtccgagcag atctggtgat ctgtgccgac 420
acctgtctag accagttcac cgaacatgga cattgcggag ctctgacccc tgatggctgc 480
gtcgataatg actccacctg gccgctatat gccgcgatgg cggctctccc agcgcgcgcc 540
ggagcccaca tggctctccc gtcggggatg atggacggtc agatcgctgt cattcgcgat 600
gctctggatc gcgagggatt caccgacgtg tccatcatgg cctattcagc taagtacgct 660
tcagccttct ttggccctct cggggacgcc gttagactgt cccctaaagg tgatcgcaaa 720
gcttaccagc aggatccatc gaaccgcccg gaaggcctac gcgaaccctc tctcgacctc 780
gccgagggag ccgatctggt catggtgaag cccgctccc actacctcga cgtgtgtgctc 840
gatgtcgccg aggtgtcgca ggtaccggtt gccgcttacc aggtctctgg agaatacgcc 900
atgctcgagg ccgctgcgcgc caacggttgg atcgatcgcc gacggtgcac cggagagtcc 960
ctgacctcga tctgtgcctg cgggtgcgat ctctgtttga cctactgggc aatcgaaagg 1020
gcccagatgt tccgcgagga cctctga 1047

```

```

SEQ ID NO: 5      moltype = DNA length = 1005
FEATURE          Location/Qualifiers
source           1..1005
                 mol_type = genomic DNA
                 note = hemC reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 5
atgtctgcga tccgtctggg aaccctgtgc agcactctgg ctactactca atcagagatg 60
gttgcgaggg tgctcagatc ggaaggggctg gacgtcgatc tgacgacgat cactactcac 120
ggggacacct cgacggcctc gttagcggcg atgggaggaa tcgggggtct cgcctcggcg 180
atcgtgcgca cctcctcaca gggagaagcc gacattgccc tccactcatt caaggacctc 240
ccgacaggac tccgctgagg cctggcgatc ggtgcgcttc cagctcgagc agaccgcgt 300
gacgttttgg ttgcccgtga tggggtcacc cttcacgacc tcccgcagga agctagcgtc 360
ggaacgggct cgccacgtcg ggcgcacag ttgttgccgg tcagaccaga cctgacgatc 420
gtcgacatcc gtggaaacct cgacaccggt cttgggtcgtg tcaagggaact cggccactac 480
gcccagggtg ttggcaagga ggacctcgac gccgtcgtgc tcgcccctc tggactagct 540
cgactgggtc atagcggcac cgtcactgag ttcctcgacc cgtcggtagt gctgcctgcc 600
cctgcccagg gagcgctagc cgtcgaatgc cggacgcgtg attcgcgtca tggcaagctc 660
gccaaggcgt ttggacggat tgacgaccgc agaaccaggg tggccgccac tgcggaaagt 720
gcggtgctct cccatctgga ggcgggggtg gctgccccca tcggagcatt ggctcagctc 780
aagccagttt cgggaagtct ccacaaagt ttaactctgg atgttgcgt agctactgtc 840
gatggctccc gcatcattcg ggaacaggtc agcgtcgaat taccgcgga ggtggattca 900
gcactggctg ctgcgcatac gctgggggtg acggctgccc aggagttgct ggacgatggt 960
gccgcccagc ttgcagatct caaagcttca ggaagcactc gatga 1005

```

```

SEQ ID NO: 6      moltype = DNA length = 726
FEATURE          Location/Qualifiers
source           1..726
                 mol_type = genomic DNA
                 note = hemD reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 6
atgaccacaa accctgcccga tctcgtcaag ccccggttg tcgttccccg cgacgactcc 60
caccgacaaat tctgtagcac ttgaaagcag gccggttatg gactgatcca cacagcgtcg 120
acgcgcaccg tcaccttgcc ggaacgcagag ggggtcacta cccctgacct gtggcatgcc 180
gactggctgg tcgtcacctc gaagaccacc gtccgactgc tccctacccc gttgccgaat 240

```

-continued

```

cccaacatca aagtagccgc ggtcggtgtc gctacctccg ctgcgctgcg cacacgtggg 300
attgacgtcg acttcgtccc cgacgatcac agcggcgctg gtctgggtcg cgagtggcca 360
ggcggaaccg ccagcatcct cctaccaacg tcccagttgg ccgccgacac cgtaccgctg 420
ggactaacga agatcggtcg cacggtgaac cgattggagg tctacaccac cgtgcccgtc 480
gagaaactac ctgagggtcat cgcccagtcg tggccgtctg tcgggggtcg cgctgcact 540
gcctcctcgg tgggacgtgc cctcgctcag ctggtggagc cgctgggatg gcgtcatcag 600
aagctcgtcg cctcggcga gcccaactgca agaactcgc aggagctcgg tcacccagcg 660
catgcggtgg cctcctcccc tactcctgaa gccgtccttg aggcgggtga gtccttggtc 720
gcctga 726

```

```

SEQ ID NO: 7      moltype = DNA length = 1281
FEATURE          Location/Qualifiers
source           1..1281
                 mol_type = genomic DNA
                 note = hemE reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 7
atgacggtgc aatgccaggg caagtcggtc aacctgcccg gccgcctcgg aaacgacgcc 60
aagccctcgc tcggcatgat ccacgggtcaa gacacacaa cccacgaagc gttattcaac 120
catgctgtaa cgacgaaccc cagtacata gaagctcata actcgcccaa taatcgtccc 180
atgggcgcca tccgcacgtg gtggggcaca atcggggtga tgactgagac cactaccttc 240
acggcagatt caccgcttct gcaagccatg actgggcacc gccctaccgt cgccccgggtg 300
tggtttatgc gccaaagcagg ccgttcgcta ccggagtagc gcgaagcccg agagggcacc 360
acgatgctcg agtcttgct caacctgag ctgcgccgcc agattacctg ccaacctgtg 420
cgccgccacc acgtcgacgg ggccgttctc tactccgaca tcatggtgcc gctagccctg 480
cgggggcgag aggtgcgcat tgagccagga gtccggcccg ttctcgacga gcccgtcgcc 540
actgcctcgc acgttgaccg cattactcga cgtcatgtcg aggacctcgc ggctatcgag 600
gaggtgcac gcctggcgat caccagatta ggcgacgtca caccgctcat cggtttcgcg 660
ggtgccccgt ttactatcgc cgctacctc gtccagggcg gccctcccg cgacctcgc 720
tctgccccgg cgatgatgca ctgagacca cagacctggg caaggtcat ggagtgggtt 780
gccgacttgg atgctgcctt cctggctgct cagctgcgcg gcggggcgcg tgcctaccag 840
ctctttgatt cctgggtggg gtccgtgggc gcgagcgatt atcgagccag cgtgcaccag 900
ttctctcagc gggctctggc tacctcgat ccgaacatcc cggttggtca ctttggcgtc 960
aatgccaccc atctgctgct tagttttgcg caggtggcag ctactgcgtc acggtacca 1020
gtacttgggt tcgaccacgg gattagccct gatgaggcct gcgccactct tatggtctcc 1080
gggatggaga tgcgggttca gggcaacatc gatcccgctc tgctcttcgc cggtatggaa 1140
ccgctggaac ccgccacccg cgatatcgct acggcaggac gtcaggcccc tggacacgtc 1200
gtcaatctgg gtcacggcgt gccctccaac accgacccca acgtcttgac acggctgggtg 1260
gagctgggtc actccctgtg a 1281

```

```

SEQ ID NO: 8      moltype = DNA length = 2052
FEATURE          Location/Qualifiers
source           1..2052
                 mol_type = genomic DNA
                 note = hemH reference
                 organism = Cutibacterium acnes

```

```

SEQUENCE: 8
gtgacgcgta acccctacgc tgcceccacg gaccggttgg ccccttatgg cgccgtgctt 60
gtcgtttctt ttggcggtcc cggttcccc gaagagggtca tgccttttct gagaagggtc 120
tcccacggtc gcactccctga agaacgcttg gctgatgtcg cccggcacta tgaccgttcc 180
ggcgggtgta gcccgattaa cgatgcgacg gacgttttgc tcaacgcgat tgggaatgaa 240
ctgcgcgcgt ccggagatcg agttccagtg ctgttaggga accgtaacgg taccaccttc 300
cttgaggaag cgctgactga catgcacgcg cacggggtag gtccgggtgt agcgggtgtc 360
acctctgcct accccagttc ctccgggtgc cgtcagtagc gcgaggagat cgcgactgct 420
ctggctcacg tcggtatcac cgatatgcag gtccacaagg taccaccttc caacgaggcg 480
ccgggattca ttccgcgtaa ccgggaagct ctgatgcagg cgtttatgcg gatccacct 540
actcctctcg aagctacccg cgtcgttttc gtcactcact ccactccctga ctctatgcag 600
gacgcacccg gcgctgggca gccgggaacg gactacattt cccagcacia agcgggttgt 660
gagaagggtc ccggccaggt gcgtcagggt ttccgggaata tgcgcgagtg ggacttggcc 720
tactgctcga ggtcgggggc tcccacgac ccgtgggttg aaccggacat tatcgatcac 780
ctccgcaatc ttcccgagca gggcgtgcag tctgttgtcg tgcctccaat tggcttcgtt 840
gctgaccaca tggaggtcgt taatgacctt gattacgagg ccgcccaggc agccaagggtg 900
tccgggtcgc ccttcacgcg gccggcgacc gctggcacc atcccgctt tatcgctgat 960
ctgcgcgggc tcatcttgct ccaggccgcg gcggctcgtg gtgaggggtg caatctaacg 1020
tctgtggcgg ctcctcgcgc cgctggctgt tgcgcagcgt acccgagcgc tggagacatt 1080
ccaaccgtct ccggtagtga tgcagagtct gtacgtgccg gtgcccagtg agtcgatgct 1140
gaaccaggag gtgcgggttt tgtgcccagc ggctcagcga gtgcagtcga tgcgccgggg 1200
ccagaggcgg tcgaactgga gactcctcct tcccctaca acccggtgac taaggagacc 1260
cccatgtctg accattcgag tgcgtattcg gtgattgagg gaccacgcga cgacgaagtt 1320
cccgccggct cctacaccgc accgaccgat ccccgcgaca ccccggtcat tcccaggag 1380
gtcaacgcct ccagcaaatg ggcgatgtac tccgtattcc gcgtgcgac cgcgcttcc 1440
gccgaggatg acgagcgtcg ccgccttgtt gagggttccg acgagtgggc gggacactcc 1500
ggtgtcgaca ctcggtgatg gtacgacttg tcaggcctgc gcgctaacgc cgacctgctg 1560
gtgtgggtgg tttagcagca tcggcggtg ctgcaggacg cctaccatcg tttccgcga 1620
tctggggtgg ggcgccacct tgaaccgggt tgggtcgaaag taggagttca tgcctcgtcg 1680
gagttcaaca agtctcact gcctcatgc ttgcgcgca tgcgccgag gcgttgggt 1740
gcgttttacc ccttcattcg ctccaaggag tggtagctcc ttccggcggc tgatcgttcc 1800
cgcatgctgc gcgaacacgg tatcgttggc gcagccagct ccgacgtcaa ggcgtcgacg 1860
ttggcgccat tgcgctcggg ggactacgag tggattctcg ccttggaggg tgacgacctc 1920

```


-continued

```

gcccgtgttg ttgatgtcat gaaggacett cgttacgttg aagcccgtcg ttatgtcgac 1980
gtggataccc ccttctttac cggcgagcgc gtttctccgg tgggtgtgggc cgaccgccag 2040
atgagagcct ga                                     2052

```

```

SEQ ID NO: 9          moltype = DNA length = 1311
FEATURE              Location/Qualifiers
source                1..1311
                     mol_type = genomic DNA
                     note = hemL reference
                     organism = Cutibacterium acnes

```

```

SEQUENCE: 9
atgacctcca acgccaact gttctctgcc gccaggccgc tcattcccgg cggggttgat 60
tccccggtgc gggcttacgg ctccgttaggc ggggttccgc gcttcacga gaaggcccaa 120
ggcccgtgga tctgggacgc cgaaggcacg cgtacgtcg atctgggtg cacctggggc 180
ccggctctgc ttgggcatgc ccgtcccagc gtcgtctctg cggttcagga ggccgcgtcg 240
aagggggtgt cctttggcgc ccgacccgc accgaaaccg aactagcgga tgccatcatc 300
gagcggatgg atccggtcga gaaggtgcgt ttcgtctcga caggcaactga ggcgacaatg 360
actcgggtcc gtttggcccg tggcgcaccc ggtcgtgacg tcatcgtcaa gtttgcgggt 420
aattaccacg gccattctga cgccctgctc gccgcgcgcg gttccgggtg agcgacgggt 480
ggctaccacg gttcggcgcg tgtgccagcc gcctcgactg ccgacactat tgtggtggat 540
tacaacgacg tcgcgcgcct cgacgcgggt tctcccgagc gtggcgacca gattgctgca 600
gtgatcgtgg agtcctgccc ggccaatatg ggagttgttc cgcccgagcc cggttcaat 660
gcccggatcc gtcgcctcac aaccgagcac ggcgcctca tgaacacga cgaagtactc 720
accggattcc ggtgcagccc gtcgggattc tggggtttgc aacagcagcg cgccgaggac 780
ttcgcgcccc acattttcac cttcgccaag gtcgtggggc gaggcatgcc gctggctgcc 840
ctgggtgggc accgcgatgt catggacttg ctggctccta ccggccccgc ctaccaggcc 900
ggcactttgt ccggttaacc gctggccacc gtcgcgggtg tcaaaacccct acagctggcc 960
gacgcccagg tgtatcagcg cctcgatgct cgctccgaga aatggcgcaa tgagctcgaa 1020
tcgcacttgg acgcagccgg cgtcacgtac cgcctgcaga acgcccggaaa tttcttcagc 1080
gtattcctcg gcgtcgactc cccggtgcgc aattacgaca acgccaagtc tcagaatgcc 1140
gaggettaca ccgcgttctt ccacgcgatg ctagatgccg gggtcacact gccgccatcg 1200
tgctttgagg cctgggttct ctcggagcgt cagcagcagc aagctttcga ggttttcgcg 1260
gcgcgcctgc cgagggtcgc ccaggcggtc gccagggtga tcagtgcctg a 1311

```

```

SEQ ID NO: 10         moltype = DNA length = 1431
FEATURE              Location/Qualifiers
source                1..1431
                     mol_type = genomic DNA
                     note = hemY reference
                     organism = Cutibacterium acnes

```

```

SEQUENCE: 10
atgaacattt ctgatccgca gcacgtcgac gccgtcgta ttggcggggg catgggtggc 60
ctcgtggccg cctggcagtt caccaggag ggtctgcacc cggtaactgt cgaatcccgc 120
gggtacactg gtgggctgat tgccagggtc acgctggccg gagtcagcta cgacatccgg 180
gccgaggggt gggcggtccg tctaccagac accgctgaac tggccaccga gttgggggtg 240
acggttgaga agccgaccaa gcgcccgct tgggtgtact tcgatgacgc ctgcttcgcg 300
atgcccgata atgccattct tggcatcccg gcgaatttgt cggaccacgc cgtcgtcatc 360
gctctgggag cggaaaaggc tgctcgcgcc gctgagctcg atcgcgcgcc catgcctgag 420
gatcttcctc gccactctgg tactcctctg gcctcccga tgggtccaga agtaactcga 480
cgtctcgtca ccccaatcgc cgggggaatc cagcccgccg atccacatct actgtcagcc 540
gacgtcgtct ccccggtct gagggctgcc gccaccggca ccggatcgct agcccgggga 600
gttgcgctgt gccgtcacg gatgccagcg gggcccgca tgcgtcggt agccggtggc 660
atgttcacga tgcgggccga gctgcgtcgt caggcagaa acggccggcg gacgaccatg 720
acccgagtag gagctcgagg actaacccgc cagcaagatc agtggctcgt ggagaccgct 780
ggaacctctc gcaatccgga tccatcttgg ccccccgtca ccaacggcga gactcgtttc 840
ttcctgaccc ctggggtcgt cgtagcctgt tcgggtggac ctgccacaaa actgctcagc 900
aatgtcatcg agaccgttgg acccgagatg gcgccaggcg cgccgatcgg ccacgtcaat 960
ctcgccctgc aggcaccgga gctagactcc agcccacgcg gaaacggcat gctggtggcg 1020
cccggaacga ccaccgttac tgccaaggca cttaccaca tgtcgcgcaa gtggcccagc 1080
ttgagagaat catgtcccga aggtctccac ttgctacggg tttcctatgg tcgatccggt 1140
gaaaacgaat tctcttttga catcgaccag cgctagcggc acgctccgt cctactcggt 1200
gtcgatctct cggcggagca ggtcgtagac tccagatca ttcactggac aggttccctc 1260
gcaccgacaa ctccacagac ccgagcatgg cgcgacggtc tgcaccgcca gctagaggca 1320
ttgaaccagt ctgacggggg gcgaattggc ttagtcgggg cgtgggcatc gggttcgggc 1380
attgccgctc ttggtggcga gggccgcgca actgtccgtc aacttgtctg a 1431

```

```

SEQ ID NO: 11         moltype = AA length = 1298
FEATURE              Location/Qualifiers
source                1..1298
                     mol_type = protein
                     organism = synthetic construct

```

```

SEQUENCE: 11
MTKTFDSEFF NLYSLQKTVR FELKPVGETA SFVEDFKNEG LKRVVSEDER RAVDYQKVKE 60
IIDDYHRDFI EESLNYFPEQ VSKDALEQAF HLYQKLKAAK VEEREKALKE WEALQKKLRE 120
KVVKCFSDSN KARFSRIDKK ELIKEDLINW LVAQNREDDI PTVETFNNFT TYFTGFHENR 180
KNIYSKDDHA TAISFRLIHE NLPKFFDNVI SFNKLKEGFP ELKFDVKED LEVDYDLKHA 240
FEIEYFVNFV TQAGIDQYNY LLGGKTLEDG TKKQGMNEQI NLFKQQQTRD KARQIPKLIP 300
LPKQILSERT ESQSFIPKQF ESQELFDSL QKLHNNCQDK FTVLQQAILG LAEADLKKVF 360
IKTSDLNALS NTIFGNYSVF SDALNLYKES LKTKKAQAEF EKLPAHSIHD LIQYLEQFNS 420

```

SLDAEKQOST	DTVLNYFIKT	DELYSRFIFS	TSEFTQVQP	LFEALSSK	RRPPESEDE	480
AKQGEPEQI	KRIKAYLDTL	MEAVHFAKPL	YLVKGRKMIE	GLDKQDSFYE	AFEMAYGLE	540
SLIIPYINKA	KRSLRKPFP	ADAKKFINFDN	NTLSSGDWAN	KETANASILF	KKDGYYLGI	600
MPKGKPLFD	YFVSSEDEK	LKQRRQKTAE	BALAQDGBESY	FEBKIRYKLLP	GASKMLPKVF	660
FSNKNIGFYN	PSDDILIRTN	TASHTKNGPT	QKQHSKFVEFN	LNDCHKMIDF	FKSSIQKHPE	720
QSGPFTTDS	TDSDFMSAF	YREYENQGVV	ISFPDKIKETY	IQSQVEQYLN	YLPQIYNKDF	780
SPYSKGKPNL	HTLYWKALFE	EANLNNUVAK	LNGEAEIFFR	RHSIKASDKV	VHPANQAIDN	840
KNPHTETQS	TFEYDLVKDK	RYTKQDFPHL	VPISLNFKAP	GVSFKNDKVC	GPLKGMPDVN	900
IIGIDRGRHS	LIIFYTTVMNQ	GELLQVQFEN	TLMSDGKHVN	DYQKLDKKKE	QERDARAKSW	960
TTVENIKELK	EGYLSHVVHK	LAHLIIKYNA	IVCLEDNFGF	FKRGRKFVEK	QVYQKPEKAL	1020
IDKLNLYLVFK	BEKEGVEGHY	LTAYQTATFP	ESPFKLKGQS	GLTFYVPADY	TSKIDPTTGF	1080
VNFLLDRYQS	VEKARQLLSD	FNAIRNSQVQ	NYFBEIDYKQ	KILTPKRKYVQ	QSKWVICTYS	1140
DVRYQNRNRQ	KGHWETEEVN	VTEKLKALFA	SDSKTTTVID	YANDDNLDIV	ILEQDKASFP	1200
KELLWLKLT	MTLRSHSKTS	EDDFILSPVK	NEQGEFYDSE	KAGEVWPKDA	DANGATHIAL	1260
KGLWNLOOIN	OWEKGKTLNL	AIKNODWFSF	TOEQPFYOR			1290

```

SEQ ID NO: 12      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

SEQUENCE: 12
 aqaagagctt tcgctcttca 20

```
SEQ ID NO: 13      moltype = DNA   length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct
```

SEQUENCE: 13
ttctcaaaact cgttacgggt 20

```

SEQ ID NO: 14      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

SEQUENCE: 14
acgaacgccg qcaqcattaa 20

```
SEQ ID NO: 15      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct
```

```

SEQUENCE: 15
acgatgacgt cacqcccagt                                     20

```

```
SEQ ID NO: 16      moltype = DNA  length = 20
FEATURE            Location/Qualifiers
source             1..20
                  mol_type = other DNA
                  organism = synthetic construct
```

SEQUENCE: 16
cgctctactc ggcgtggacc 20

```
SEQ ID NO: 17      moltype = DNA  length = 132
FEATURE            Location/Qualifiers
source             1..132
                  mol_type = genomic DNA
                  organism = Cutibacterium acnes
```

SEQUENCE: 17
ctgttcggtg tctctacatc gacgattcgc cgcgatgtcg atgccctctc ggatgaatcc 60
aagatctgga agatttcggg gggagacgtc atgatccgac gacacgagcc tacctggcac 120
qacaaqqaac aa

```
SEQ ID NO: 18      moltype = AA  length = 44
FEATURE            Location/Qualifiers
source             1..44
                  mol_type = protein
                  organism = Cutibacterium acnes
```

SEQUENCE: 18
LFGVSTSTIR RDVDALSDDES KIWKISGGDV MIRRHEPTWH EKEO 44

```
SEQ ID NO: 19      moltype = DNA  length = 134
FEATURE            Location/Qualifiers
source             1..134
                  mol_type = genomic DNA
                  organism = Cutibacterium acnes
```

-continued

SEQUENCE: 19
 ctgttcggtg tctctacatc gacgattcgc cgcgatgtcg atgccctctc ggatgaatcc 60
 aagatctgga agatttcgga gggggagacg tcatgatccg acgacacgag cctacctggc 120
 acgagaagga acaa 134

SEQ ID NO: 20 moltype = AA length = 31
 FEATURE Location/Qualifiers
 source 1..31
 mol_type = protein
 organism = Cutibacterium acnes

SEQUENCE: 20
 LFGVSTSTIR RDVDALSDS KIWKISGGET S

31

I claim:

1. A method for treating a *Cutibacterium acnes* bacteria-associated disease in a subject, said method comprising reducing the ratio of the amount of α -type *C. acnes* bacteria to the amount of β -type *C. acnes* bacteria in the subject by administering to the subject a nucleic acid encoding a programmable nuclease, prime editor, or base editor designed to specifically target said α -type *C. acnes* bacteria at a nucleotide selected from one

of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9 and wherein the first nucleotide in the sequence is indexed as 0, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10 and wherein the first nucleotide in the sequence is indexed as 0,

wherein the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1; and wherein β -type *C. acnes* bacteria are *C. acnes* bacteria which are not α -type *C. acnes* bacteria.

2. The method according to claim 1, wherein said method comprises specifically reducing the amount of α -type *C. acnes* bacteria on the skin of said subject.

3. The method according to claim 1, wherein said method comprises administering to said subject a therapeutically effective amount of a phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, or engineered bacteria comprising said nucleic acid encoding a programmable nuclease, prime editor, or a base editor designed to specifically target said α -type *C. acnes* bacteria.

4. The method according to claim 3, wherein said programmable nuclease is selected from the group consisting of CRISPR-Cas nucleases, TALENs, zinc finger nucleases, and engineered meganuclease.

5. The method according to claim 4, wherein said programmable nuclease is a CRISPR-Cas nuclease.

6. The method according to claim 5, wherein said phage, recombinant phage, packaged phagemid, plasmid, DNA- or RNA-containing vesicle, or engineered bacteria, further encodes a guide RNA designed to specifically target said α -type *C. acnes* bacteria.

7. The method according to claim 6, wherein said guide RNA is designed to specifically target the hem locus of said α -type *C. acnes* bacteria selected from the group consisting of SEQ ID Nos: 13-16.

8. The method according to claim 3, wherein said administration is a topical administration.

9. The method according to claim 1, wherein said nucleic acid encodes a base editor or a prime editor.

10. The method according to claim 1, wherein said method comprises increasing the amount of β -type *C. acnes* bacteria in said subject on the skin of said subject.

11. The method according to claim 1, wherein said *C. acnes* bacteria-associated disease is a *C. acnes* bacteria-associated inflammatory disease.

12. The method according to claim 1, wherein said *C. acnes* bacteria-associated disease is acne vulgaris.

13. A method for treating a *C. acnes* bacteria-associated disease in a subject, said method comprising specifically reducing the expression of at least one hem locus protein in α -type *C. acnes* bacteria in said subject by administering to said subject a nucleic acid encoding a programmable nuclease, prime editor, or base editor designed to specifically target said α -type *C. acnes* bacteria at a nucleotide selected from one

of the following nucleotide variants in the hem locus:

in the hemA gene: T173, A479, T506, A614, T629, C671, T971, A989, T1085, and T1094, wherein the positions are defined with reference to sequence SEQ ID NO: 3 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemB gene: C142, A155, A176, A183, C224, C245, G248, T281, G311, C314, T346, C374, G377, C452, T626, C683, A693, A720, C738, C794, A920, T944, T992, G1022, and T1031, wherein the positions are defined with reference to sequence SEQ ID NO: 4 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemC gene: G77, A96, C149, A233, T236, T255, A257, A260, C317, A347, T425, T437, A464, A486, A487, C488, A524, A545, A590, T605, A644, A665, C710, C713, A737, A770, A776, T785, T802, G830, C842, T845, G855, C893, A896, C897, G902, G908, A941, and A947, wherein the positions are defined with reference to sequence SEQ ID NO: 5 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemD gene: T11, T41, T50, A150, T155, G159, G176, T257, C266, T278, G323, A324, T362, T367, T374, G387, A398, A399, T407, G422, G425, G432, A446, A459, T464, G595, T617, T643, A644, and C680, wherein the positions are defined with reference to sequence SEQ ID NO: 6 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemE gene: T241, T476, A530, T704, A728, G767, T834, A867, A872, A882, and A957, wherein the positions are defined with reference to sequence SEQ ID NO: 7 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemH gene: T11, T23, A48, T89, C188, G338, T464, T470, G536, T548, T608, G626, T632, T641, T671, T674, C683, C689, A740, G745, G787, A797, T893, A898, T1034, T1039, T1046, C1071, G1095, C1109, T1153, T1358, G1496, C1511, C1527, G1532, T1598, T1712, A1850, G1901, A1926, C1931, and C1997, wherein the positions are defined with reference to sequence SEQ ID NO: 8 and wherein the first nucleotide in the sequence is indexed as 0,

in the hemL gene: T107, C197, T229, A242, C310, A338, A374, T389, G392, T442, C467, G470, C482, G518, and C641, wherein the positions are defined with reference to sequence SEQ ID NO: 9 and wherein the first nucleotide in the sequence is indexed as 0, and

in the hemY gene: A89, A236, A354, C383, C392, A395, T398, T465, T474, C521, T538, C566, A593, A767, T782, G863, G920, T927, A932, G935, C979, G989, C1019, T1055, G1091, T1136, G1143, G1152, T1166, T1172, A1176, T1184, C1199, C1205, G1268, G1278, T1295, T1318, A1336, G1343, C1346, G1355, T1359 and C1395, wherein the positions are defined with reference to sequence SEQ ID NO: 10 and wherein the first nucleotide in the sequence is indexed as 0,

wherein the nucleic acid sequence of the hem locus is at least 97% identical to the sequence SEQ ID NO: 1.

14. The method according to claim 1, wherein the programmable nuclease recognizes nucleotide variant A144 in the hemB gene.

15. The method according to claim 1, wherein the programmable nuclease recognizes nucleotide variants C389 and T392 in the hemL gene.

16. The method according to claim 1, wherein the programmable nuclease recognizes nucleotide variants T1199 and T1205 in the hemY gene.

17. The method according to claim 1, wherein the programmable nuclease recognizes nucleotide variant A144 in the hemB gene.

18. The method according to claim 13, wherein the programmable nuclease recognizes nucleotide variants C389 and T392 in the hemL gene.

19. The method according to claim 13, wherein the programmable nuclease recognizes nucleotide variants T1199 and T1205 in the hemY gene.

* * * * *